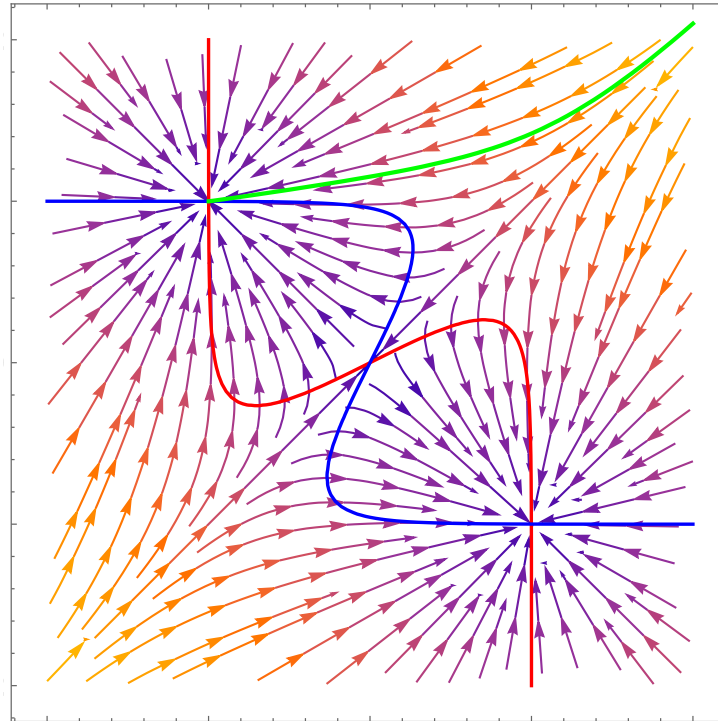


Modelling decision-making with dynamical systems

Konstanz summer school of collective behaviour – KSCB 2025

Marco Fele



Adapted from notes and work by Alessio Franci and Anastasia Bizyaeva – all mistakes mine

Topic summary

1. **Why dynamical systems are bad for modelling decision-making**

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2. **Introduce a baseline model of decision-making**

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6. **Multi-agent decision-making and the ring attractor**

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6. **Multi-agent decision-making and the ring attractor**

Focus on conceptual understanding rather than implementation

Answer questions

Code to replicate figures in GitHub repository

What are dynamical systems

We want to describe ant trails



What are dynamical systems

We want to describe ant trails



Microscopic level



What are dynamical systems

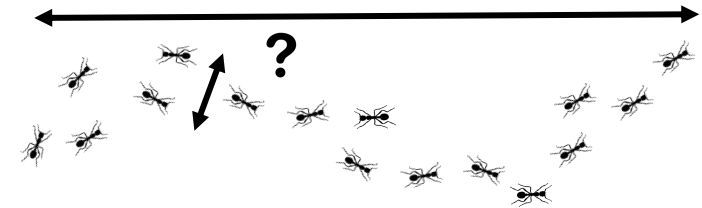
We want to describe ant trails



Microscopic level



Macroscopic level



What are dynamical systems

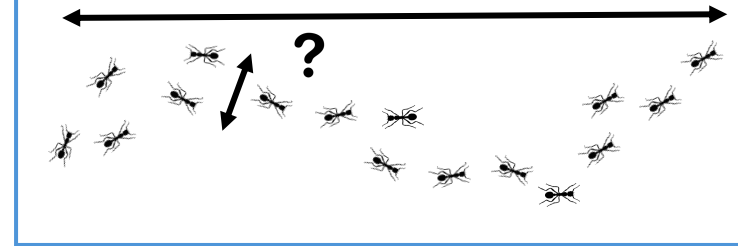
Dynamical system describe
the macroscopic level



Microscopic level



Macroscopic level



$$\dot{x} = f(x, p)$$


What are dynamical systems

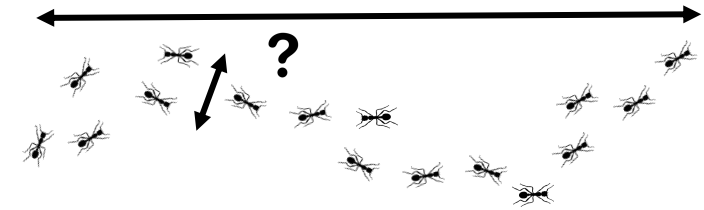
Dynamical system describe
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Microscopic level



Macroscopic level



$$\dot{x} = f(x, p)$$

State variables $x \in \mathbb{R}^n$

Parameters $p \in \mathbb{R}^k$

What are dynamical systems

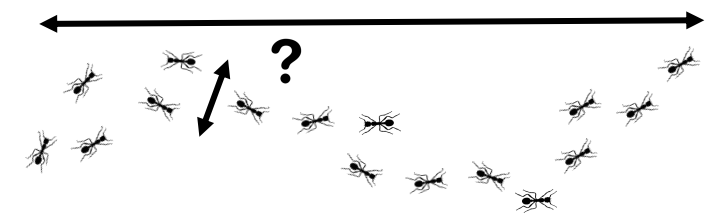
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$$\dot{x} = f(x, p)$$

State variables $x \in \mathbb{R}^n$
Can change

Parameters $p \in \mathbb{R}^k$
Don't change

What are dynamical systems

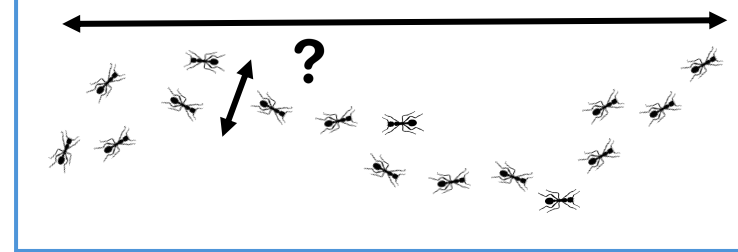
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Microscopic level



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$$\dot{x} = f(x, p)$$

State variables $x \in \mathbb{R}^n$
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Parameters $p \in \mathbb{R}^k$
Don't change

Important assumptions:

- Continuous state variables
- Time evolution of state variables is deterministic

What are dynamical systems

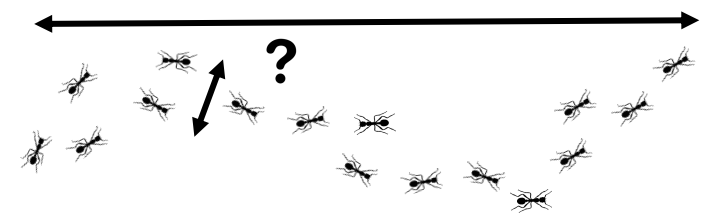
Problems



Microscopic level



Macroscopic level



$$\dot{x} = f(x, p)$$

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Can change

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Important assumptions:

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2.

1.

3.

4.

Part 1

The problems with dynamical systems



1. The problem with continuous state variables

A colony of *Temnothorax*
ants must choose a new
nest

A

B

1. The problem with continuous state variables

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A

B

Tandem running



1. The problem with continuous state variables

A colony of *Temnothorax* ants must choose a new nest

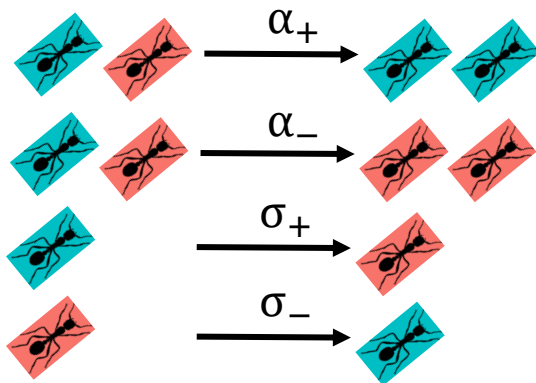
A

B

Tandem running



Voter model



1. The problem with continuous state variables

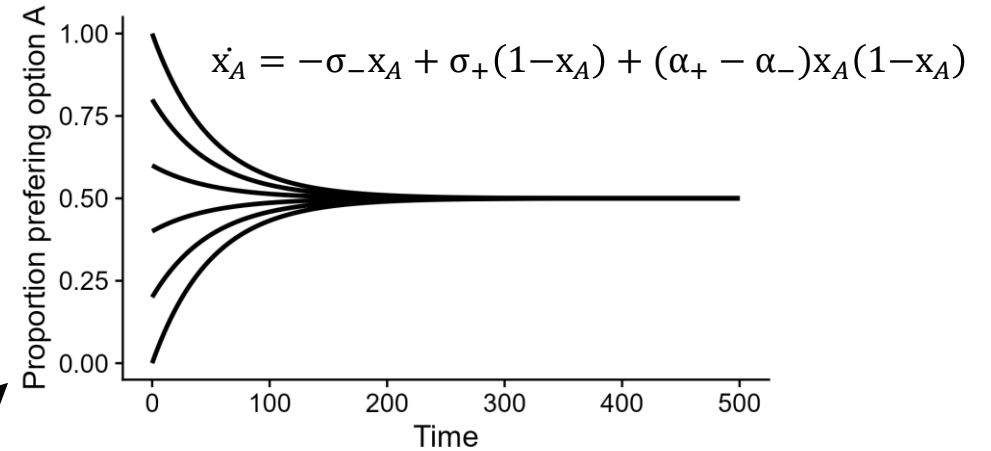
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Tandem running

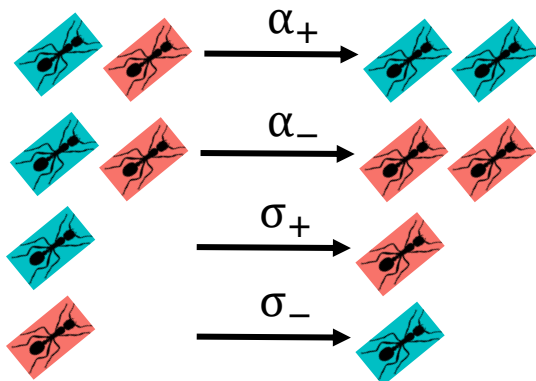


Assuming infinite population size



Tomorrow - how to derive dynamical systems from microscopic rules

Voter model



1. The problem with continuous state variables

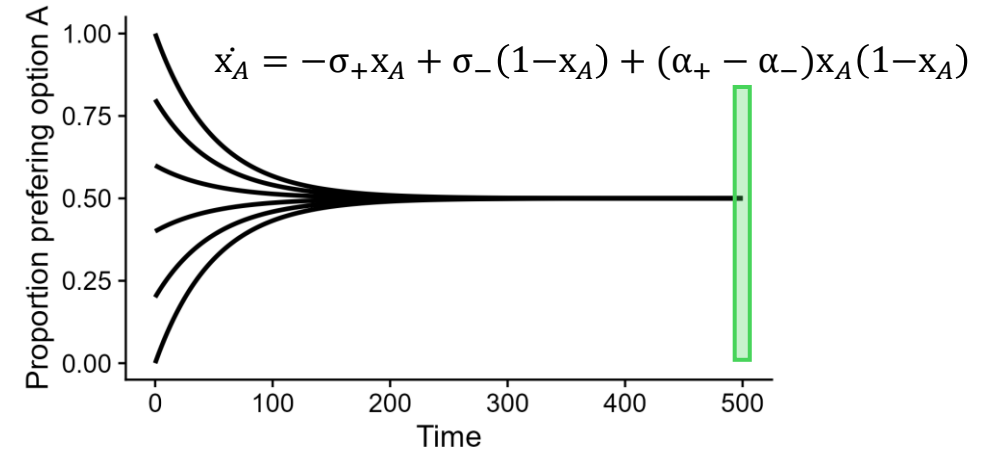
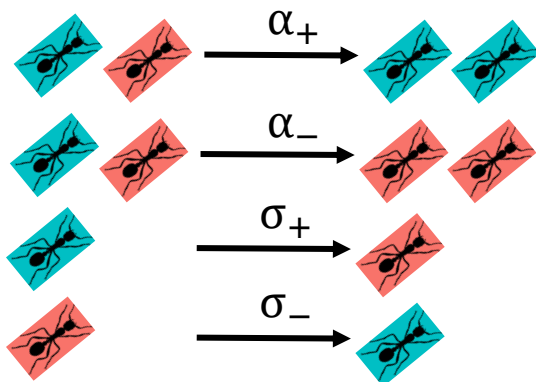
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Tandem running

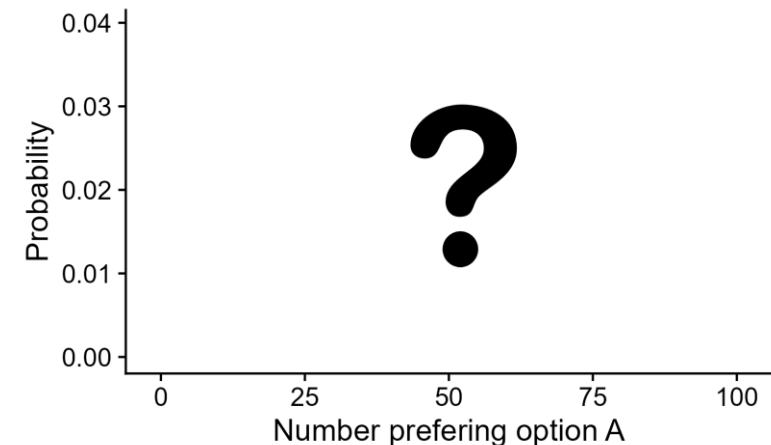


Voter model



At the **end of the task**, with 100 ants, what is the shape of the probability mass for the number of ants preferring option A:

- a) Flat
- b) Approximately gaussian (centered at 50)
- c) Bimodal



1. The problem with continuous state variables

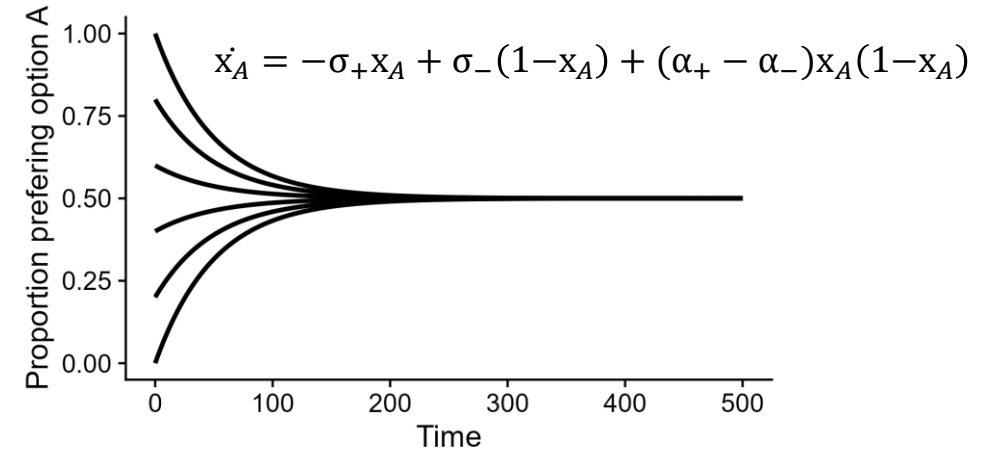
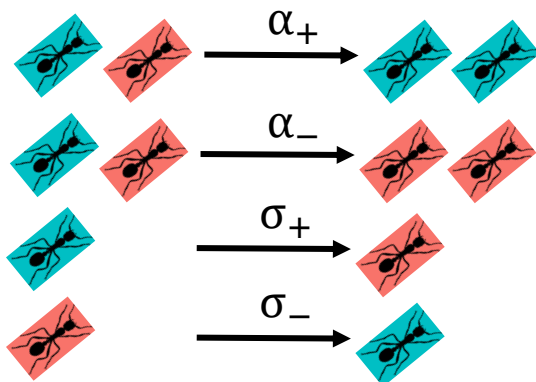
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Tandem running

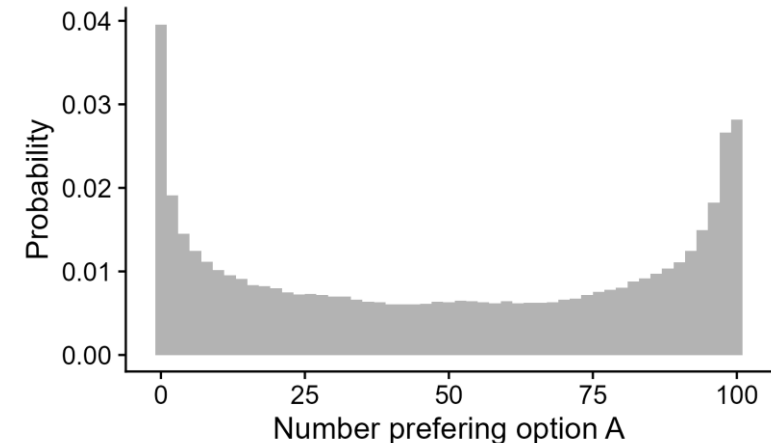


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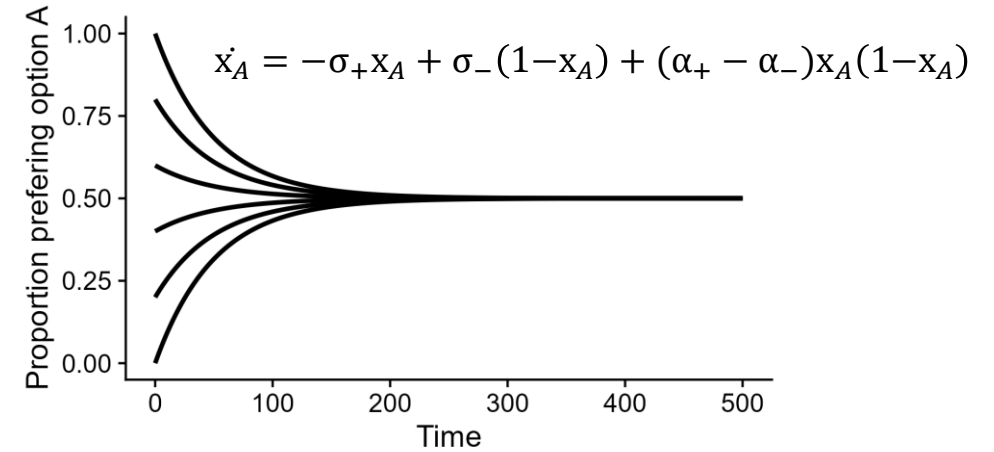


1. The problem with continuous state variables

A colony of *Temnothorax* ants must choose a new nest

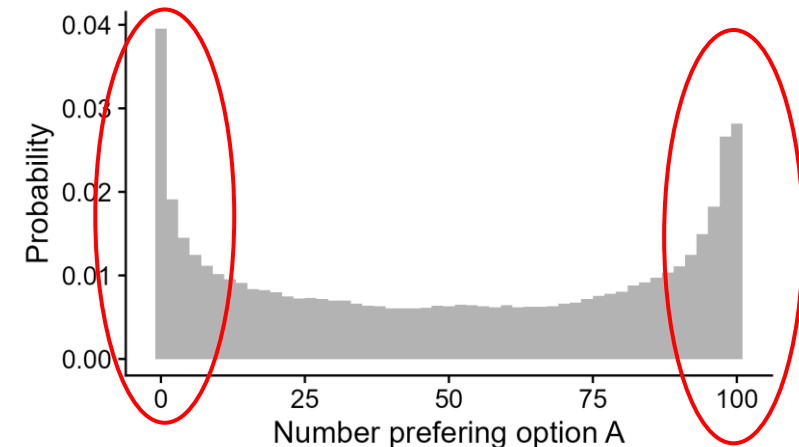
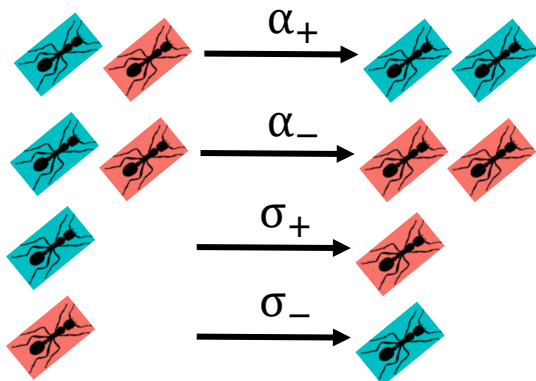


Tandem running



Very relevant for collective decision-making:
the colony mostly agrees!

Voter
model

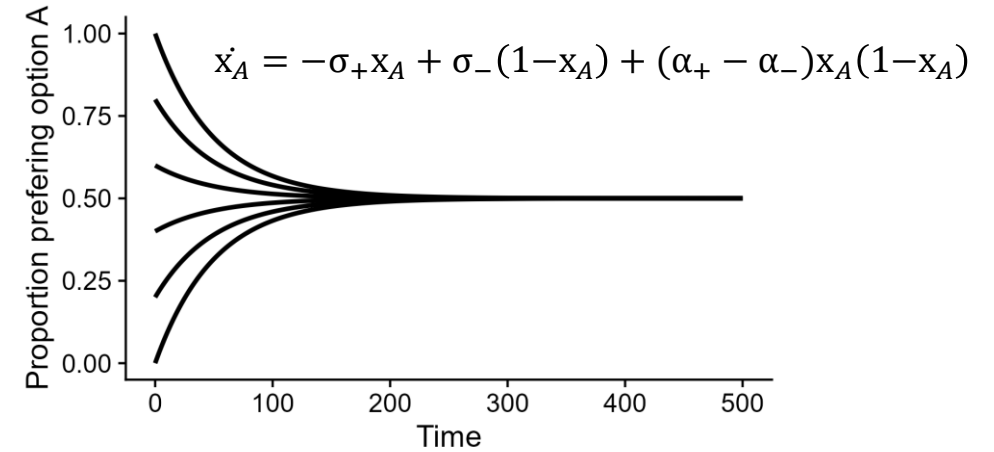


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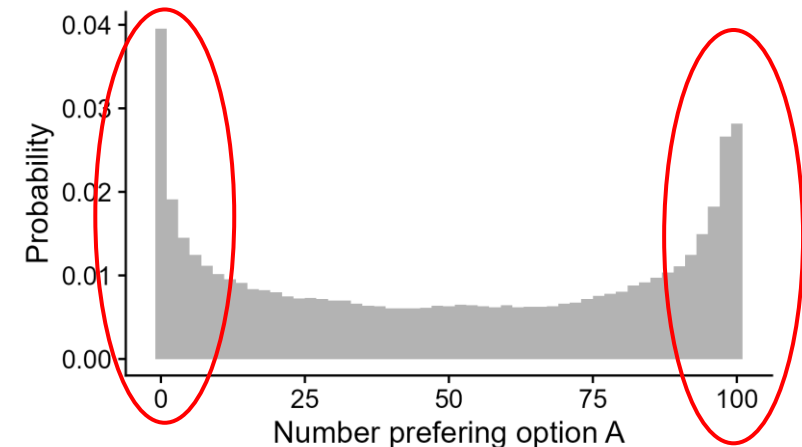
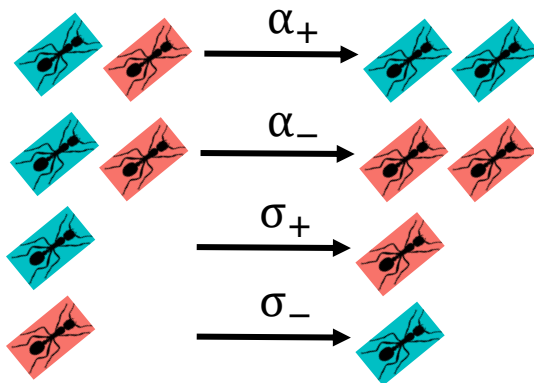


Tandem running



Macroscopic models can miss
dynamical outcomes
Noise induced effects

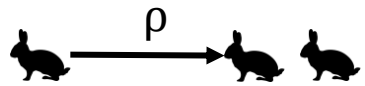
Voter
model



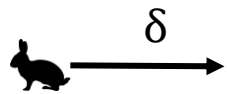
2. The problem with macroscopic descriptions

Logistic growth model

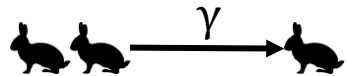
Clonal reproduction



Death



Density dependent death



r : population growth rate

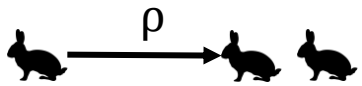
K : carrying capacity

$$\dot{x} = rx \left(1 - \frac{x}{K}\right)$$

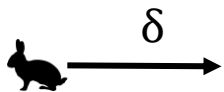
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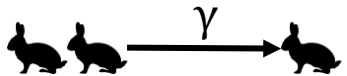
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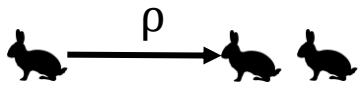
By increasing the rate of clonal reproduction, the population will change:

- a) Faster
- b) Same speed
- c) Slower

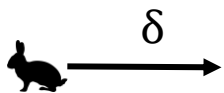
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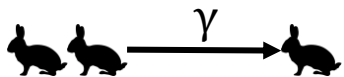
Clonal reproduction



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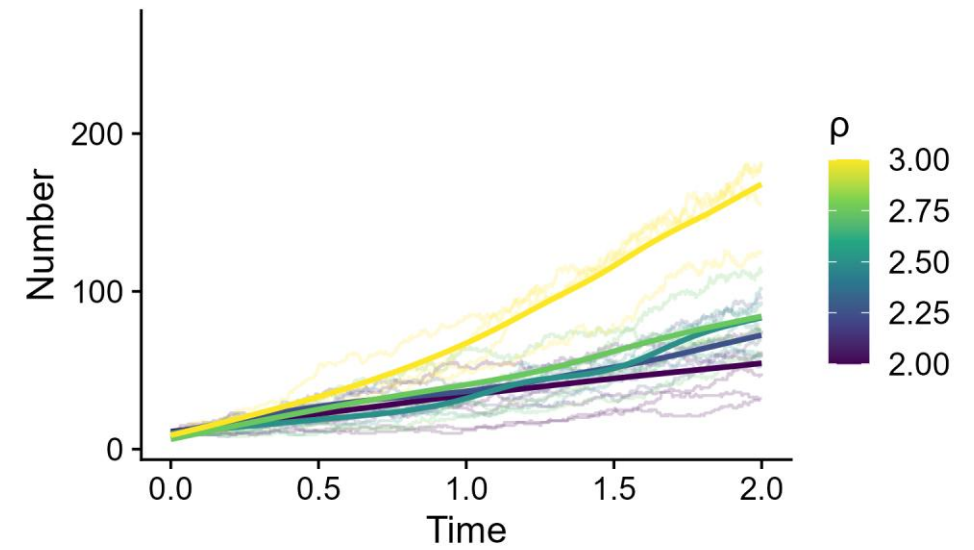


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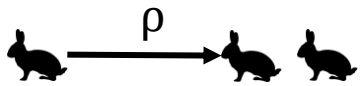
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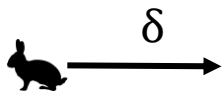
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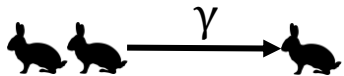
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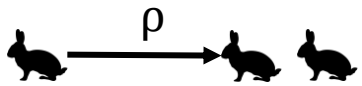
By increasing the rate of clonal reproduction, the population will saturate:

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- c) At a higher number

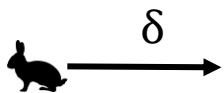
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Logistic growth model

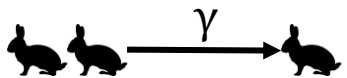
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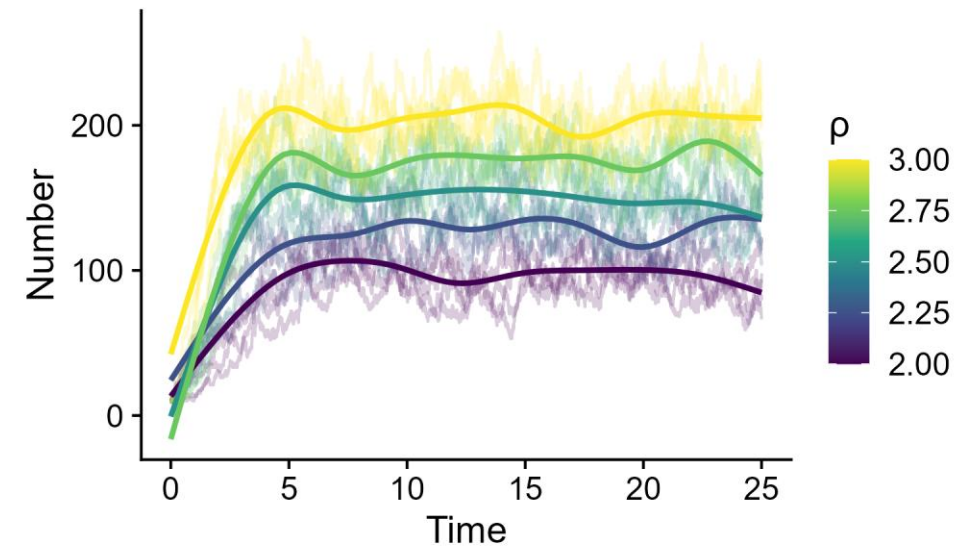


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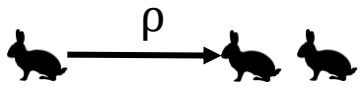
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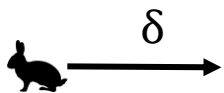
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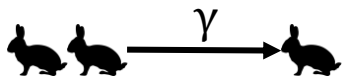
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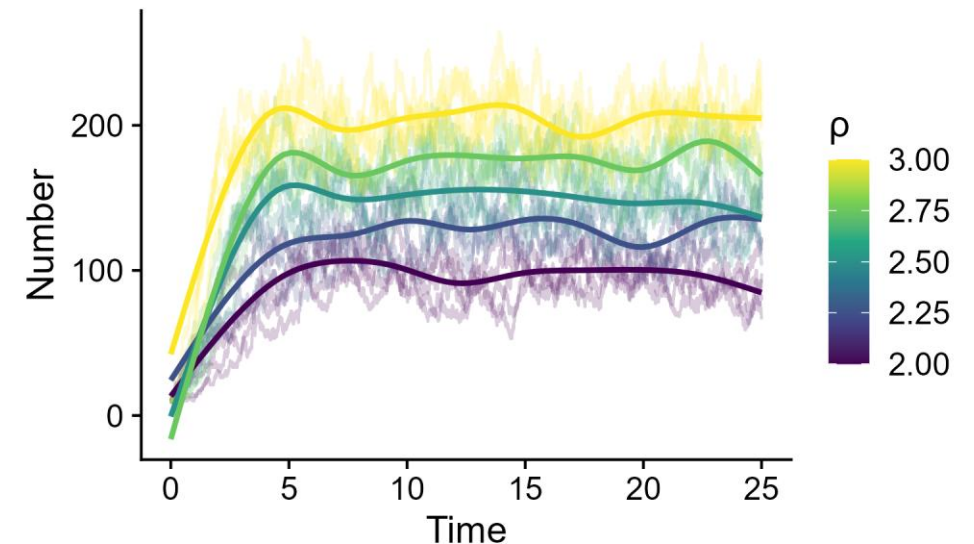
$$\dot{x} = r x \left(1 - \frac{x}{K}\right)$$

$$K = \frac{\rho - \delta}{\gamma}$$

$$r = \rho - \delta$$

By increasing the rate of clonal reproduction, the population will saturate:

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Population growth rate and carrying capacity are NOT independent

3. The problem with deterministic evolution of state variables

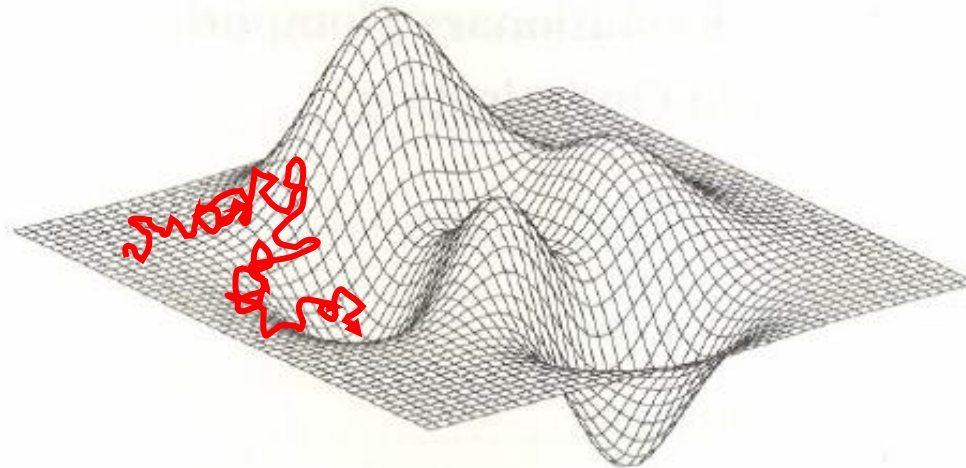
Everything that is stochastic will be missed

Accuracy (probability of choosing best option) is stochastic!

3. The problem with deterministic evolution of state variables

Everything that is stochastic will be missed

Accuracy (probability of choosing best option) is stochastic!



Dynamical systems “describe the landscape” over which decisions “move” ...

Part 2

A baseline model of decision-making



Non-linear opinion dynamics model (NOD)

Decision variable

Input or bias

$$\dot{x} = -dx + \tanh(ux + b)$$

Leak

Attention

One agent, two options

Non-linear opinion dynamics model (NOD)

Decision variable

Input or bias

$$\dot{x} = -dx + \tanh(ux + b)$$

Leak

Attention

With $d > 0$, does x go to infinity:

- a) Never
- b) Depends on parameters
- c) Always

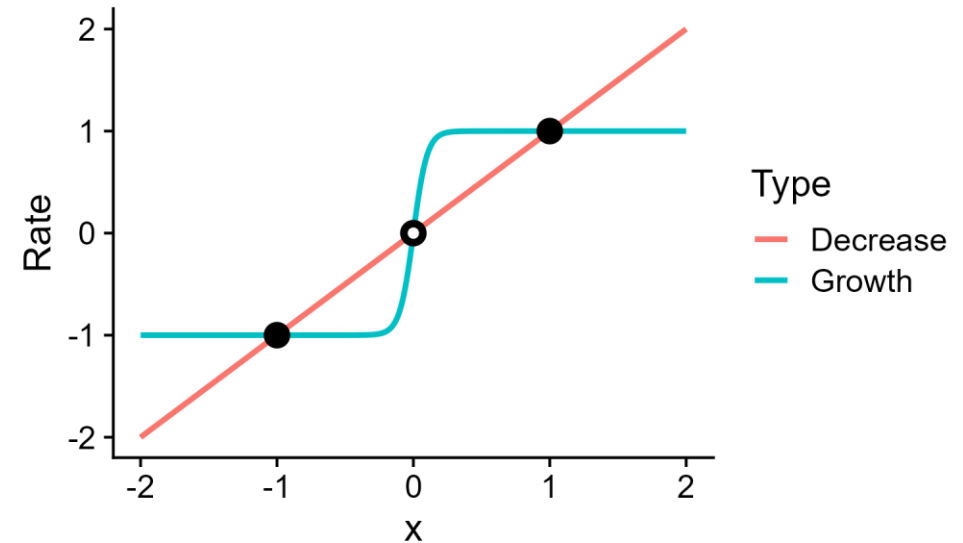
Non-linear opinion dynamics model (NOD)

Linear Constant (for big $|x|$)

$$\dot{x} = -dx + \tanh(ux + b)$$

With $d > 0$, does x go to infinity:

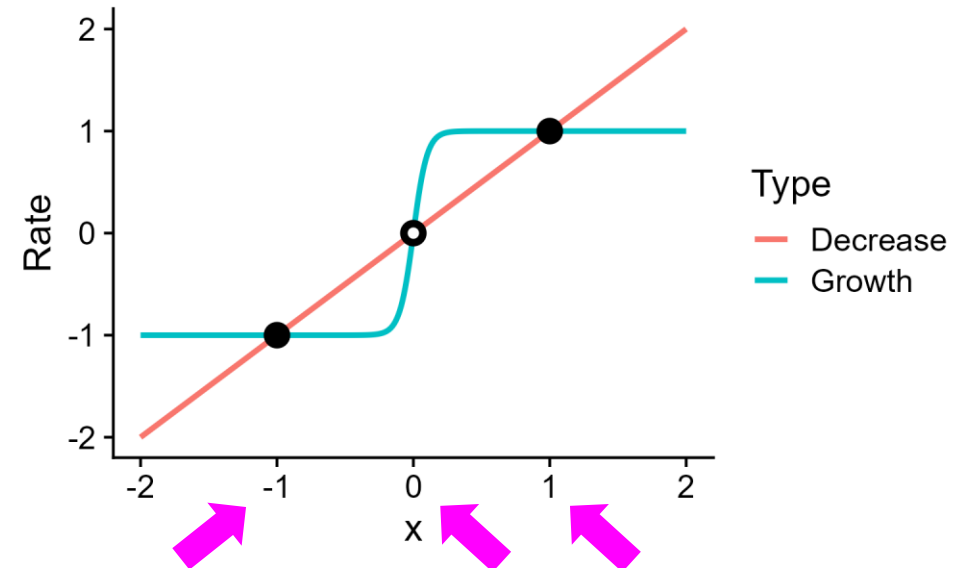
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Non-linear opinion dynamics model (NOD)

$$\dot{x} = -dx + \tanh(ux + b)$$

Equilibria $\rightarrow f(x^*, p) = 0$



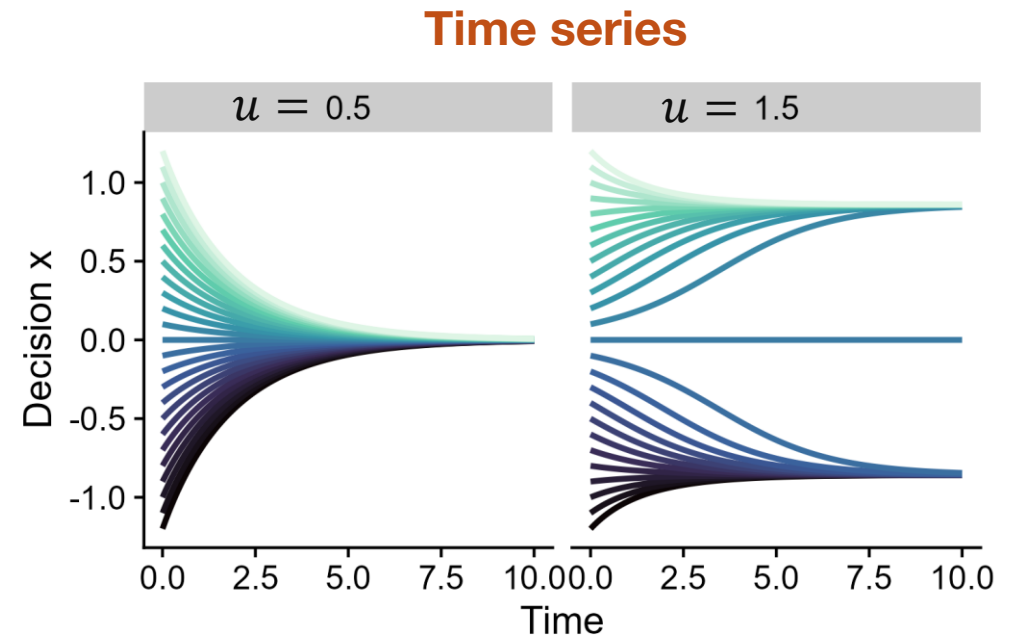
Non-linear opinion dynamics model (NOD)

$$\dot{x} = -dx + \tanh(ux + b)$$

Equilibria $\rightarrow f(x^*, p) = 0$

Qualitative analysis: explicit closed form solution for x^*

Quantitative analysis: explicit closed form solution as function of time $x(t) = g(p, t)$



Non-linear opinion dynamics model (NOD)

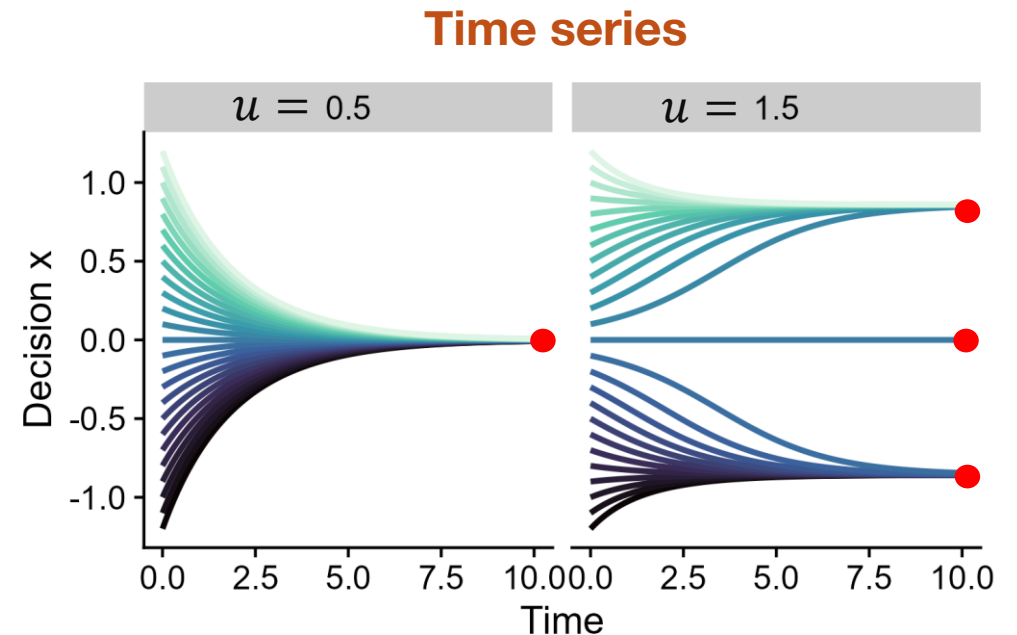
$$\dot{x} = -dx + \tanh(ux + b)$$

Equilibria $\rightarrow f(x^*, p) = 0$

Qualitative analysis: explicit closed form solution for x^*

Quantitative analysis: explicit closed form solution as function of time $x(t) = g(p, t)$

“Less information, more *informative*”



Part 3

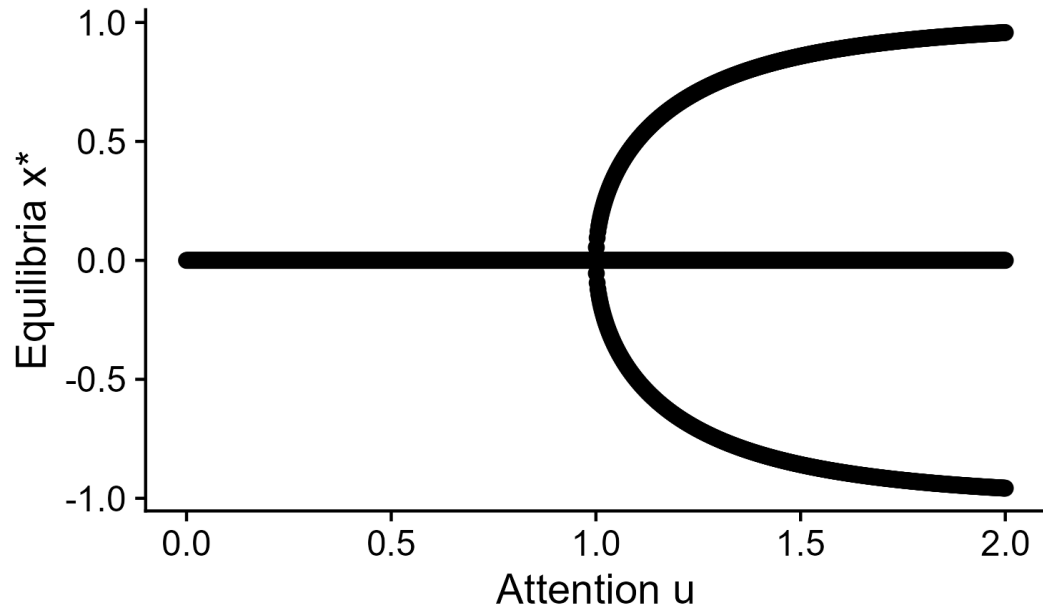
Bifurcations

1	2	3	4	5	6
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Bifurcation diagrams

Equilibria
instead of
state variable

$$\dot{x} = -x + \tanh(ux)$$



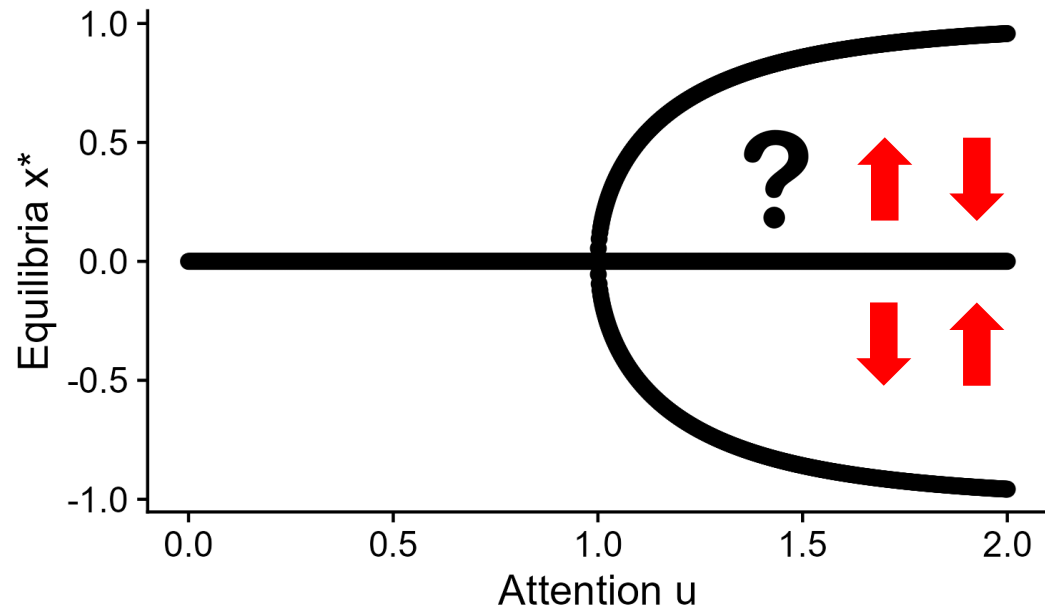
Equilibrium branch

Parameter instead of time

Bifurcation diagrams

Equilibria
instead of
state variable

$$\dot{x} = -x + \tanh(ux)$$



Parameter instead of time

Stability of equilibria

Bifurcation diagrams

Stability criterion:

1. derive the system in respect to the state variables (Jacobian matrix)
2. evaluate at the equilibrium
3. find max eigenvalue(s) λ_{max}
4. If $\lambda_{max} < 0$ **stable**, $\lambda_{max} > 0$ **unstable**

$$J(\mathbf{x}^*, \mathbf{p}) = \left(\begin{array}{ccc} \frac{\partial f_1(\mathbf{x}, \mathbf{p})}{\partial x_1} & \dots & \frac{\partial f_1(\mathbf{x}, \mathbf{p})}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n(\mathbf{x}, \mathbf{p})}{\partial x_1} & \dots & \frac{\partial f_n(\mathbf{x}, \mathbf{p})}{\partial x_n} \end{array} \right) \bigg|_{\mathbf{x}=\mathbf{x}^*}$$

Bifurcation diagrams

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$$J(x^*, p) = \left(\begin{array}{ccc} \frac{\partial f_1(x, p)}{\partial x_1} & \dots & \frac{\partial f_1(x, p)}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n(x, p)}{\partial x_1} & \dots & \frac{\partial f_n(x, p)}{\partial x_n} \end{array} \right) \bigg|_{x=x^*}$$

$$f(x, p) = -x + \tanh(10x)$$

$$f(x^*, p) = 0$$

$$J(x, p)$$

$$J(x, p)_{x=x^*}$$

Calculate $J(x, p)$

Bifurcation diagrams

Stability criterion:

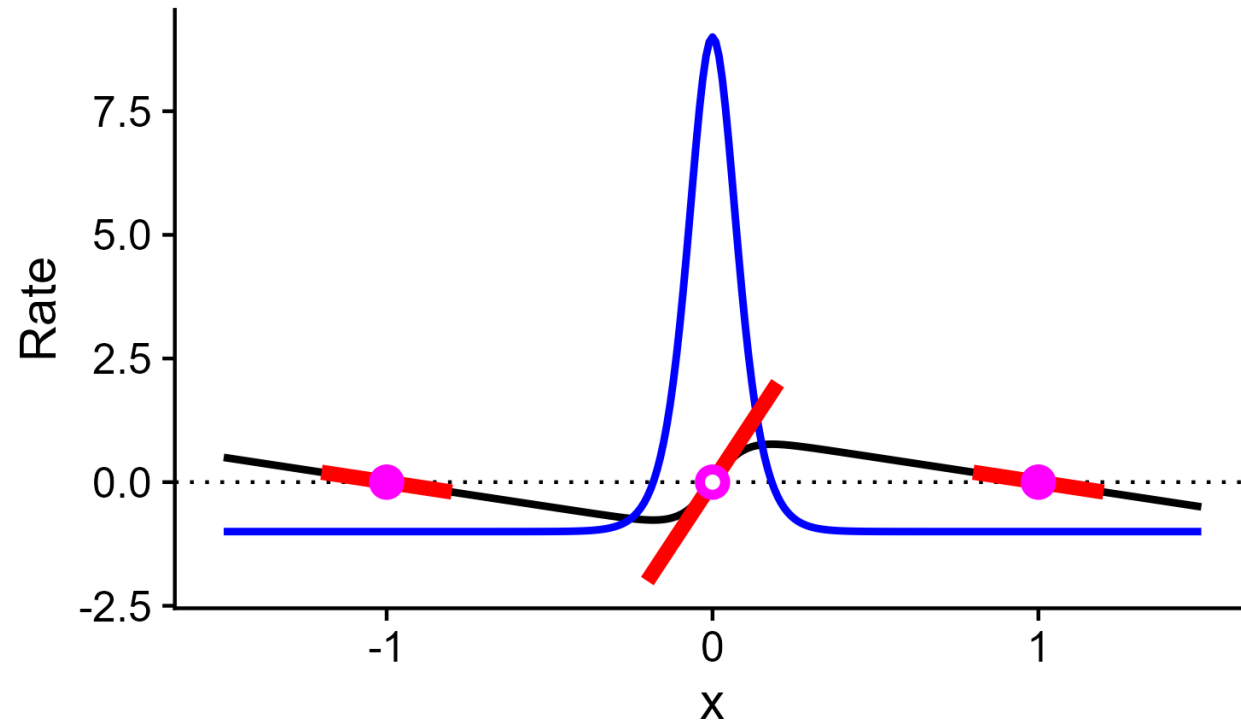
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$$f(x, p) = -x + \tanh(10x)$$

$$f(x^*, p) = 0$$

$$J(x, p) = -1 + 10\operatorname{sech}(10x)^2$$

$$J(x, p)_{x=x^*}$$



Bifurcation diagrams

Stability criterion:

1. derive the system in respect to the state variables (Jacobian matrix)
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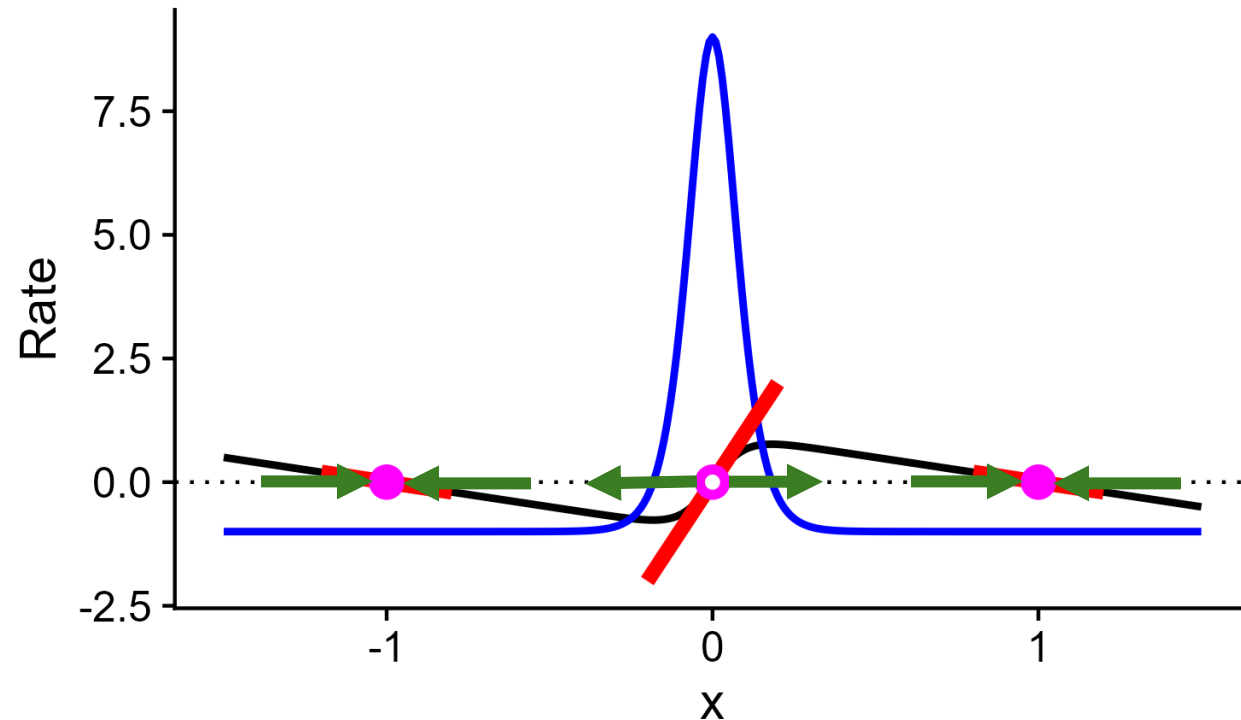
Why?

$$f(x, p) = -x + \tanh(10x)$$

$$f(x^*, p) = 0$$

$$J(x, p) = -1 + 10\operatorname{sech}(10x)^2$$

$$J(x, p)_{x=x^*}$$



Bifurcation diagrams

Stability criterion:

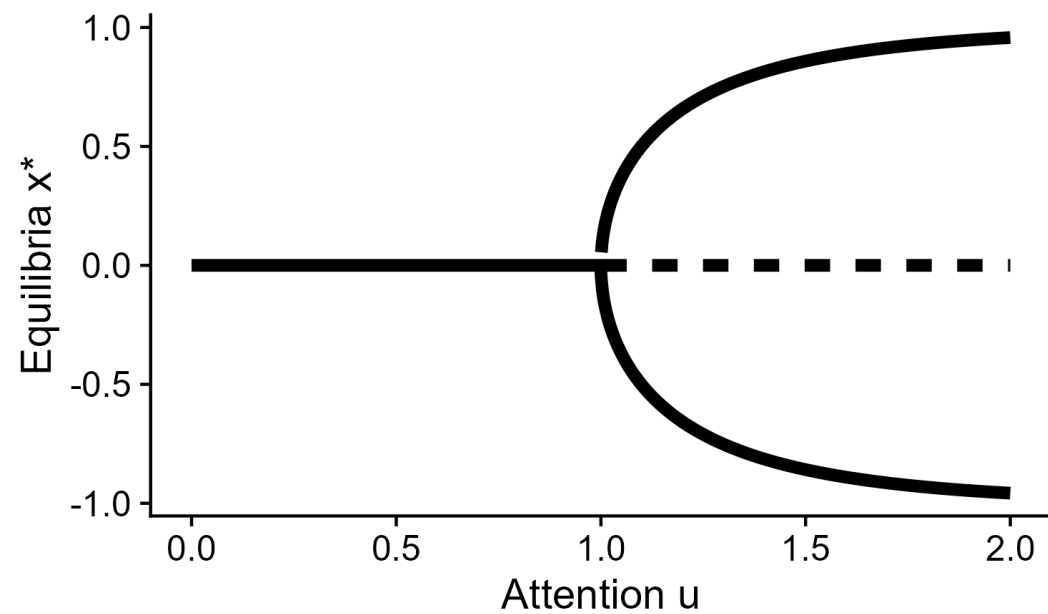
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Caution

This criteria fails when at least on eigenvalue is 0. Other tools can be used to define stability.
See end if curious

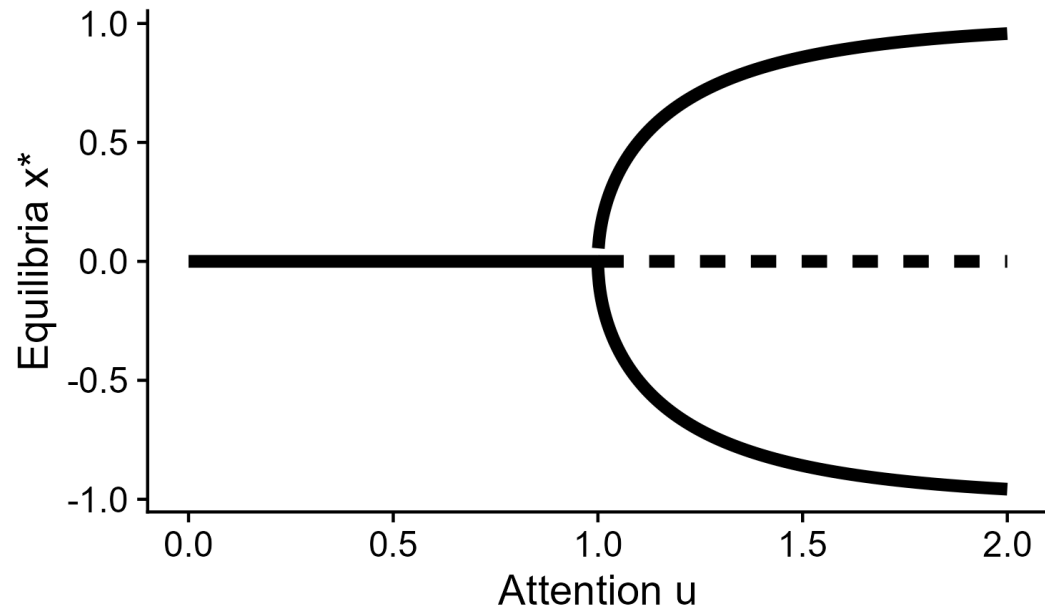
Hyperbolic equilibria: linearization succeeds
Singular/critical points: linearization fails

Bifurcation diagrams

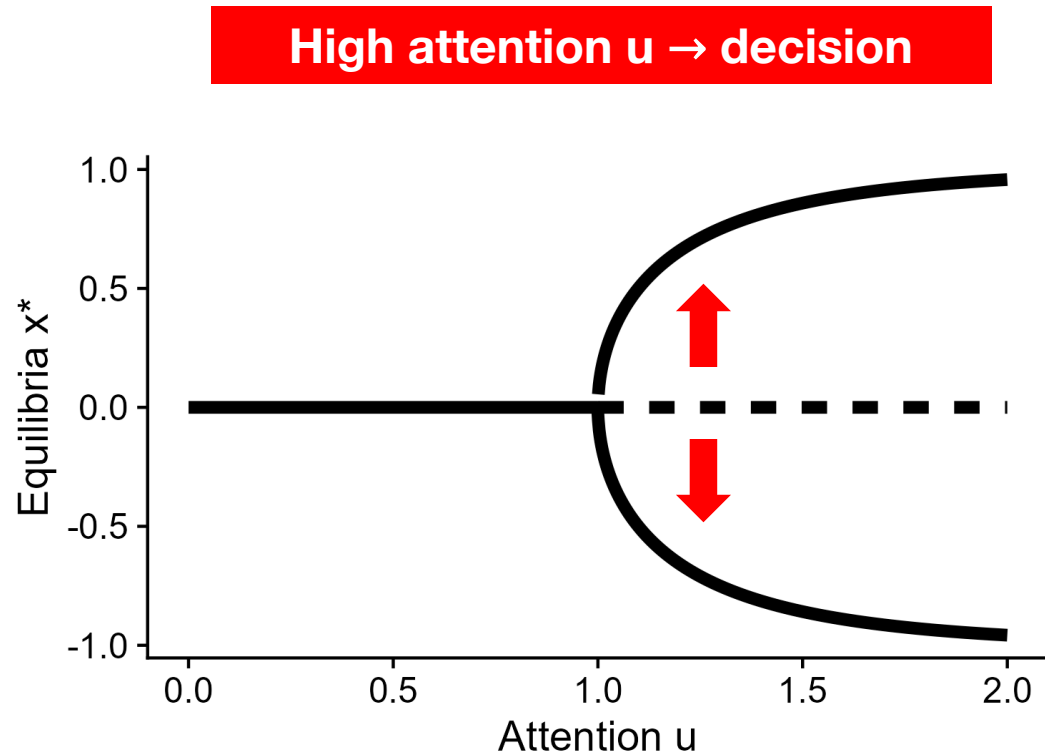


Bifurcation diagrams

What does this tell us about decision-making?

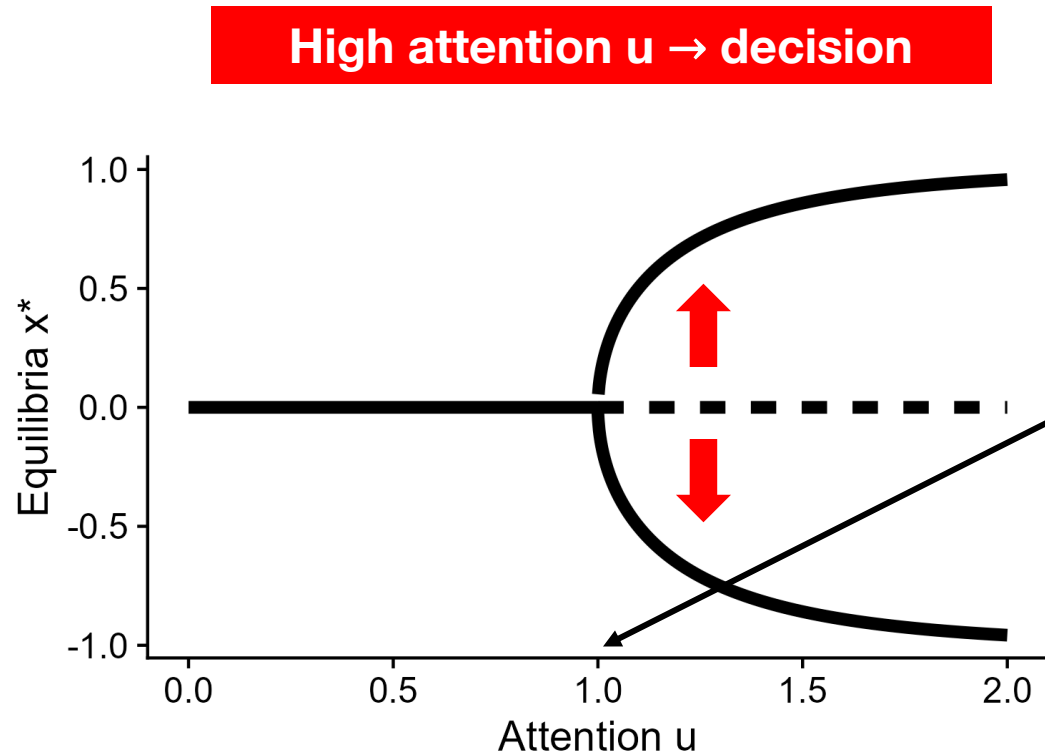


Bifurcation diagrams



In this case, multi-stability is associated with making decisions

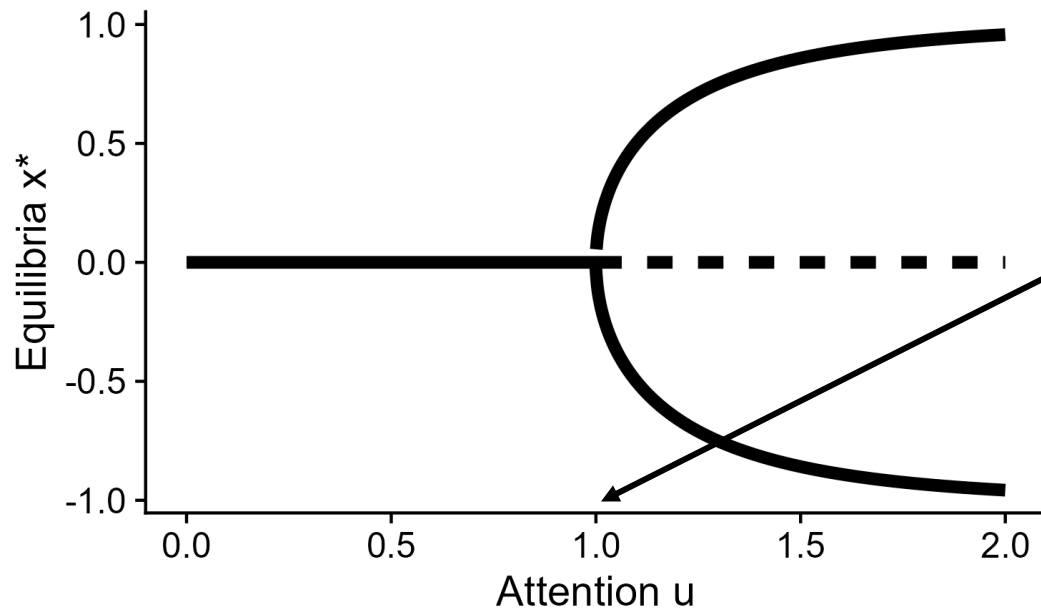
Bifurcation diagrams



Critical point u^* ($Re(\lambda) = 0$) divides different “regimes” (number/stability of equilibria) of decision-making

Could it be otherwise?

Bifurcation diagrams



**Critical point u^* ($Re(\lambda) = 0$)
divides different “regimes”
(number/stability of equilibria)
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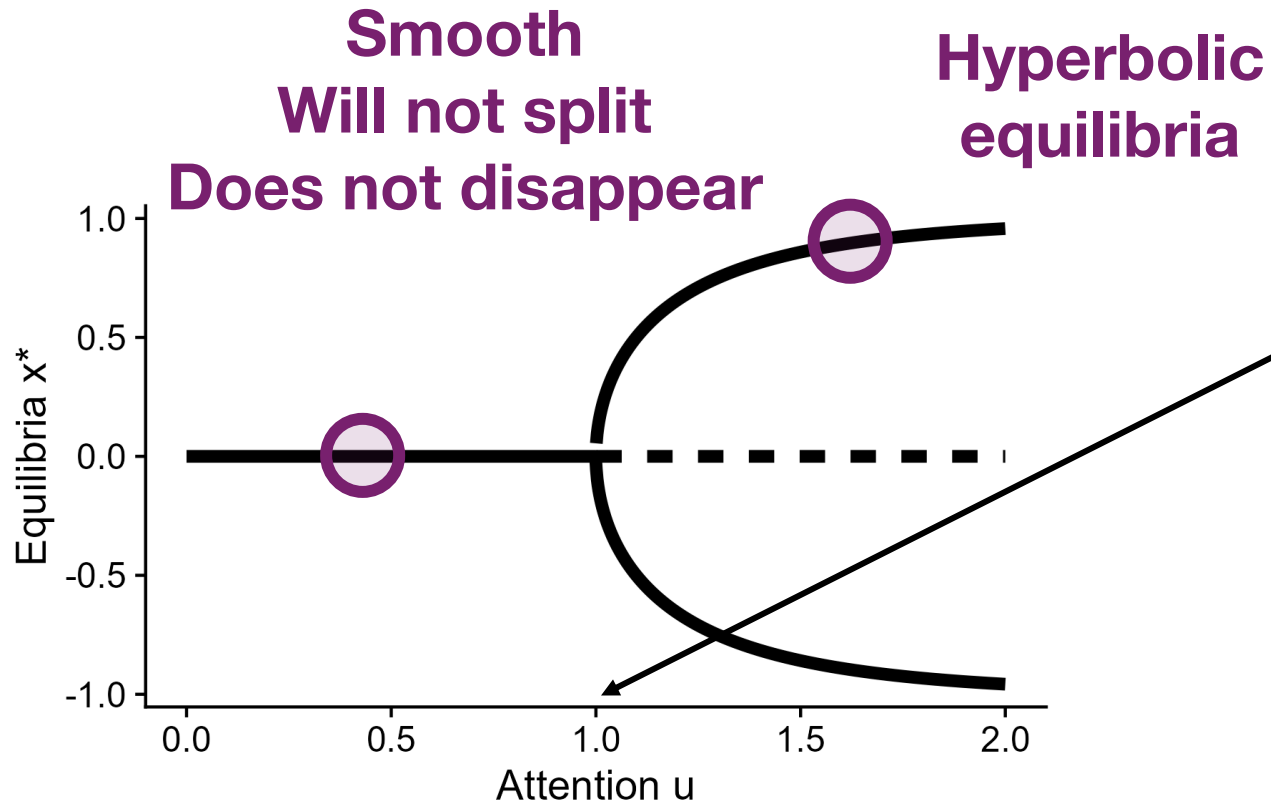
Could it be otherwise?

No*

Implicit function theorem: if $f(\mathbf{x}^*, \mathbf{p}) = 0$ and $\mathbf{x}^*$ is not singular ($\lambda \neq 0$), then locally there is a differentiable function $\chi(\mathbf{p}) = \mathbf{x}^*$

**Only considering local bifurcations; in global bifurcations (ex: homo/heteroclinic) the vector field changes without singular points*

Bifurcation diagrams



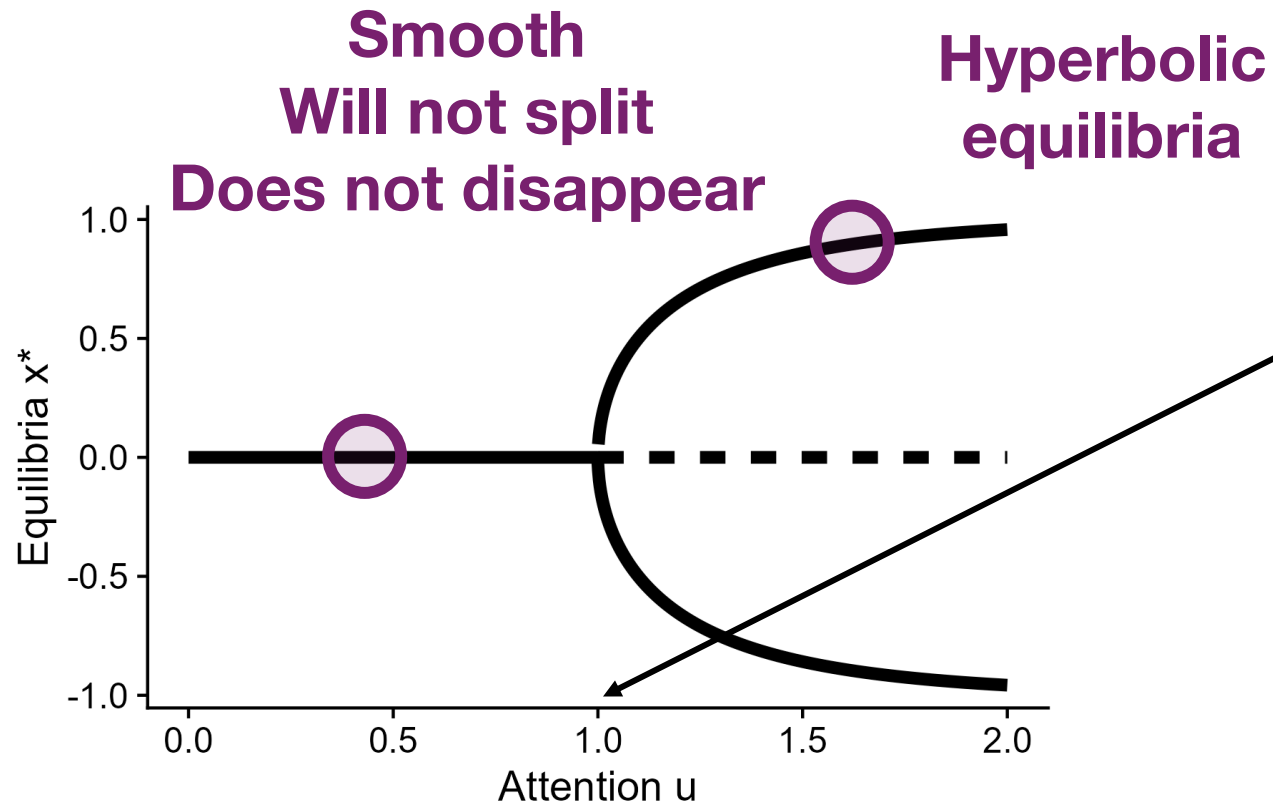
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Could it be otherwise?

No

Implicit function theorem: if $f(x^*, p) = 0$ and x^* is not singular ($\lambda \neq 0$), then locally there is a differentiable function $\chi(p) = x^*$
(NB bifurcations can occur even for non singular Jacobians, ex Hopf, but $Re(\lambda) = 0$ still holds)

Bifurcation diagrams



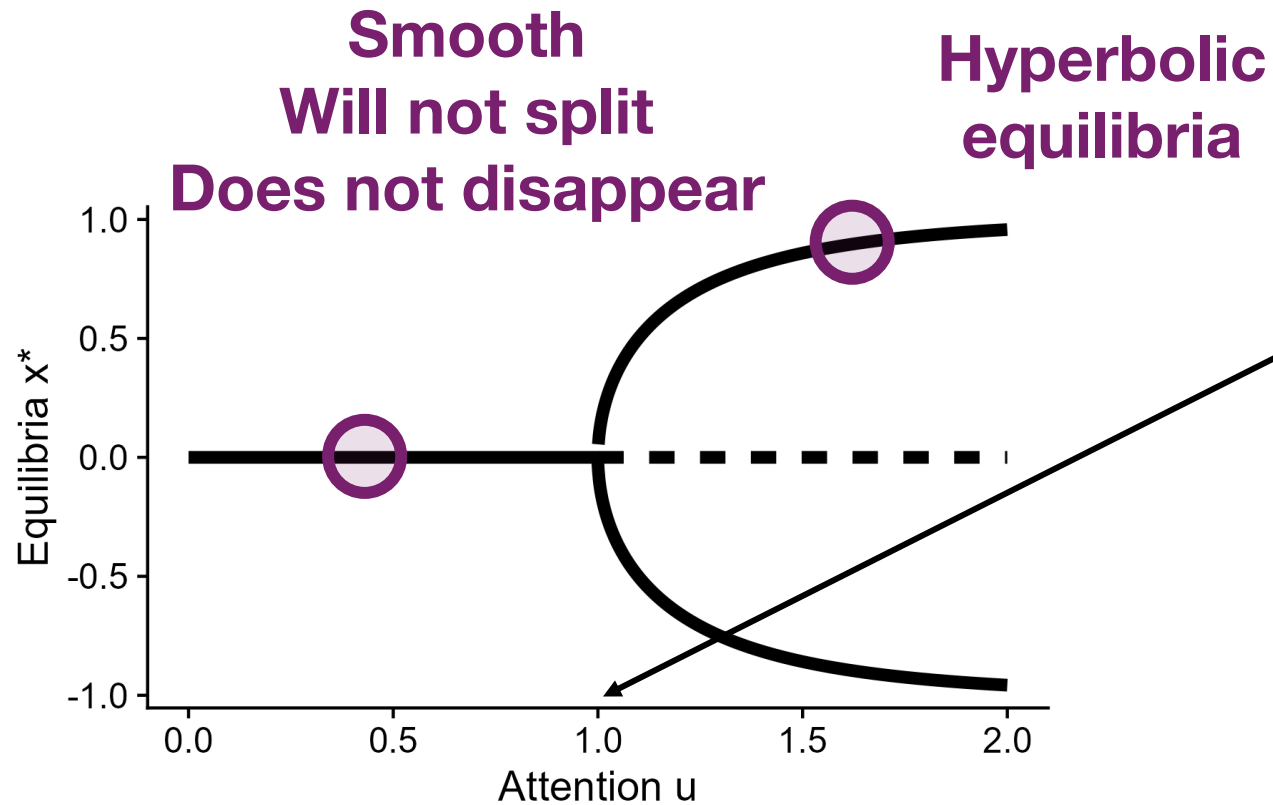
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Numerical continuation techniques use this fact to draw bifurcation diagrams. Bifurcation/qualitative “counter-part” of time integration
bifurcationKit Julia - *matcont* Matlab - *dynamica* Mathematica ...

Bifurcation diagrams



**Critical point u^* ($Re(\lambda) = 0$)
divides different “regimes”
(number/stability of equilibria)
of decision-making**

Could it be otherwise?

No

In what/how many ways can equilibria branches change?

Elementary bifurcations

At the critical point, the bifurcation diagram will have specific shapes resembling a *normal form*: these are the elementary bifurcations.

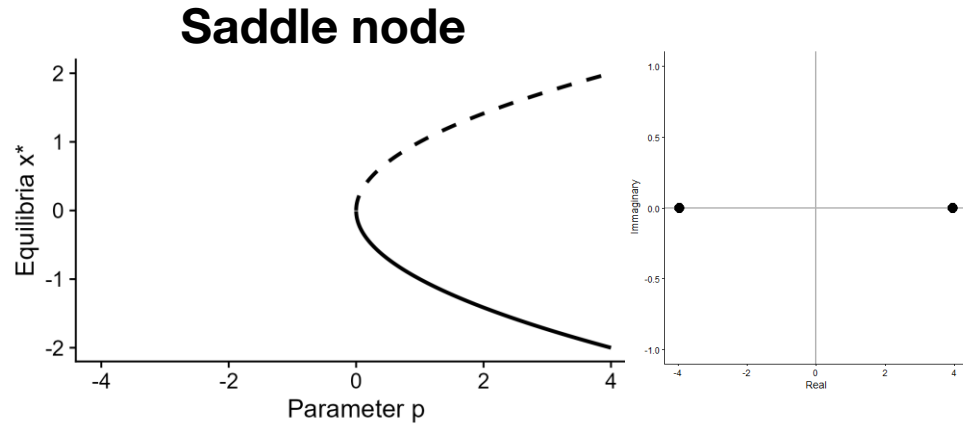
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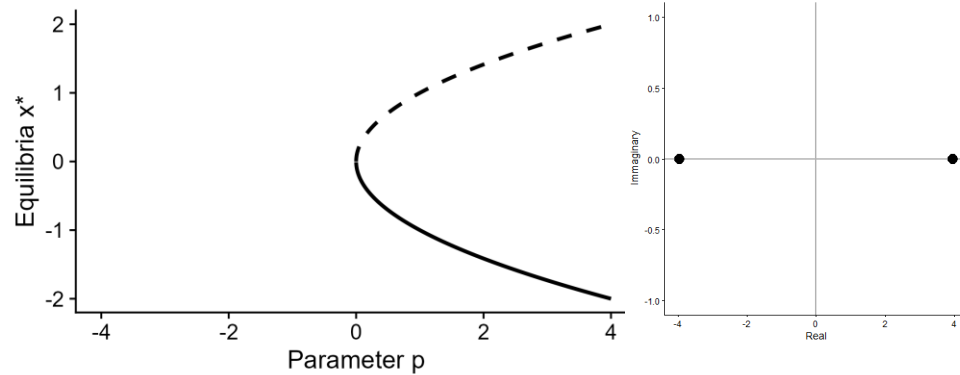


$$\dot{x} = f(x, p) = \varepsilon_1 p + \varepsilon_2 x^2, \varepsilon_i \in \{-1, +1\}$$

Elementary bifurcations

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Saddle node



$$\dot{x} = f(x, p) = \varepsilon_1 p + \varepsilon_2 x^2, \varepsilon_i \in \{-1, +1\}$$

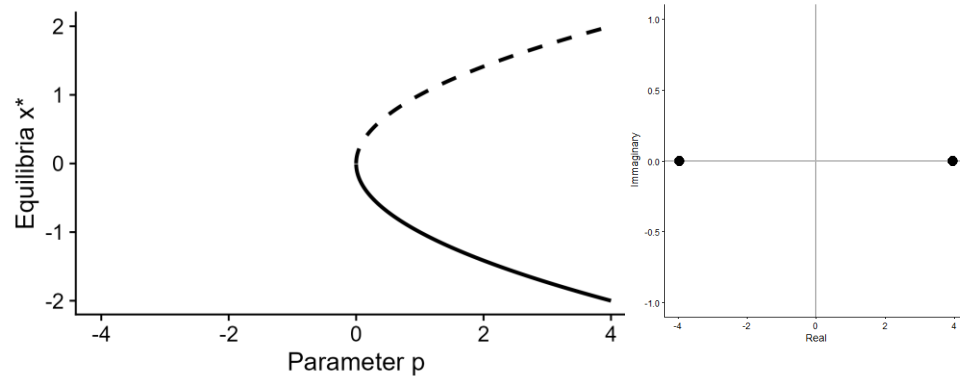
For the shown graph:

- a) $\varepsilon_1 = -1, \varepsilon_2 = 1$
- b) $\varepsilon_1 = 1, \varepsilon_2 = 1$
- c) $\varepsilon_1 = -1, \varepsilon_2 = -1$
- d) $\varepsilon_1 = 1, \varepsilon_2 = -1$

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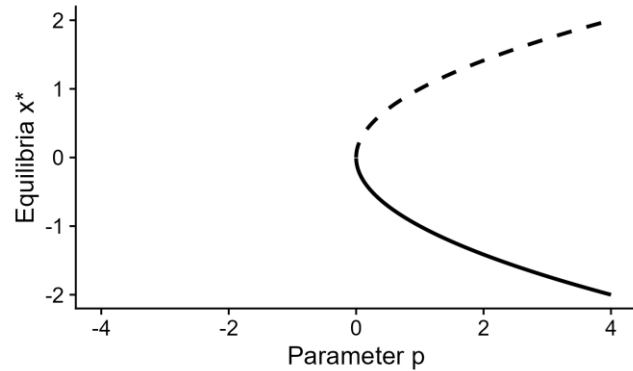
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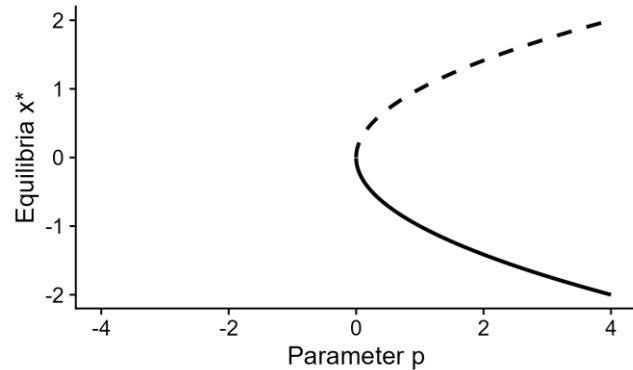
Defining conditions

$$\mathbf{a)} \quad \frac{\partial^2 f(x, p)}{\partial x^2} \neq 0 \quad \mathbf{b)} \quad \frac{\partial f(x, p)}{\partial p} \neq 0 \quad \mathbf{c)} \quad \frac{\partial f(x, p)}{\partial x} = 0$$

Elementary bifurcations

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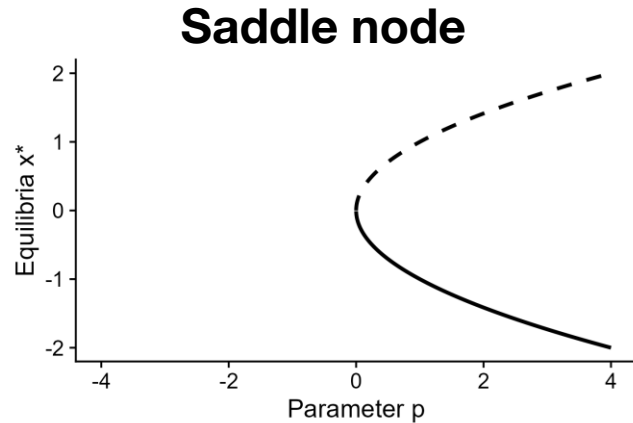
Defining conditions

Which defining condition is shared among all one-dimensional elementary bifurcations?

a) $\frac{\partial^2 f(x, p)}{\partial x^2} \neq 0$ **b)** $\frac{\partial f(x, p)}{\partial p} \neq 0$ **c)** $\frac{\partial f(x, p)}{\partial x} = 0$

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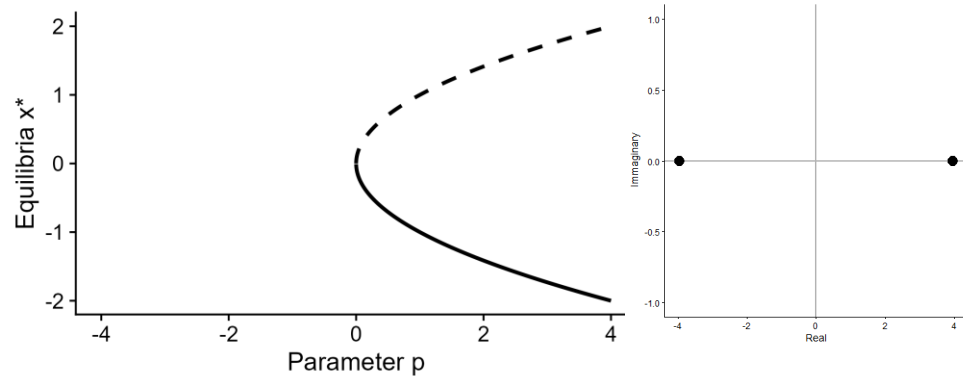
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Condition for being a critical point

Elementary bifurcations

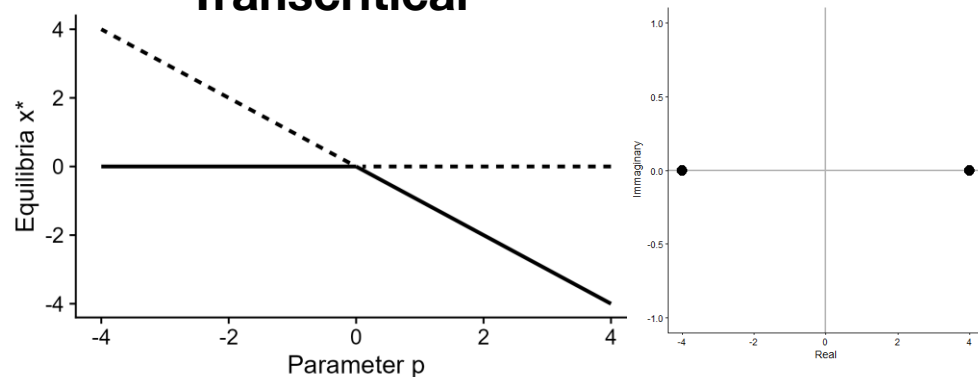
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Saddle node



$$\dot{x} = f(x, p) = \varepsilon_1 p + \varepsilon_2 x^2, \varepsilon_i \in \{-1, +1\}$$

Transcritical

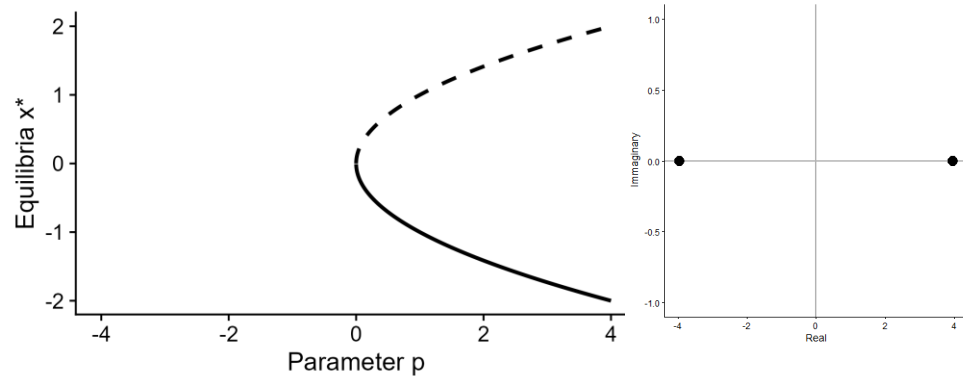


$$\dot{x} = f(x, p) = px + \varepsilon x^2, \varepsilon \in \{-1, +1\}$$

Elementary bifurcations

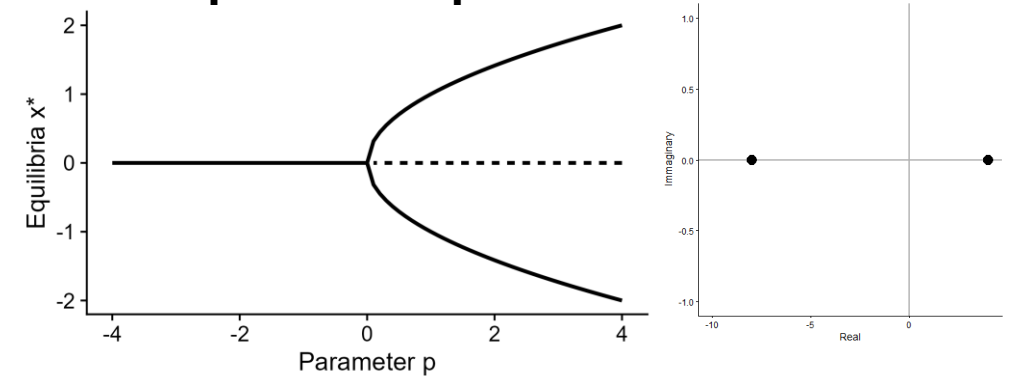
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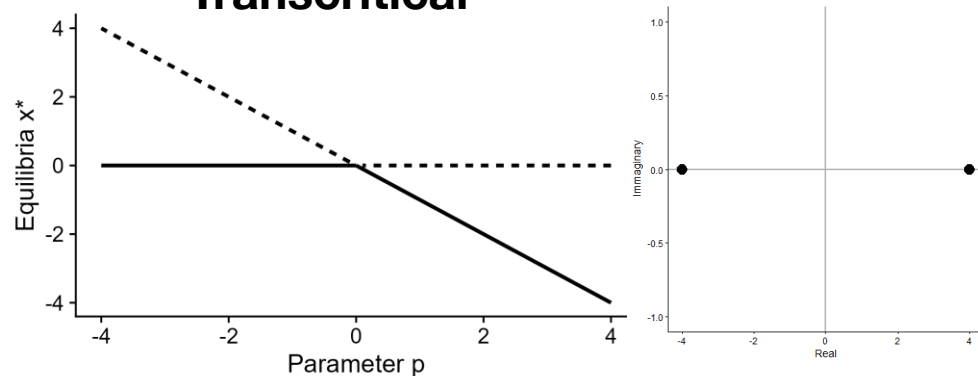
$$\dot{x} = f(x, p) = \varepsilon_1 p + \varepsilon_2 x^2, \varepsilon_i \in \{-1, +1\}$$

Supercritical pitchfork



$$\dot{x} = f(x, p) = \varepsilon x p - x^3, \varepsilon \in \{-1, +1\}$$

Transcritical

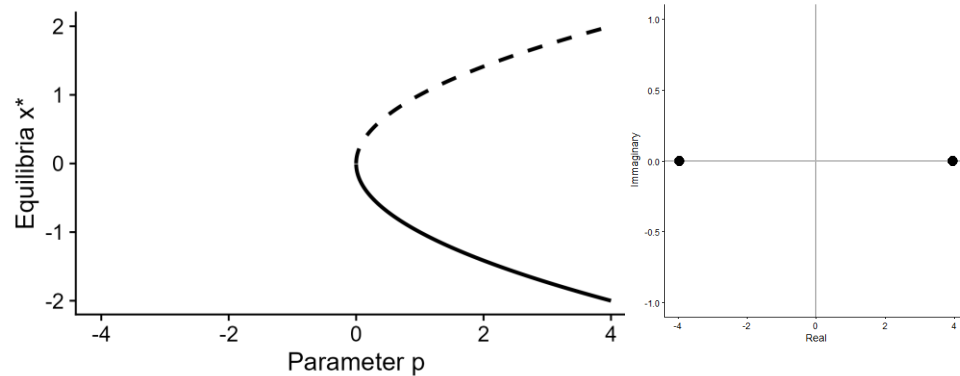


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Elementary bifurcations

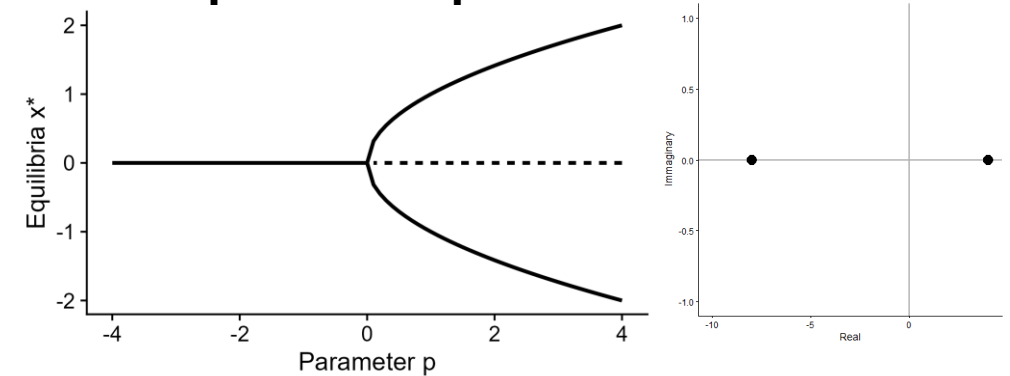
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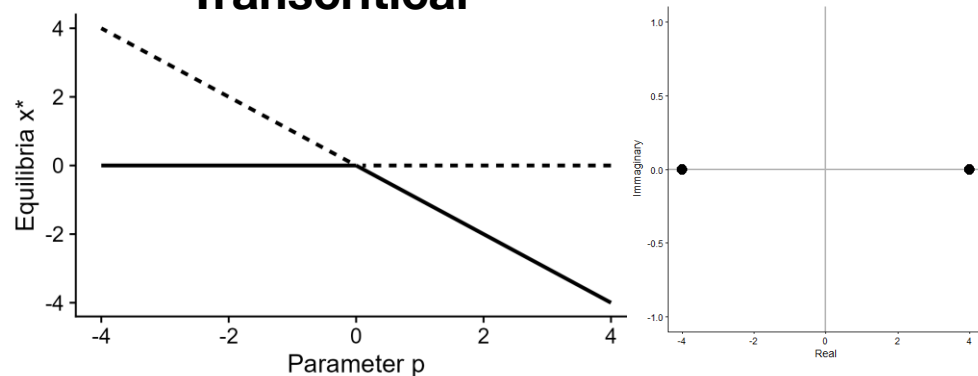
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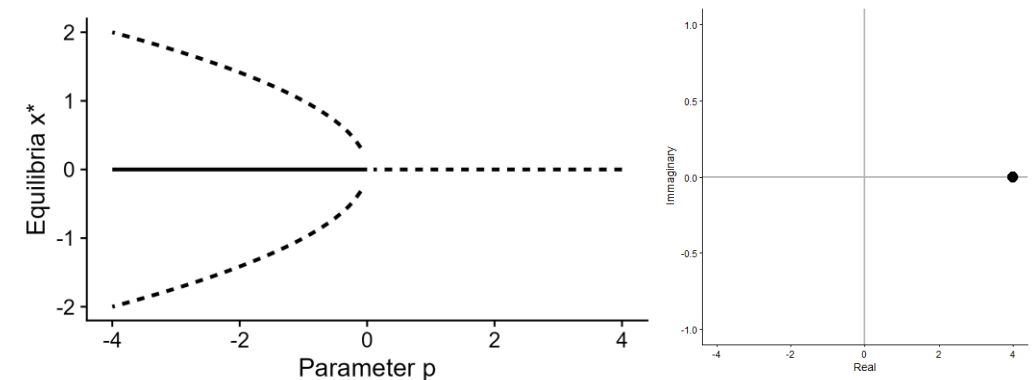
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Transcritical



$$\dot{x} = f(x, p) = px + \varepsilon x^2, \varepsilon \in \{-1, +1\}$$

Subcritical pitchfork



$$\dot{x} = f(x, p) = \varepsilon xp + x^3, \varepsilon \in \{-1, +1\}$$

What we know

- 1. Usually, equilibria will not change in number or stability**

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What we know

1. Usually, equilibria will not change in number or stability
2. When they do, it is because of singular or bifurcation points
3. At the bifurcation points, there are only specific ways in which equilibria will change (also for high dimensional systems)

Elementary bifurcations can transform into each other

Only a finite way in which this can happen!

Universal unfolding theorem

There are a finite number of ways the bifurcation of $\dot{x} = f(x, p, r = 0)$ transforms into the bifurcation of $\dot{x} = g(x, p, r \neq 0)$. r are the *unfolding parameters* $r \neq 0$ is called a *parametric perturbation*

Universal unfolding of pitchfork

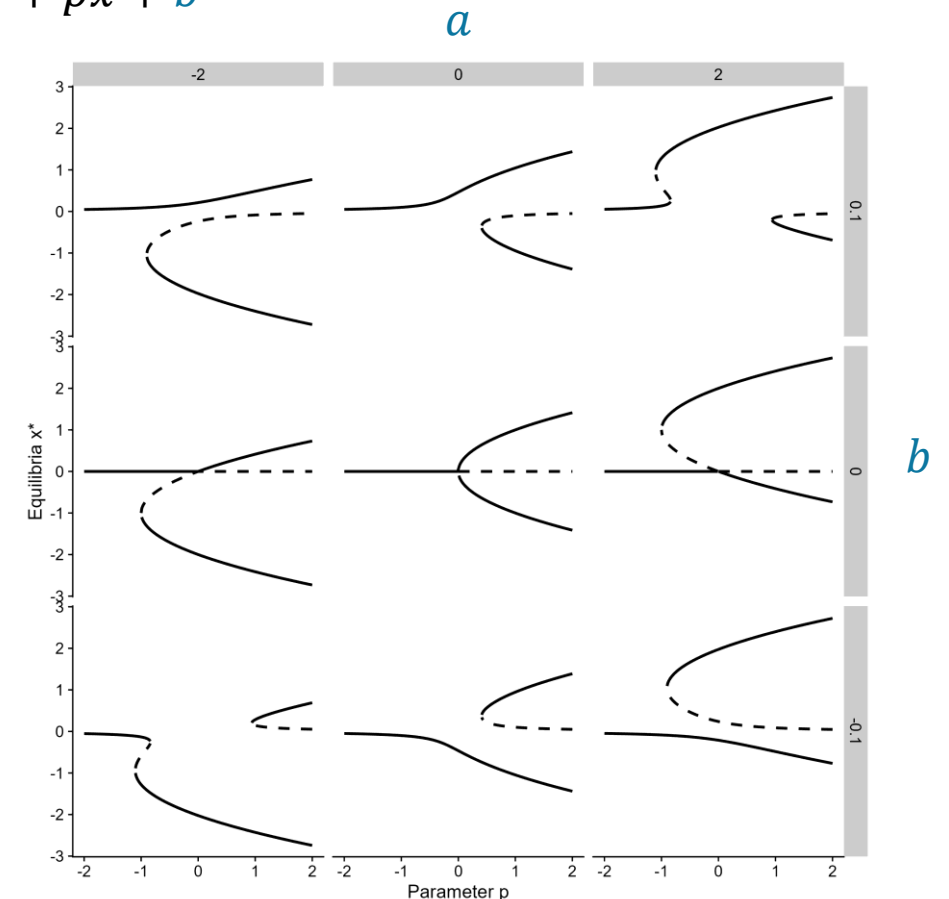
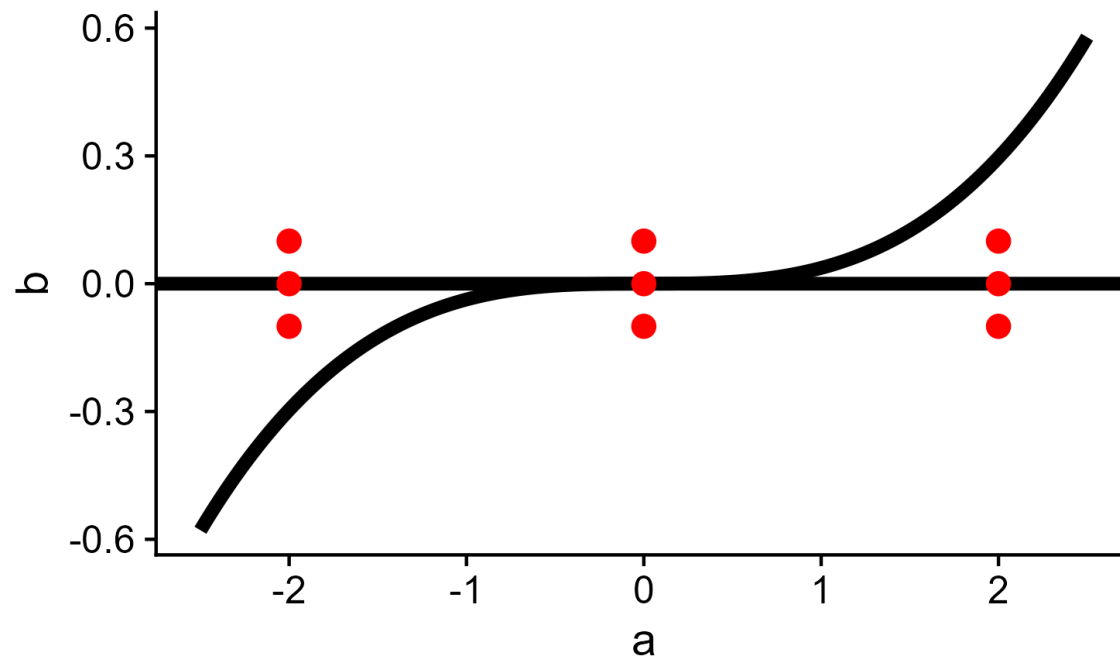
$$\dot{x} = g(x, p, a, b) = -x^3 + ax^2 + px + b$$

Universal unfolding theorem

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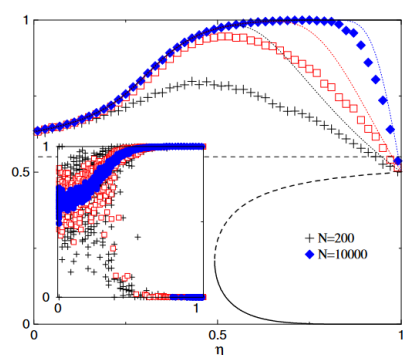
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Phase coexistence in a forecasting game

Philippe Curty and Matteo Marsili



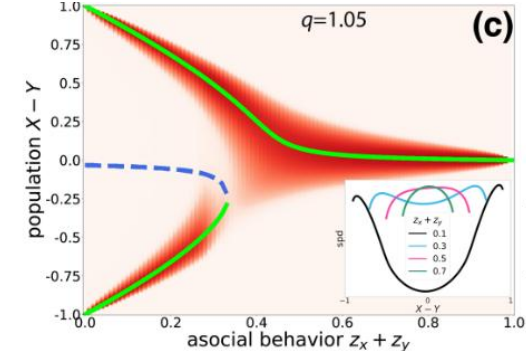
communications physics

ARTICLE

<https://doi.org/10.1038/s42005-023-01345-3> OPEN

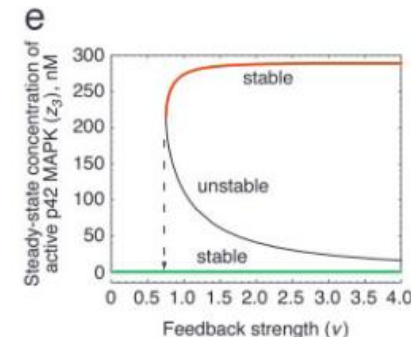
Cross-inhibition leads to group consensus despite the presence of strongly opinionated minorities and asocial behaviour

Andreagiovanni Reina^{1,2,3,4}, Raina Zakir¹, Giulia De Masi^{3,4} & Eliseo Ferrante^{3,5}



Detection of multistability, bifurcations, and hysteresis in a large class of biological positive-feedback systems

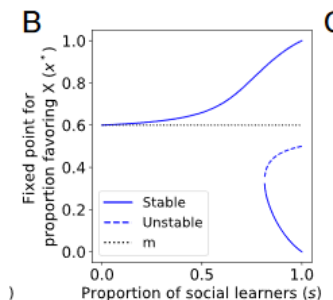
David Angeli*, James E. Ferrell, Jr.¹, and Eduardo D. Sontag^{2,5}



Dynamical system model predicts when social learners impair collective performance

Vicky Chuqiao Yang^{1,1}, Mirta Galesic^{2,3,4,5}, Harvey McGuinness⁴, and Ani Harutyunyan⁶

¹Santa Fe Institute, Santa Fe, NM 87501; ²Complexity Science Hub Vienna, A-1080 Vienna, Austria; ³Vermont Complex Systems Center, University of Vermont, Burlington, VT 05405; ⁴Zanvyl Krieger School of Arts and Sciences, Johns Hopkins University, Baltimore, MD 21218; and ⁵Sunwater Institute, North Bethesda, MD 20852

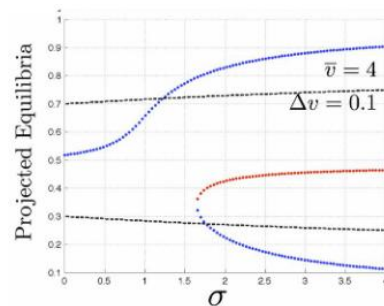


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PLOS ONE

A Mechanism for Value-Sensitive Decision-Making

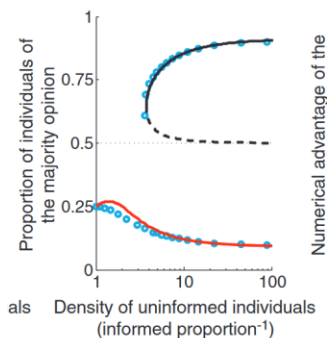
Darren Pais¹, Patrick M. Hogan², Thomas Schlegel³, Nigel R. Franks³, Naomi E. Leonard¹, James A. R. Marshall^{2*}



REPORTS

Uninformed Individuals Promote Democratic Consensus in Animal Groups

Iain D. Couzin,^{1*} Christos C. Ioannou,^{1,†} Güven Demirel,² Thilo Gross,^{2,‡} Colin J. Torney,¹ Andrew Hartnett,¹ Larissa Conradt,^{3,§} Simon A. Levin,¹ Naomi E. Leonard⁴



RESEARCH ARTICLE | EVOLUTION |

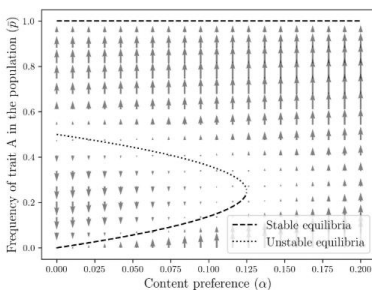
f x t in e

Weak individual preferences stabilize culture

Alberto Acerbi¹ and Benoît de Courson² Authors Info & Affiliations

Edited by Marcus Feldman, Stanford University, Stanford, CA; received June 20, 2024; accepted January 9, 2025

February 21, 2025 | 122 (8) e2412380122 | <https://doi.org/10.1073/pnas.2412380122>



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Behavioral Sciences

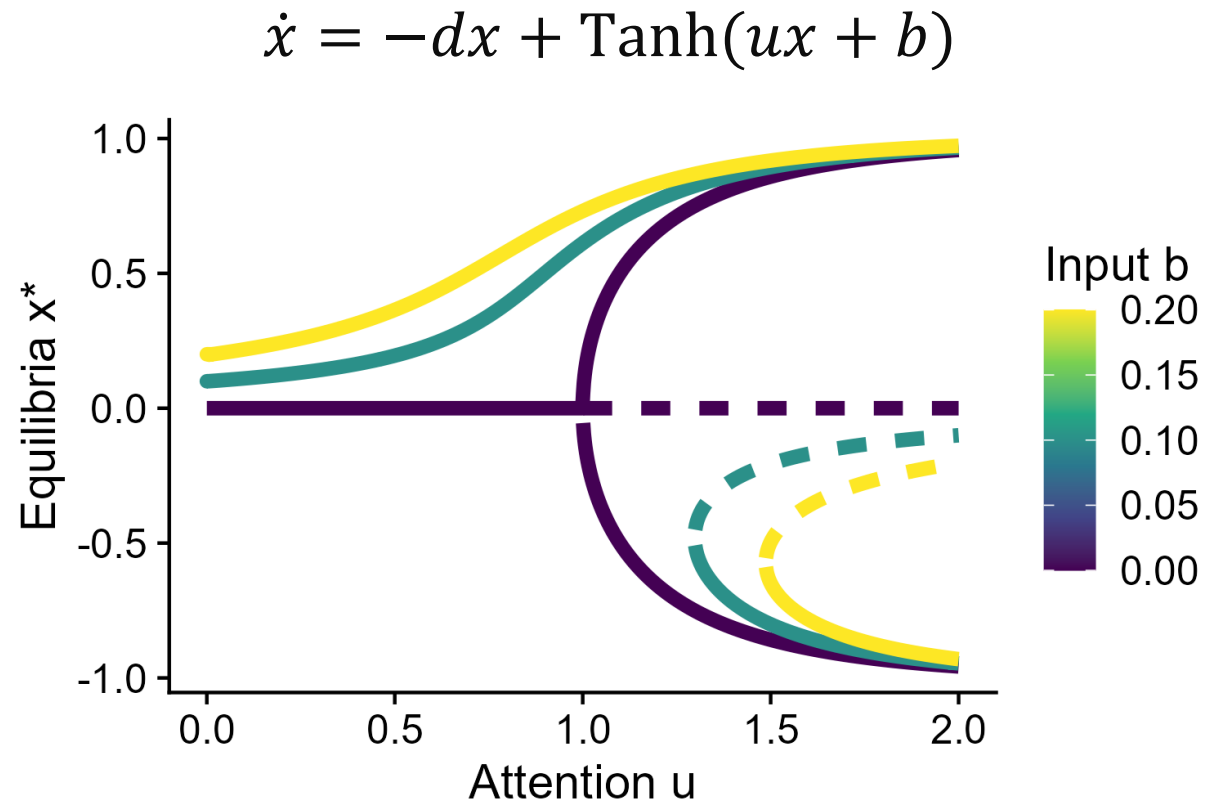
Review

Examining cognitive flexibility and stability through the lens of dynamical systems

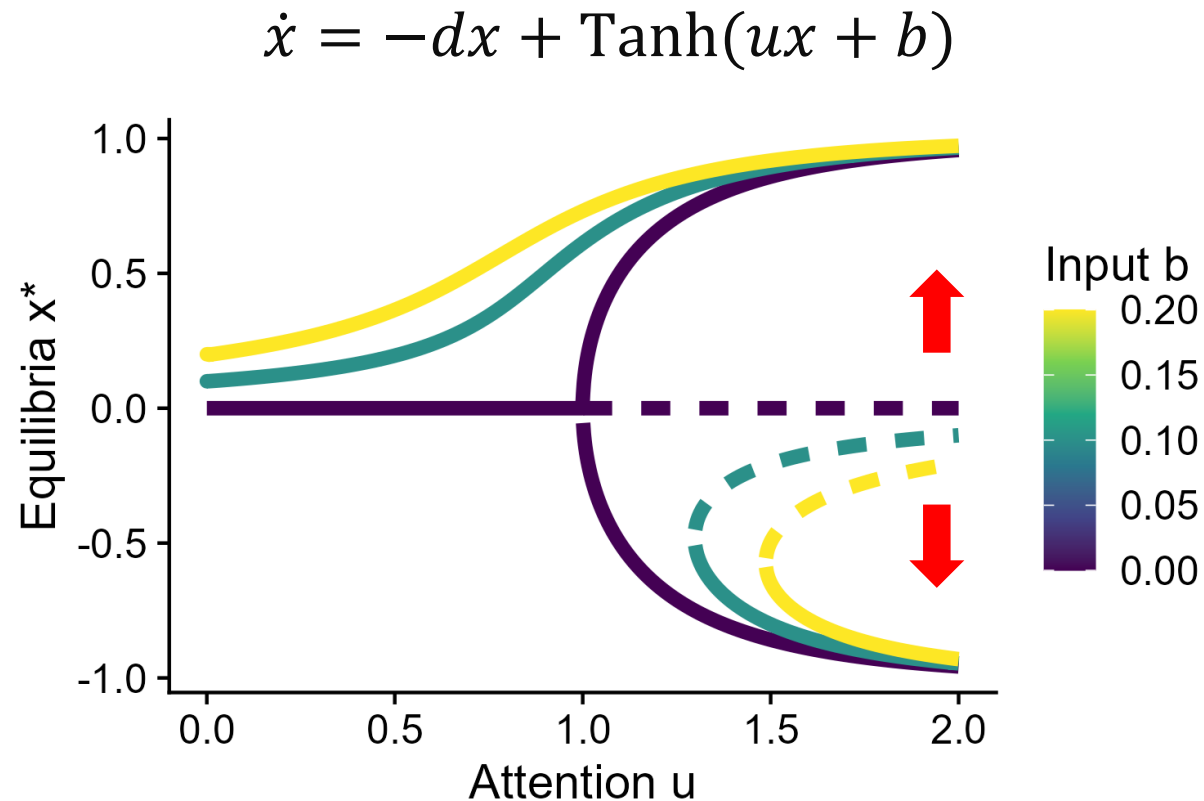
Sebastian Musslick^{1,2,*} and Anastasia Bizyaeva^{3,*}



Unfolding in the nonlinear opinion dynamics model

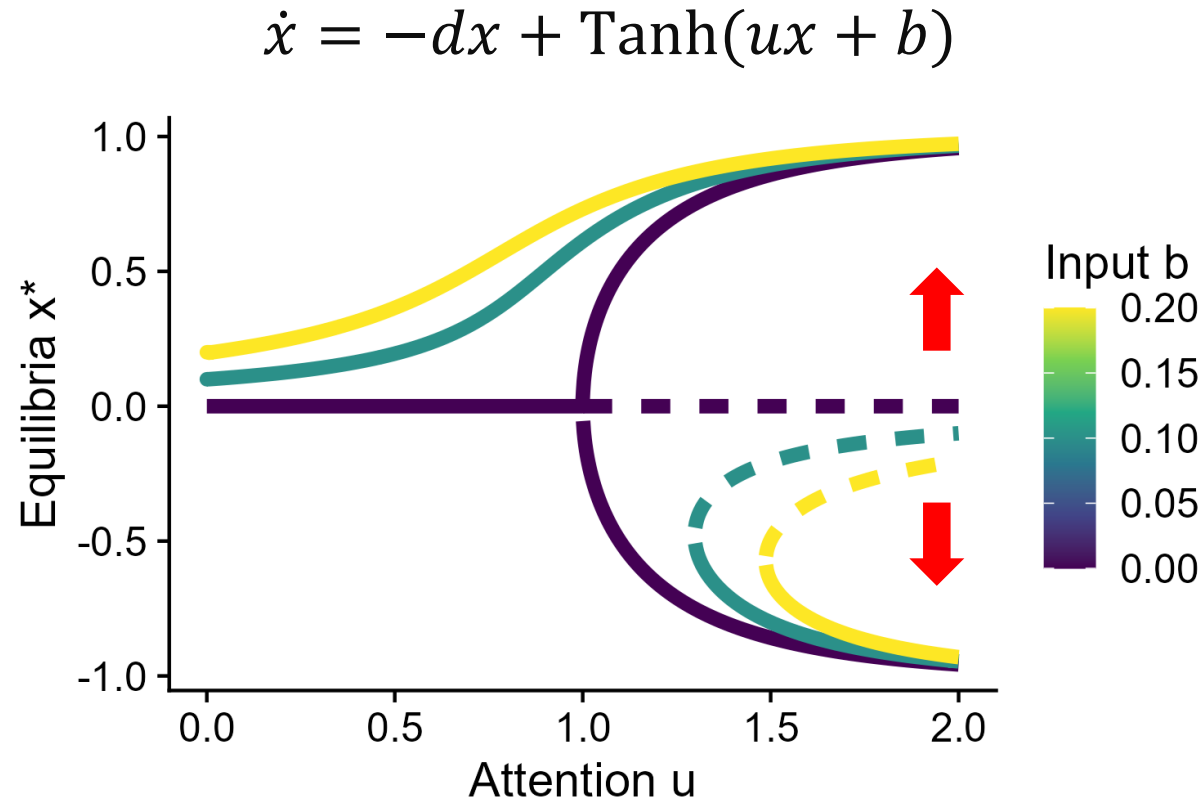


Unfolding in the nonlinear opinion dynamics model



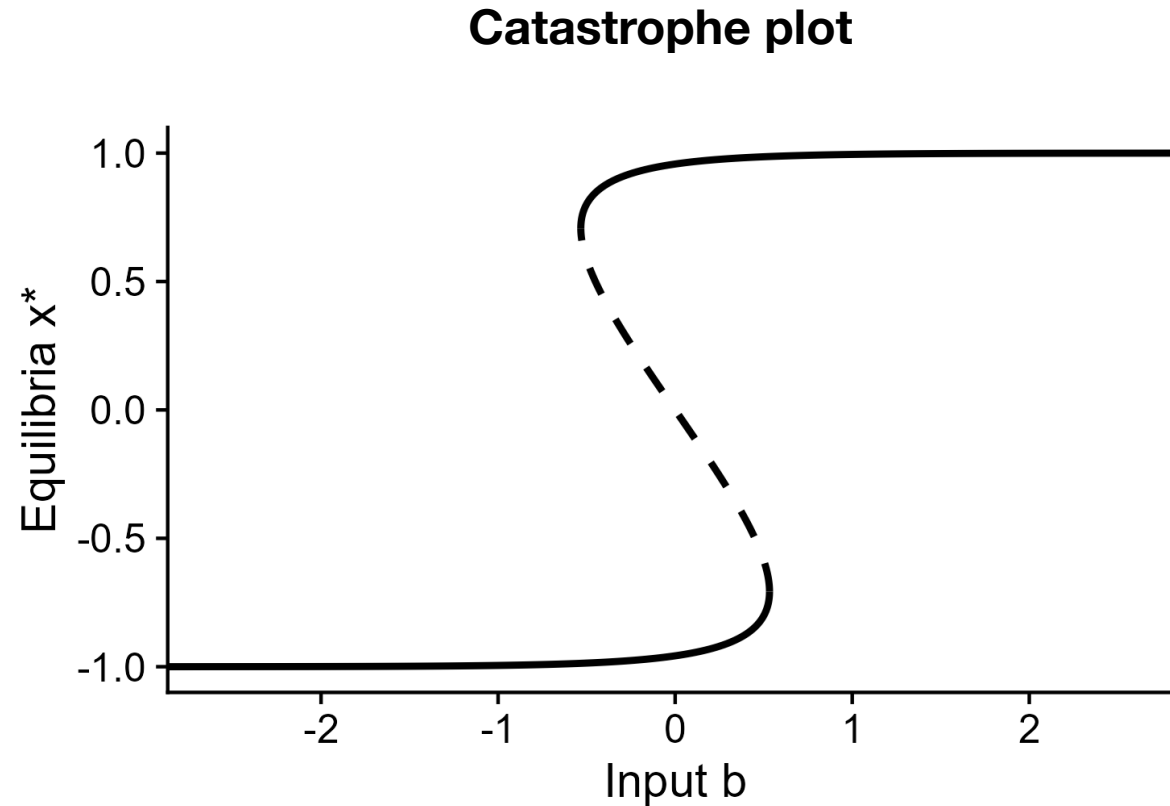
1. Input increases monostable regime for best option
2. Non-linear effects can still amplify wrong information and lead to errors
3. Attention u (strength of positive feedback) creates a trade-off between making decisions and cascades to wrong decisions

Unfolding in the nonlinear opinion dynamics model



Draw the bifurcation diagram ($u = 1.5$) for input b as the bifurcating parameter

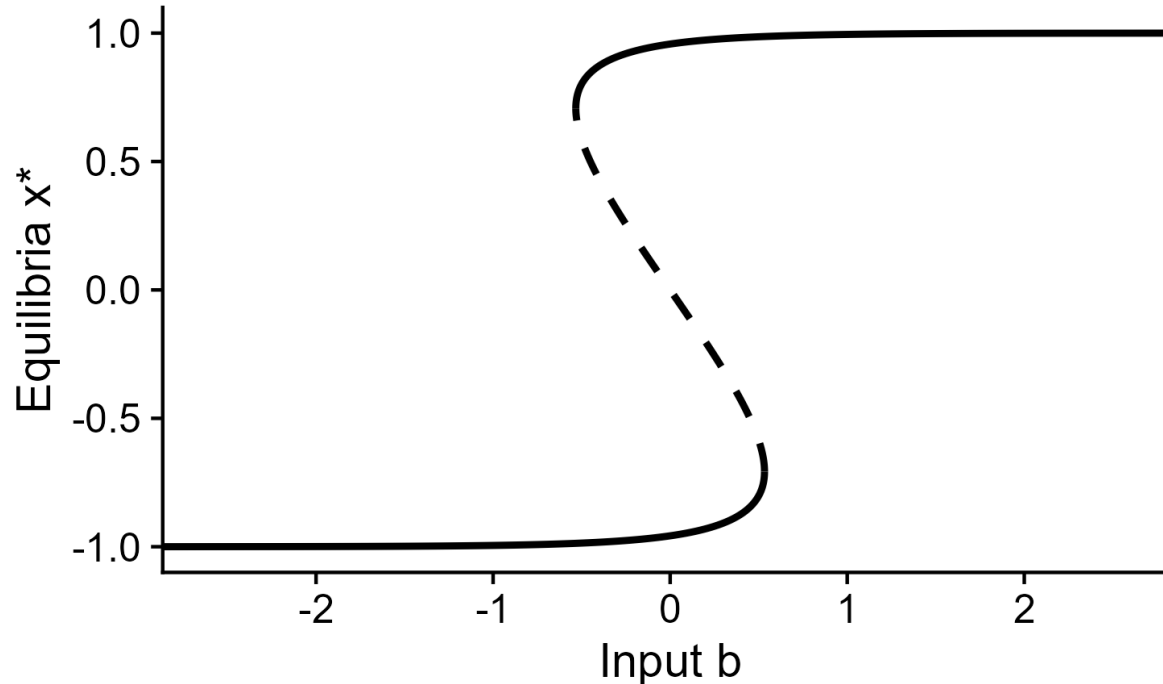
Unfolding in the nonlinear opinion dynamics model



1. **Multi stability can lead to hysteresis/memory/path dependence**
2. **Hysteresis is bad for flexible decision-making**
3. **Small changes in inputs can lead to big changes in decision**

Unfolding in the nonlinear opinion dynamics model

Catastrophe plot



Problems

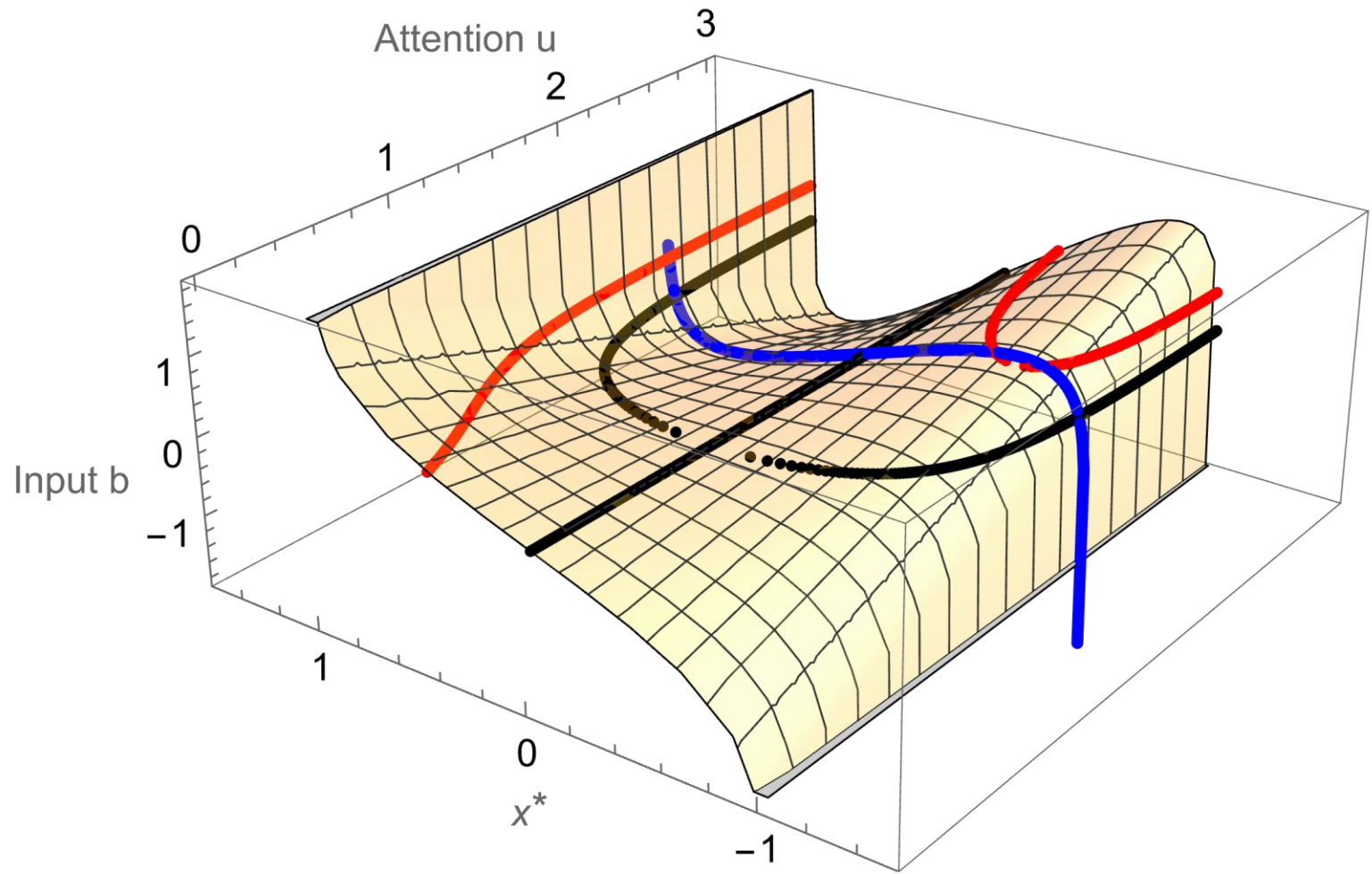
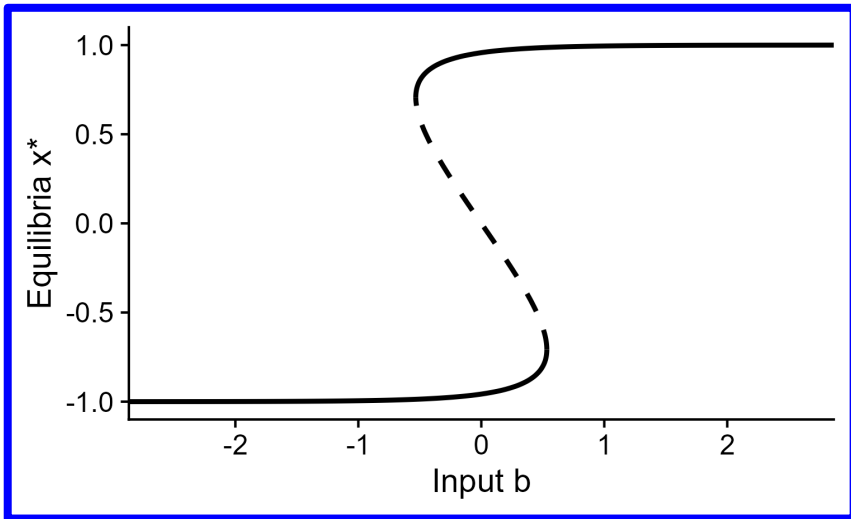
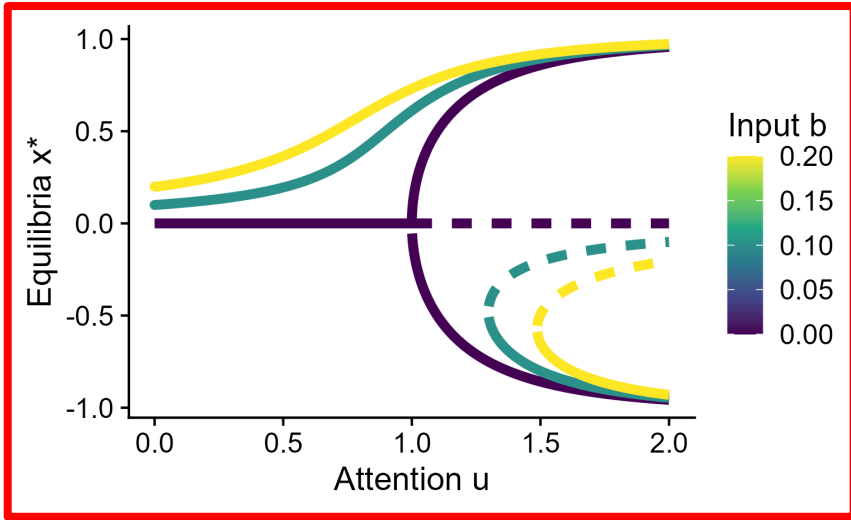
$$\dot{x} = f(x, p)$$

State variables $x \in \mathbb{R}^n$
Can change

Parameters $p \in \mathbb{R}^k$
Don't change 4.

1. Multi stability can lead to hysteresis/memory/path dependence
2. Hysteresis is bad for flexible decision-making
3. Small changes in inputs can lead to big changes in decision
4. Dynamical systems treat inputs as parameters (they must change slowly compared to the decision's dynamics)

Unfolding in the nonlinear opinion dynamics model



Part 4

Separation of timescales



State dependent attention

Decision variable

Input or bias

$$\dot{x} = -dx + \tanh(ux + b)$$

Leak

Attention

Characteristic
timescale

$$\tau \dot{u} = -\alpha u + u_0 + kx^2$$

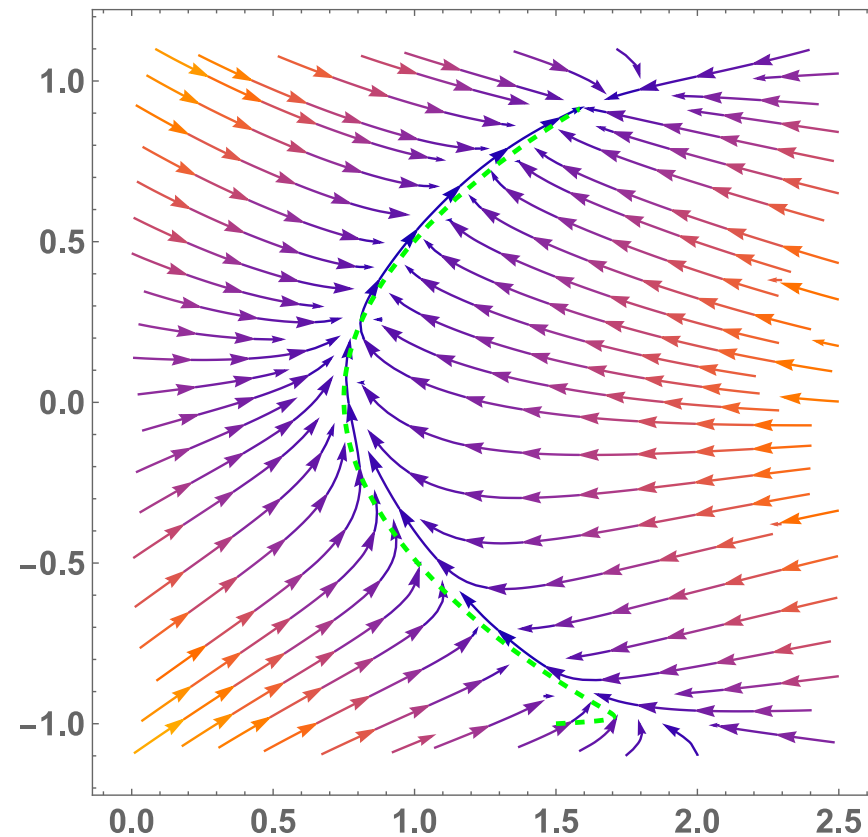
Gain

Baseline attention

Phase plane

Trajectory

Decision variable
x

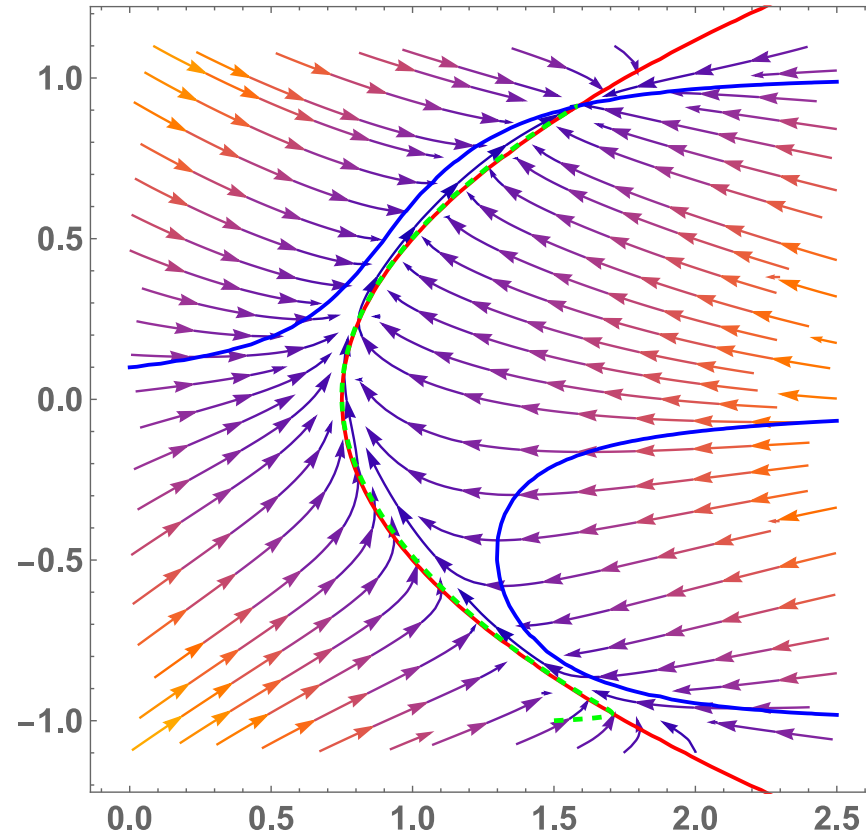


Attention
u

Nullclines

Decision variable
x

$$\dot{x} = f(x, u, p) = 0$$



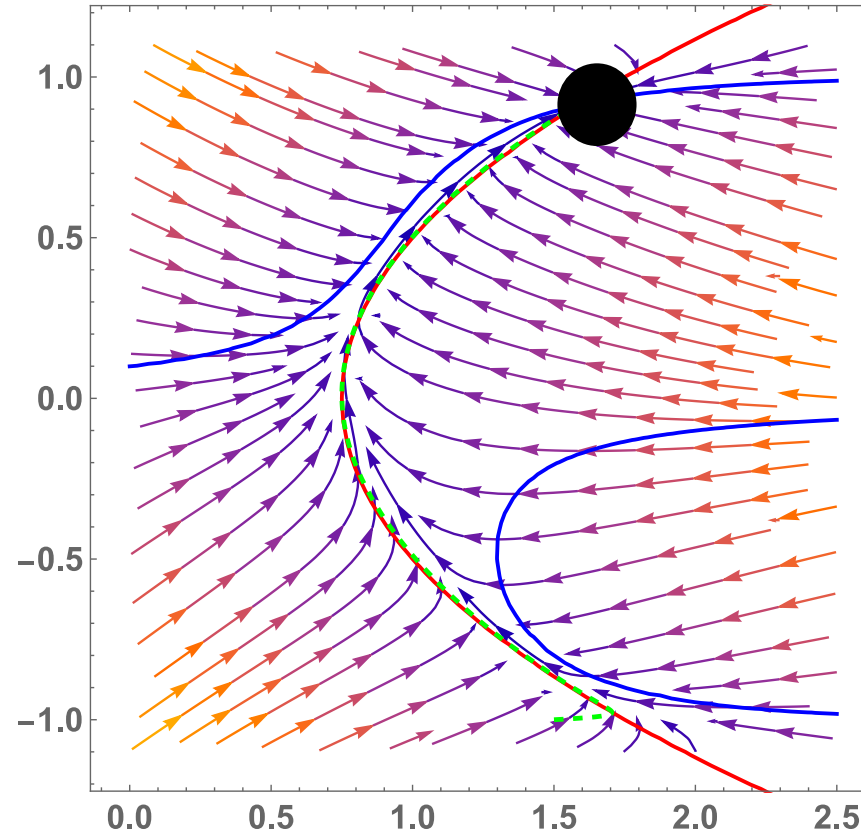
Attention
u

$$\tau\dot{u} = g(x, u, p) = 0$$

Nullclines

Decision variable
x

$$\dot{x} = f(x, u, p) = 0$$



Global equilibria
are where
nullclines cross

Attention
u

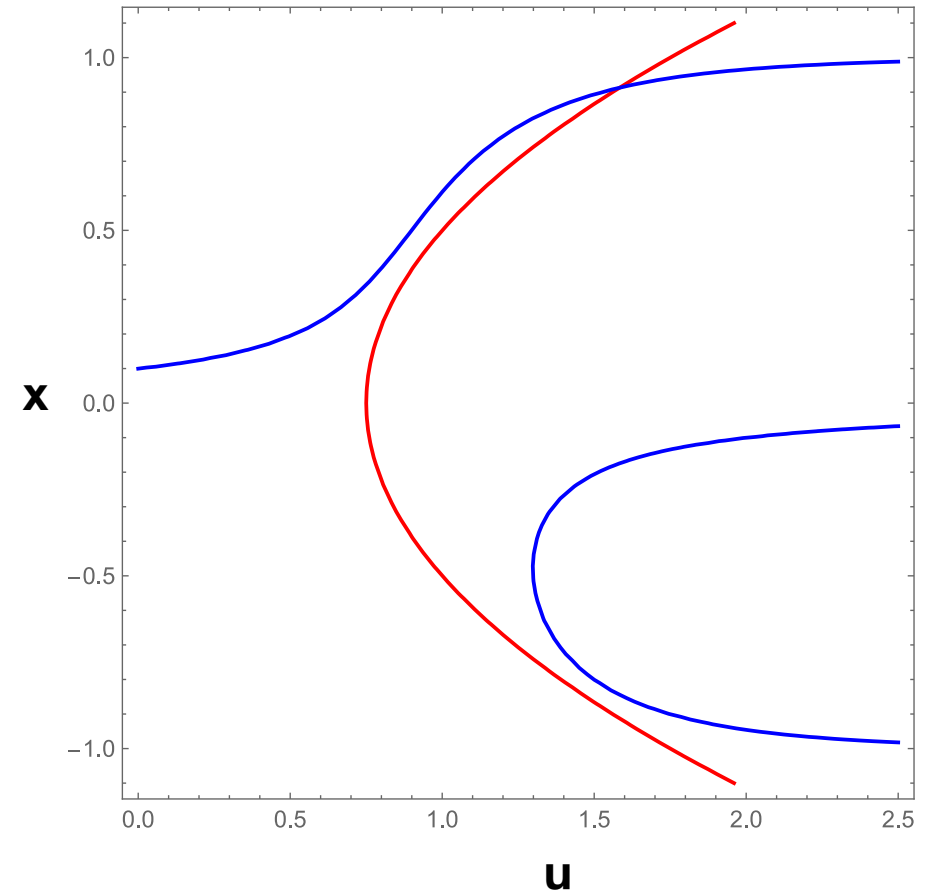
$$\tau\dot{u} = g(x, u, p) = 0$$

Separation of timescales

$$1) \dot{x} = f(x, u, p)$$

$$2) \tau \dot{u} = g(x, u, p)$$

τ is small



Separation of timescales

$$1) \dot{x} = f(x, u, p)$$

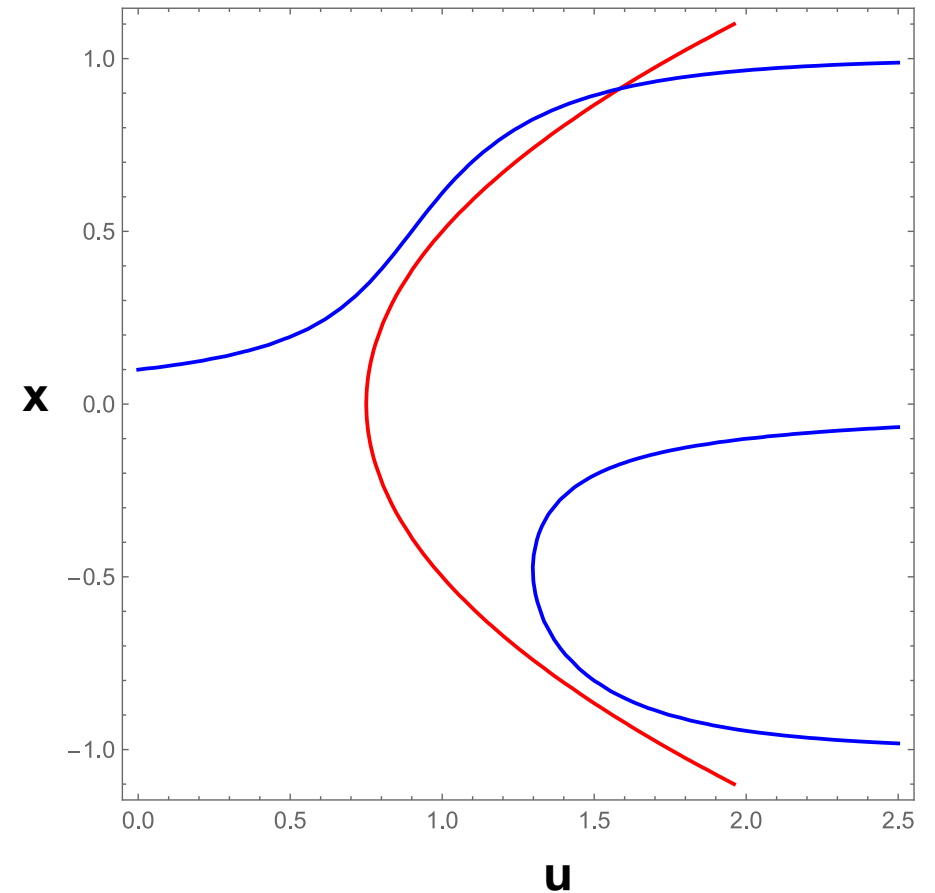
$$2) \tau \dot{u} = g(x, u, p)$$

τ is small

Which dynamic will be faster?

1)

2)



Separation of timescales

$$1) \dot{x} = f(x, u, p)$$

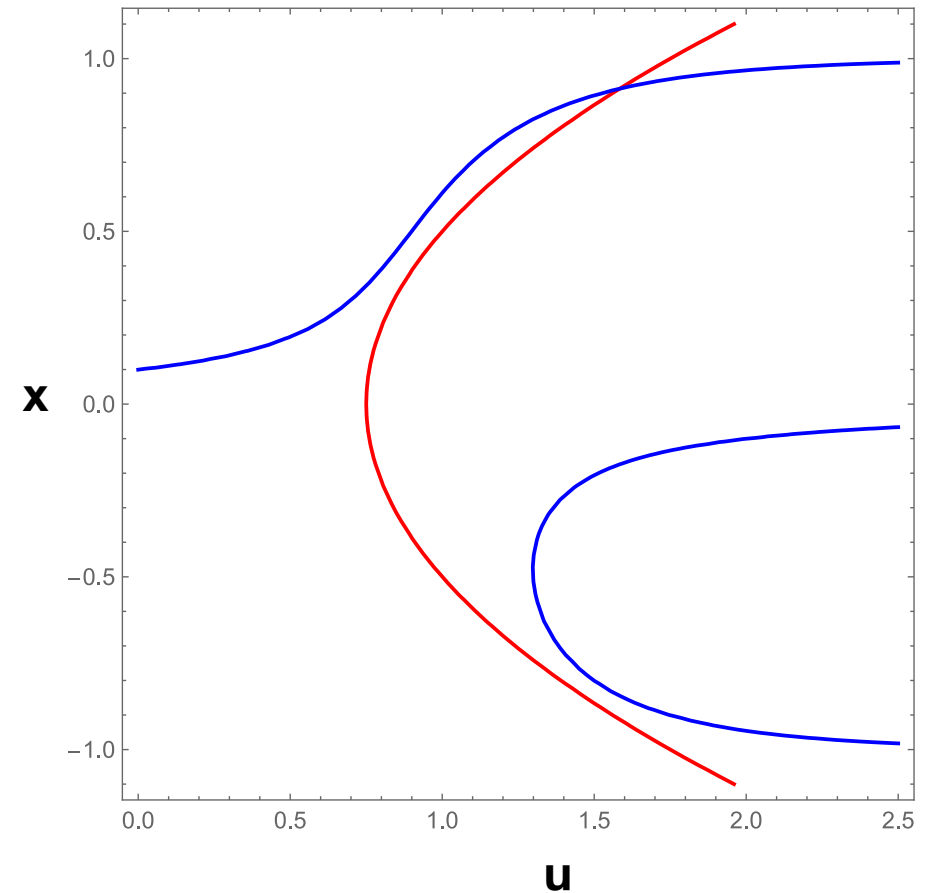
$$2) \tau \dot{u} = g(x, u, p)$$

τ is small

Which dynamic will be faster?

1)

2)



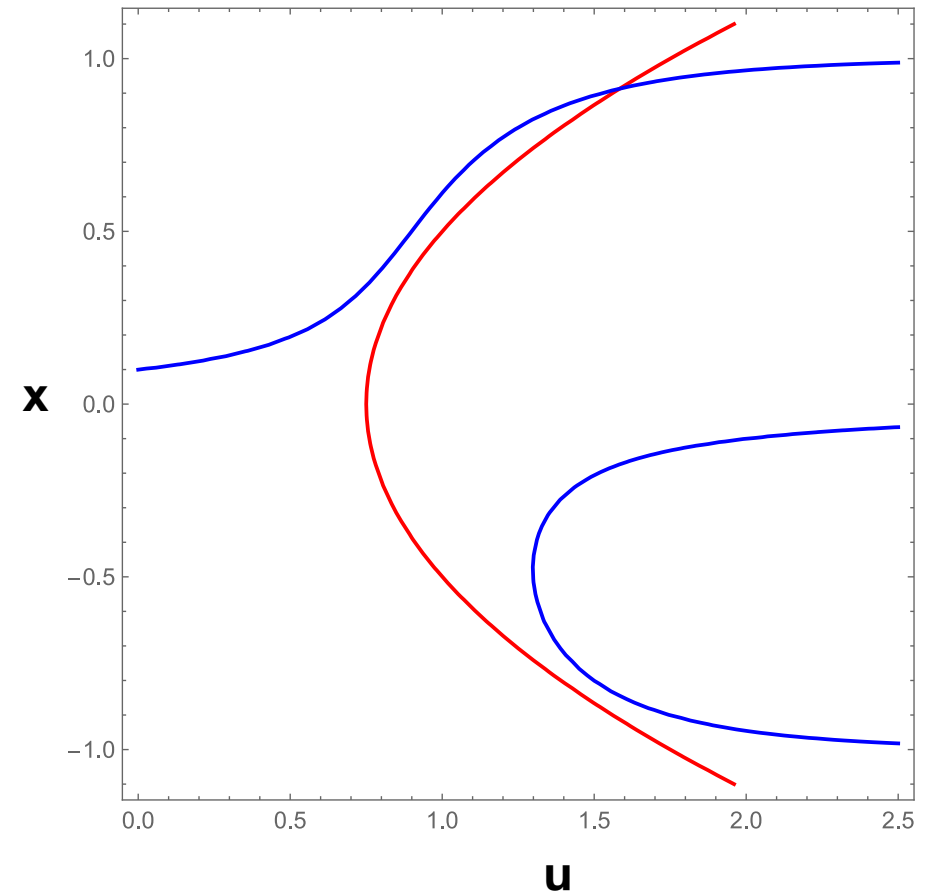
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Decompose the
dynamics along
the systems'
dimensions



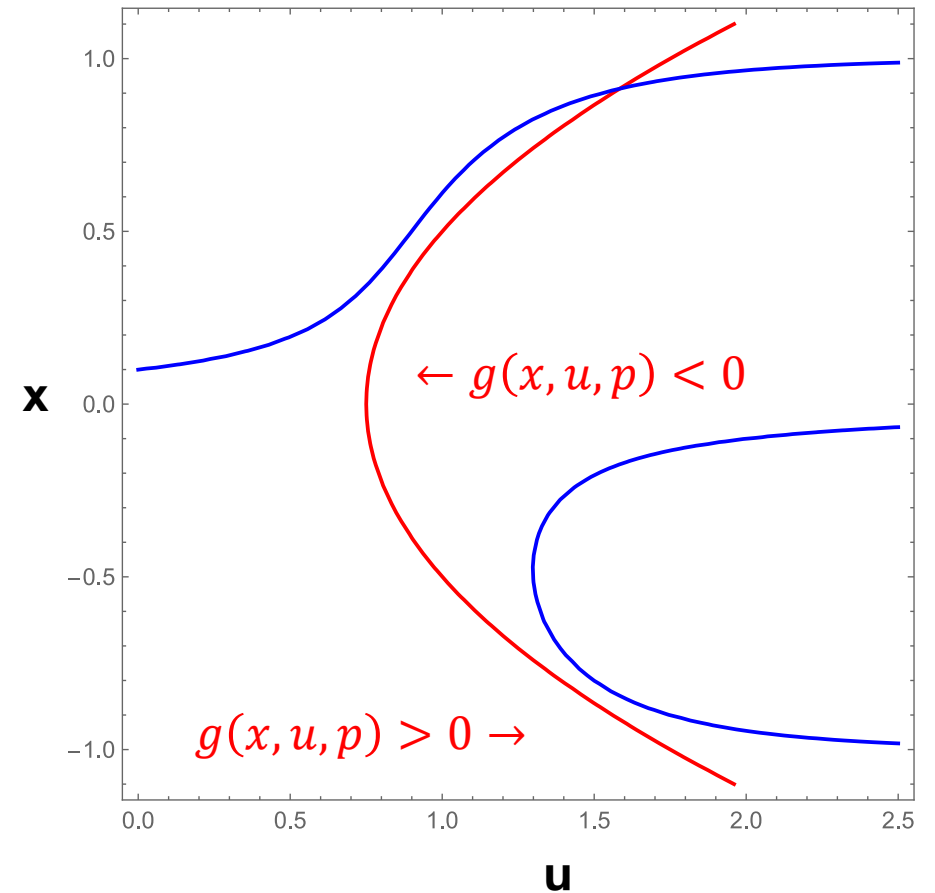
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The fast dynamics occurs along attention u

Decompose the dynamics along the systems' dimensions



Separation of timescales

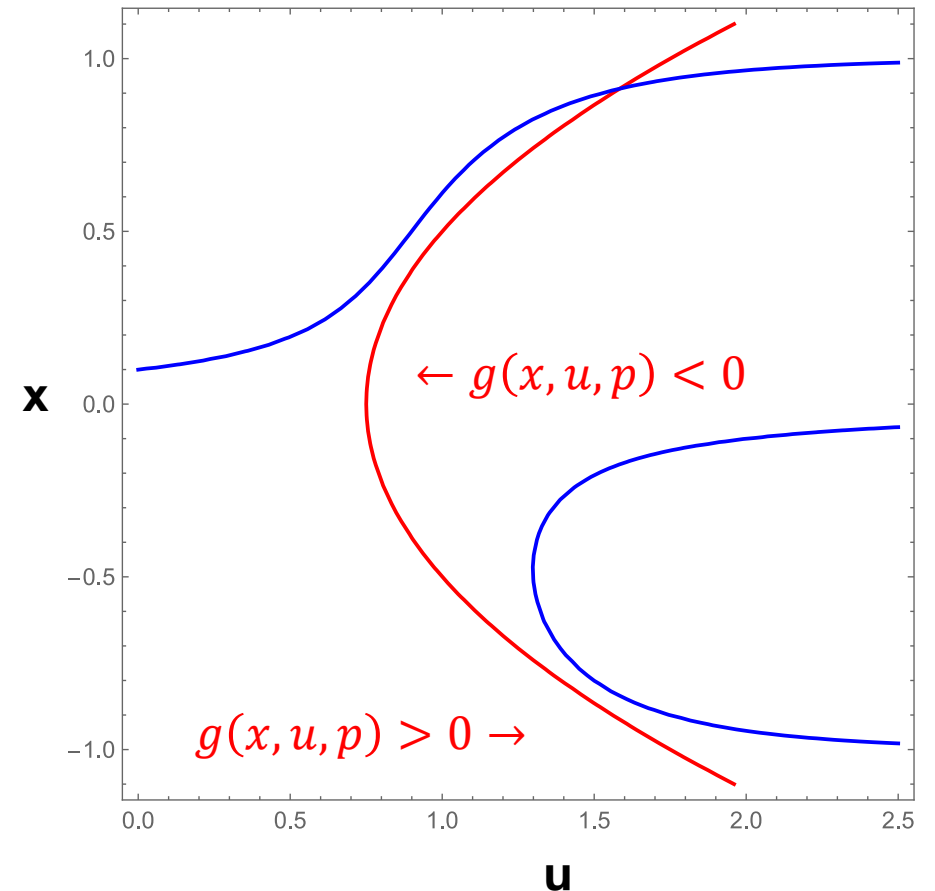
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The fast dynamics occurs
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The slow dynamics occurs
along the decision variable x

Decompose the
dynamics along
the systems'
dimensions



Separation of timescales

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$$2) \tau \dot{u} = g(x, u, p)$$

τ is small

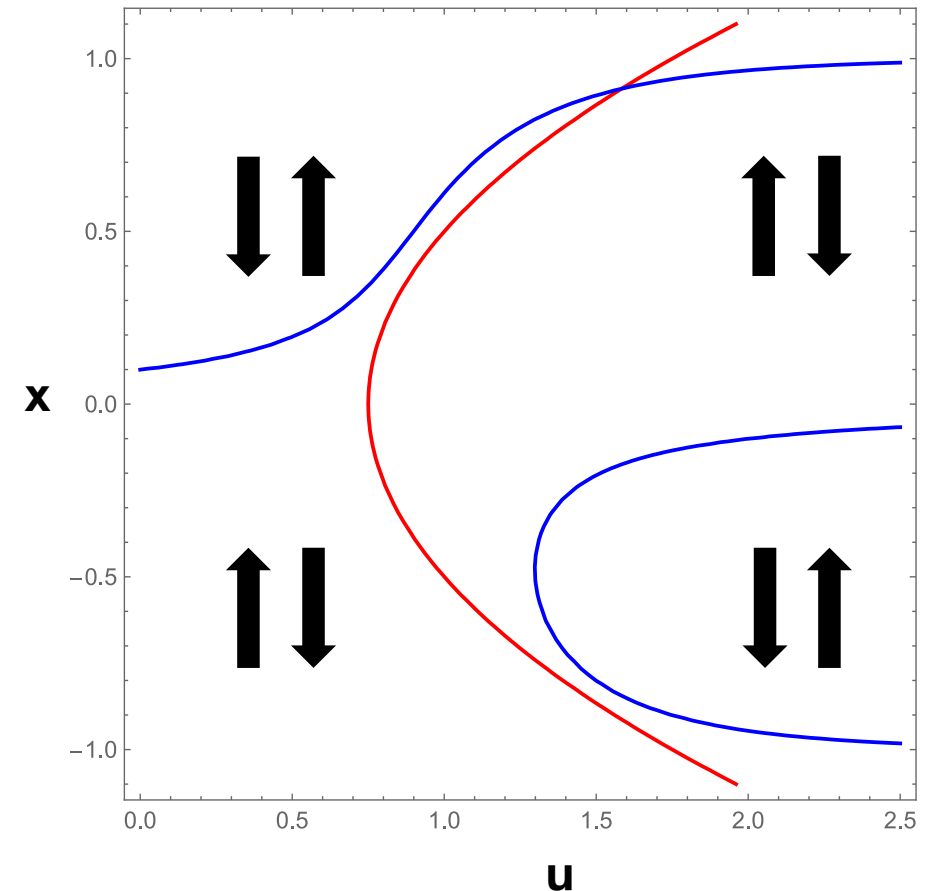
The fast dynamics occurs
along attention u

The slow dynamics occurs
along the decision variable x

Which arrows describe the slow dynamics?

1) left

2) right



Separation of timescales

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$$2) \tau \dot{u} = g(x, u, p)$$

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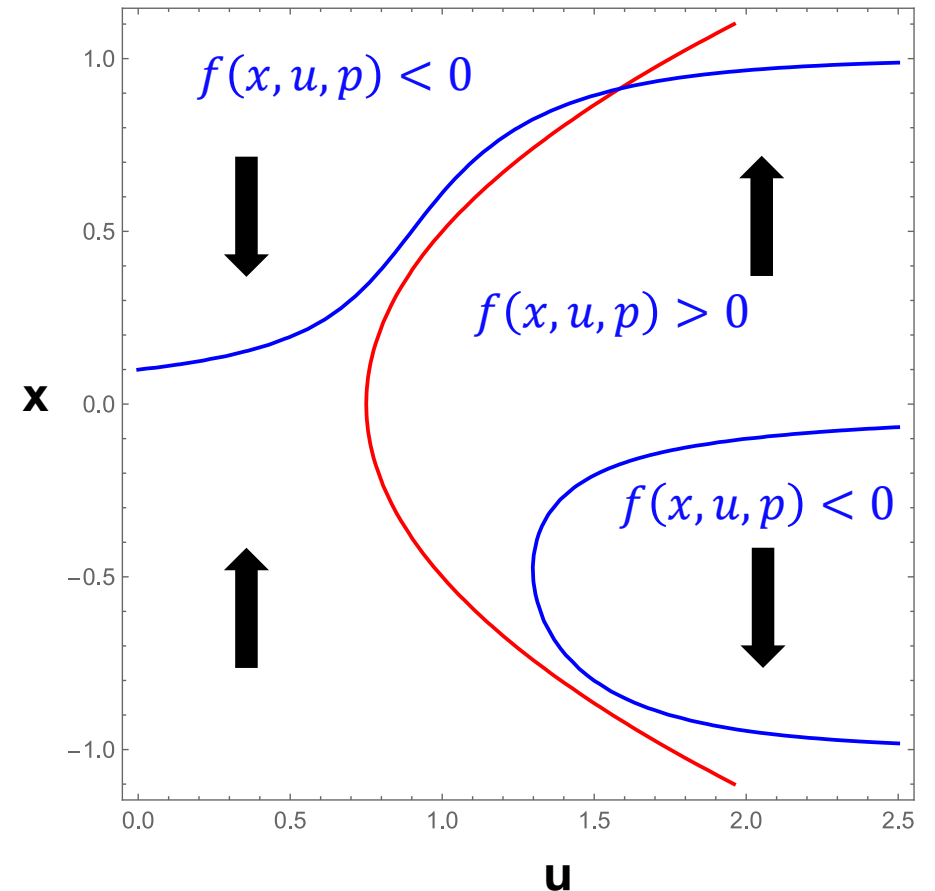
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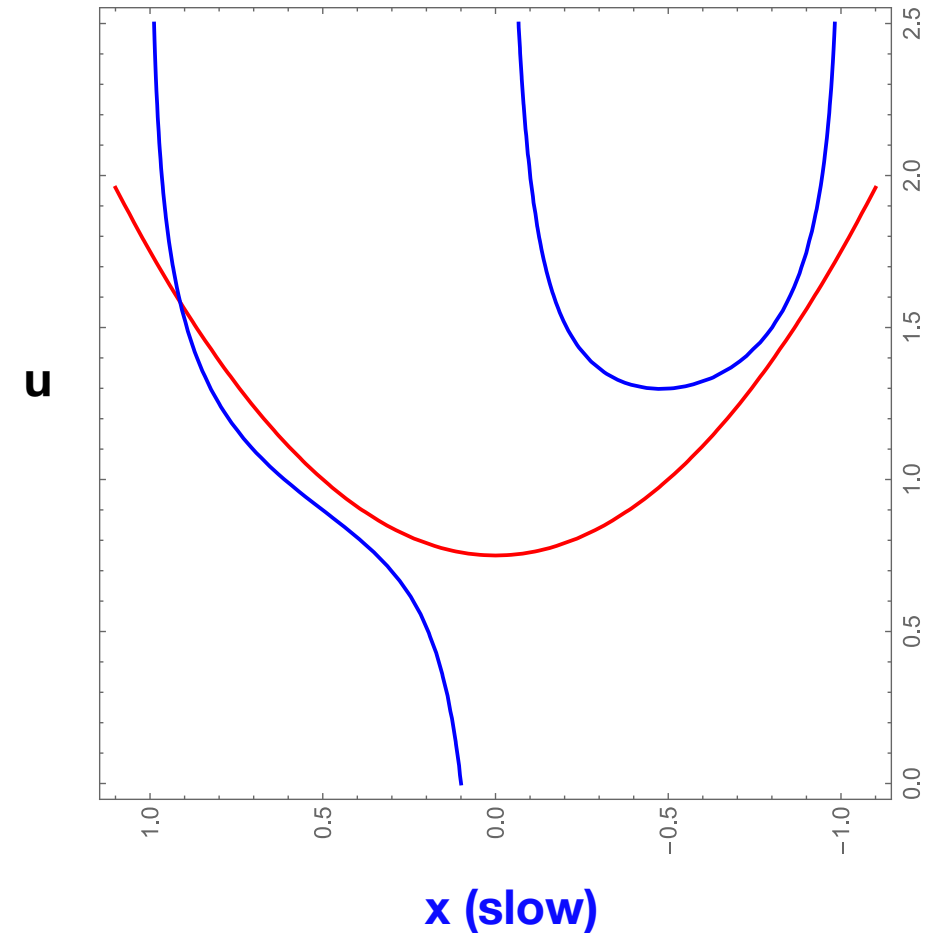


Separation of timescales

$$1) \dot{x} = f(x, u, p)$$

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You can consider the **slow variable** a parameter, along which there is an **equilibrium branch for the fast variable**

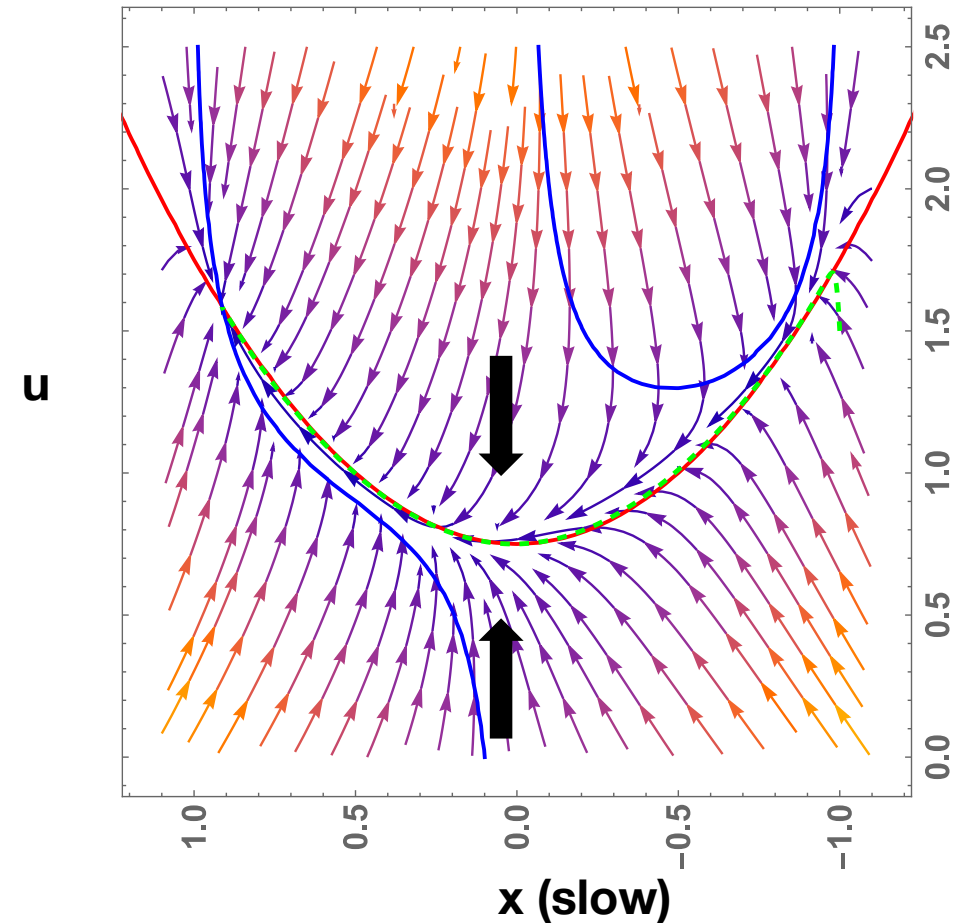


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If the state was not on the fast nullcline, the fast dynamics would take the state back to the fast nullcline

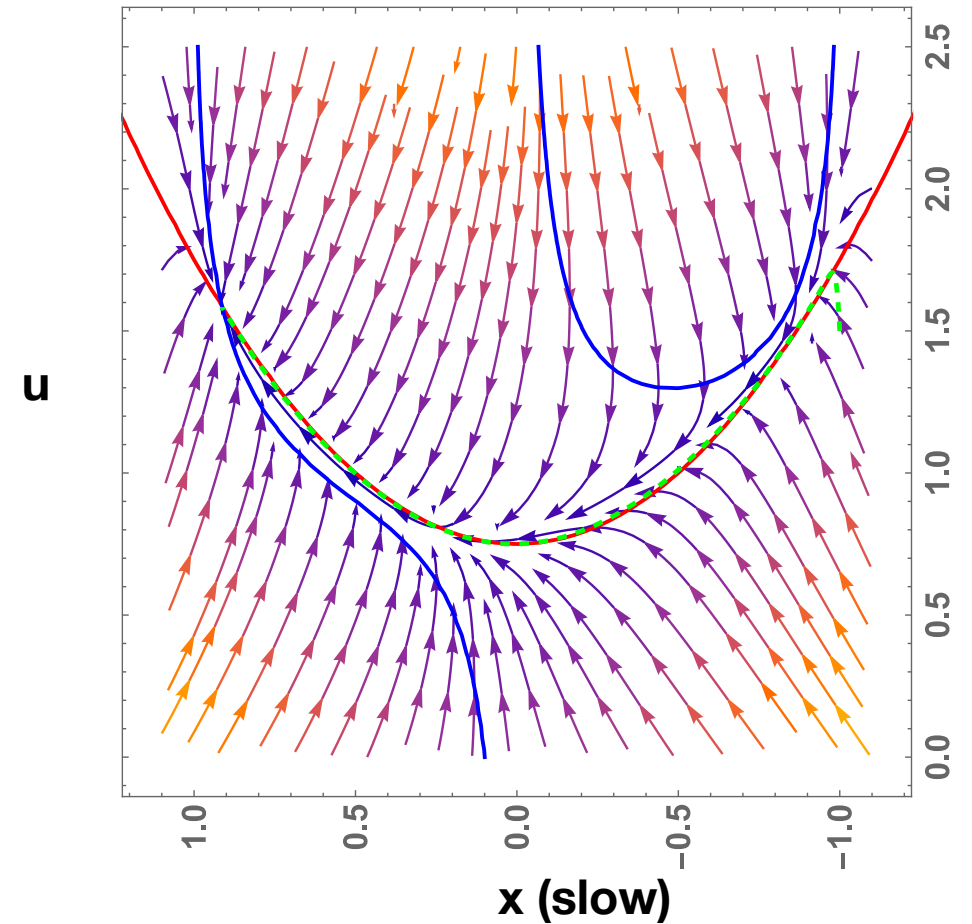


Separation of timescales

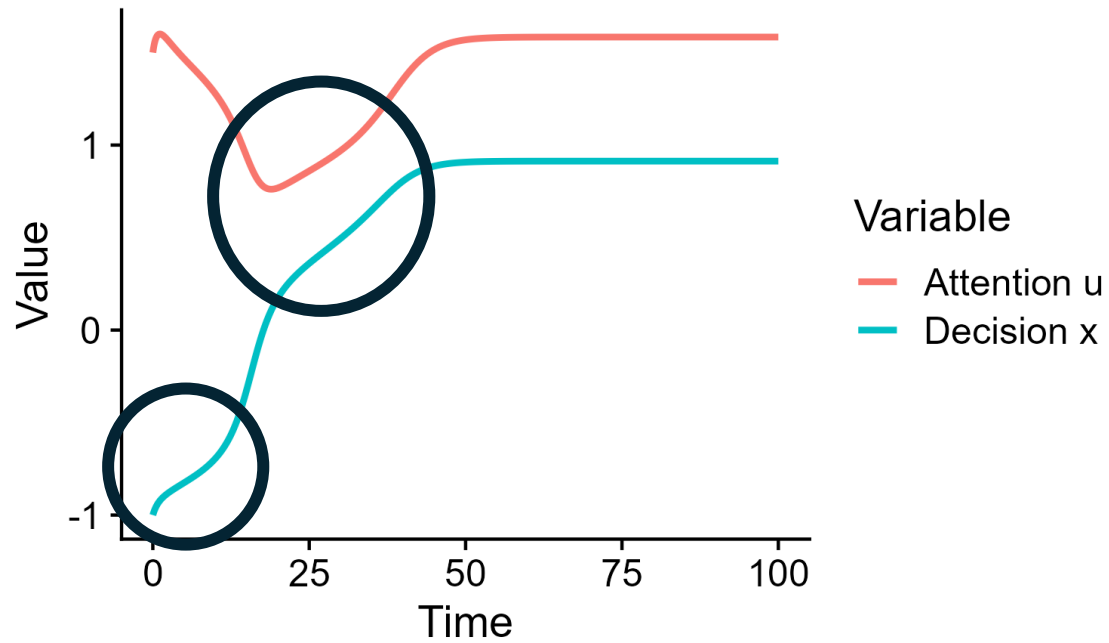
$$\begin{aligned} 1) \dot{x} &= f(x, u, p) \\ 2) \tau \dot{u} &= g(x, u, p) \end{aligned}$$

The **dynamics** mainly occur along the fast nullcline(s) in the direction indicated by the slow variable

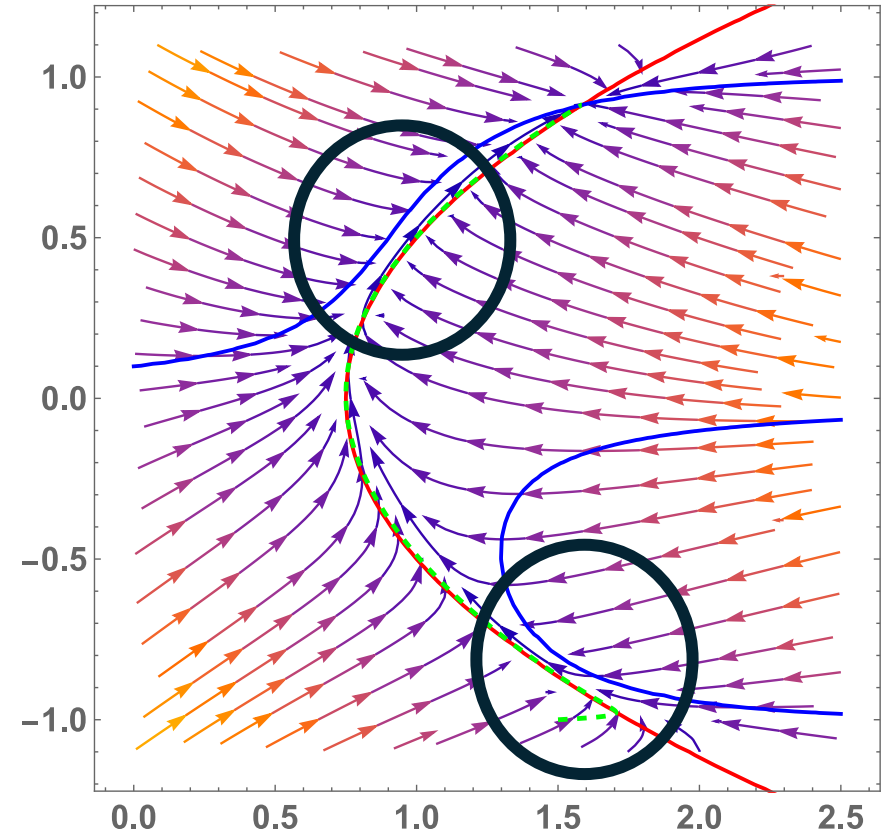
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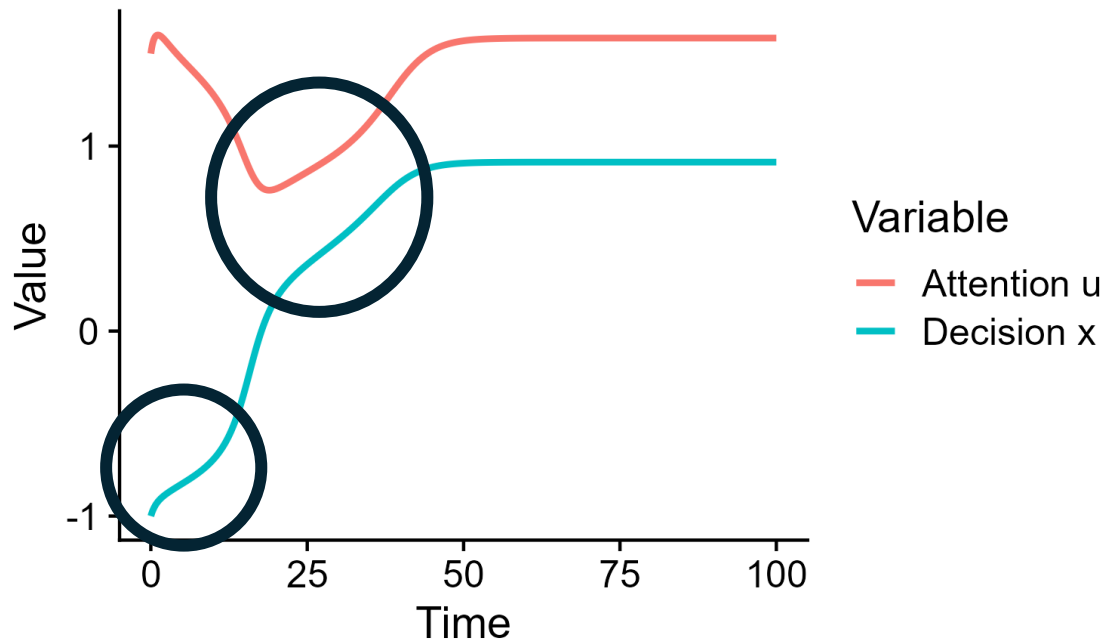
Ghost equilibria



**When passing near previous global equilibria
the dynamics slow down**

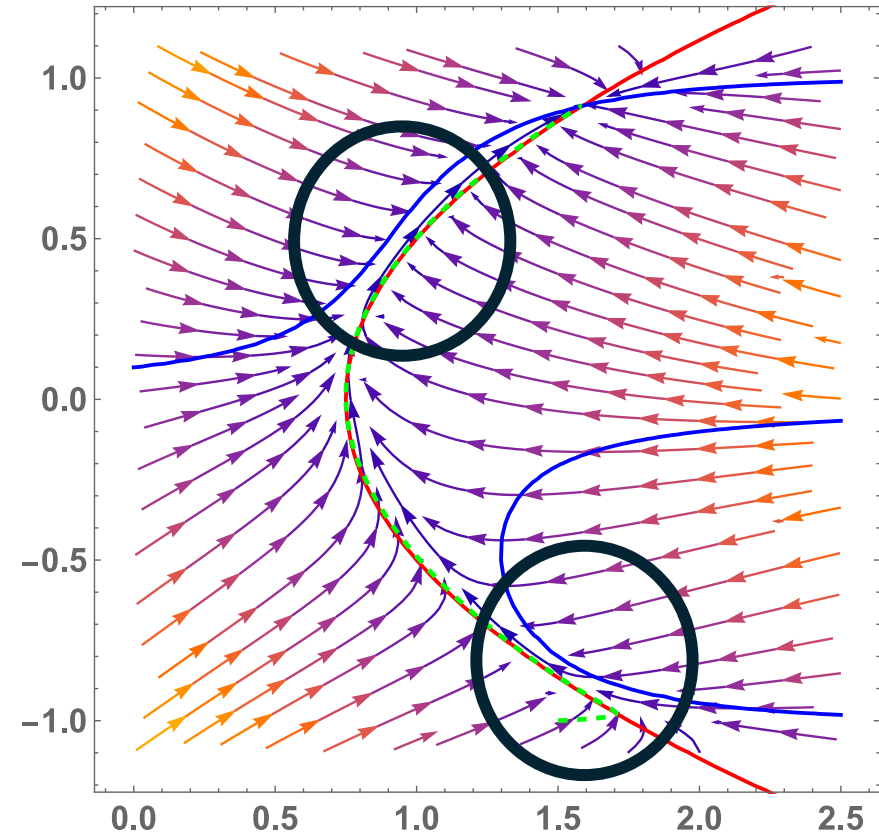


Ghost equilibria



When passing near previous global equilibria
the dynamics slow down

**Dynamics always slow down near the critical point
(critical slowing down)**



**Even if the equilibrium disappeared!
(the further away, the smaller the effect)**

Part 5

Symmetries in dynamical systems



Why do we care about symmetries in dynamical systems?

Why do we care about symmetries in dynamical systems?

1) Model development

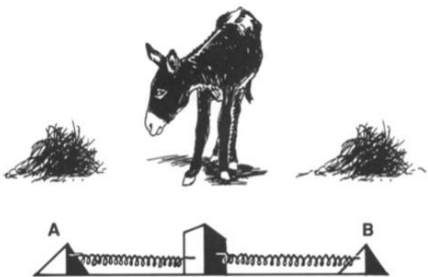
- **All decision-making systems should exhibit some symmetry. This models decision-making in absence of evidence.**

Why do we care about symmetries in dynamical systems?

1) Model development

- All decision-making systems should exhibit some symmetry. This models decision-making in absence of evidence.
- Problem of symmetry-breaking: problem of deadlock (individual) and consensus formation (collective)

Buridan's ass

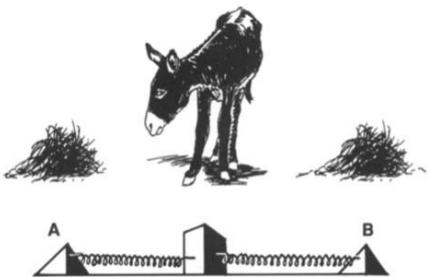


Why do we care about symmetries in dynamical systems?

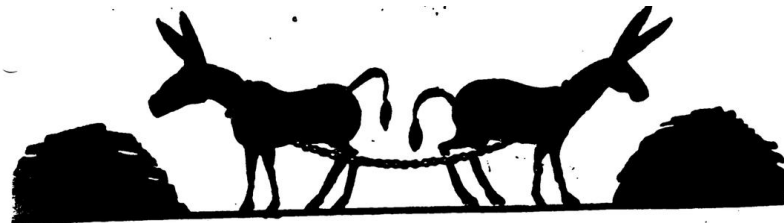
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Consensus formation

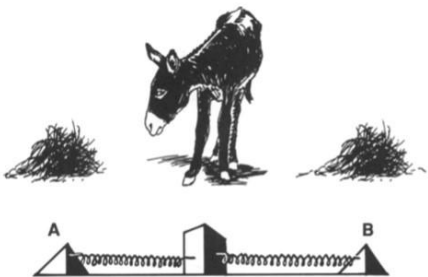


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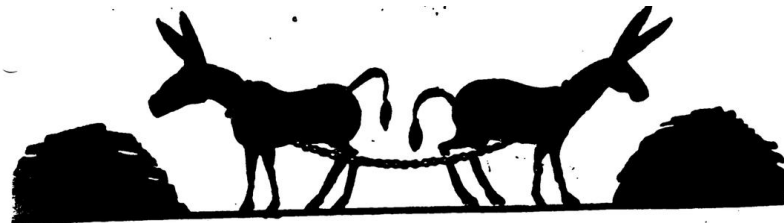
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Consensus formation

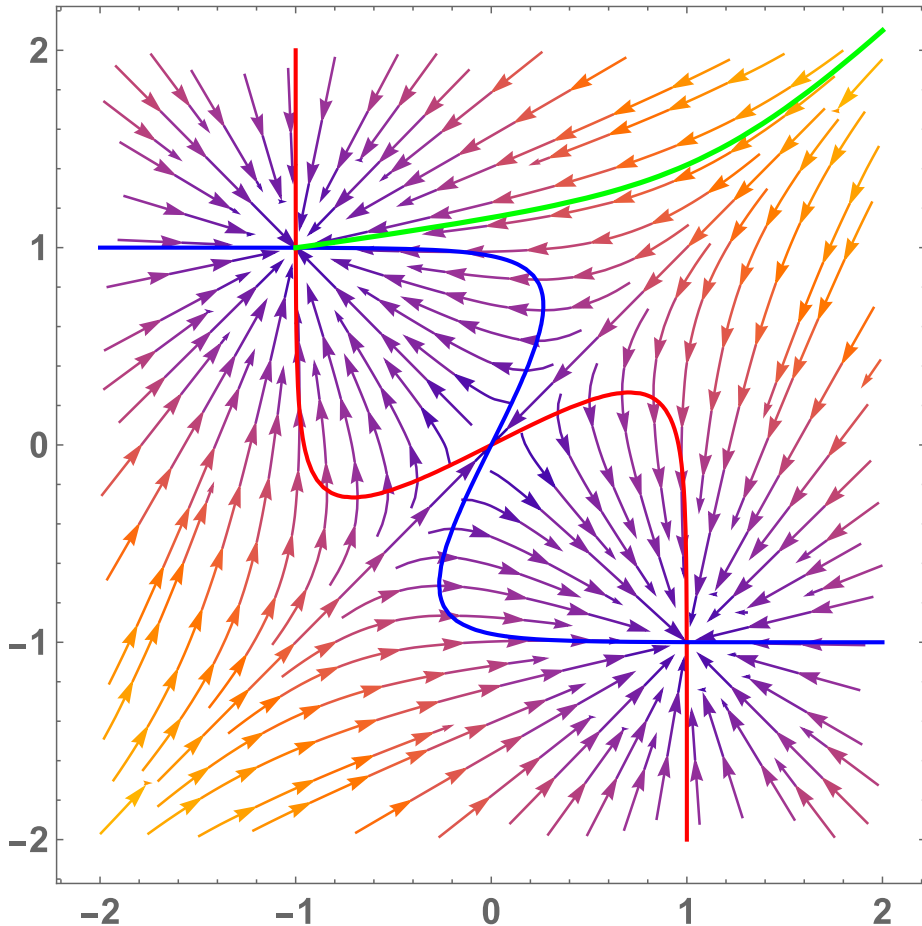


2) Model analysis

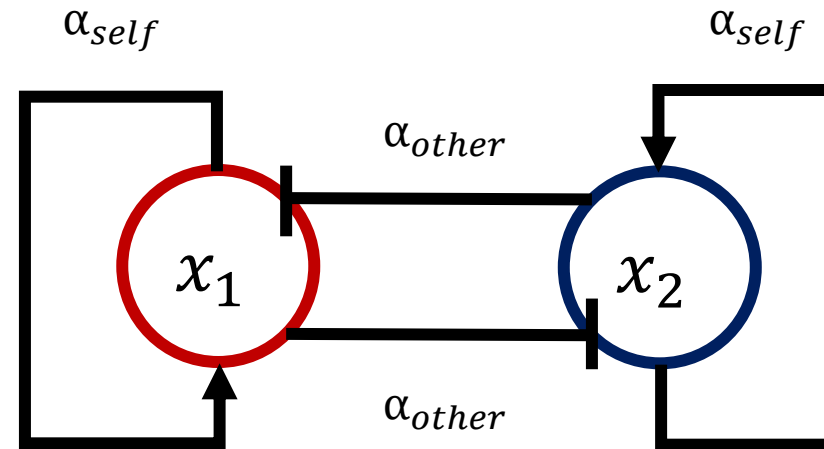
A model with symmetry can be reduced and simplified.
Example: mapping between ring attractor (many dimensions) and drift-diffusion (one dimensional)

Symmetries in dynamical systems

Equivariance: given a system $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{p})$, if for every solution $\mathbf{x}(t)$, $\gamma \mathbf{x}(t)$ is also a solution, the system is symmetric with respect to γ (for our purposes γ is a matrix) (formal definition at the end).

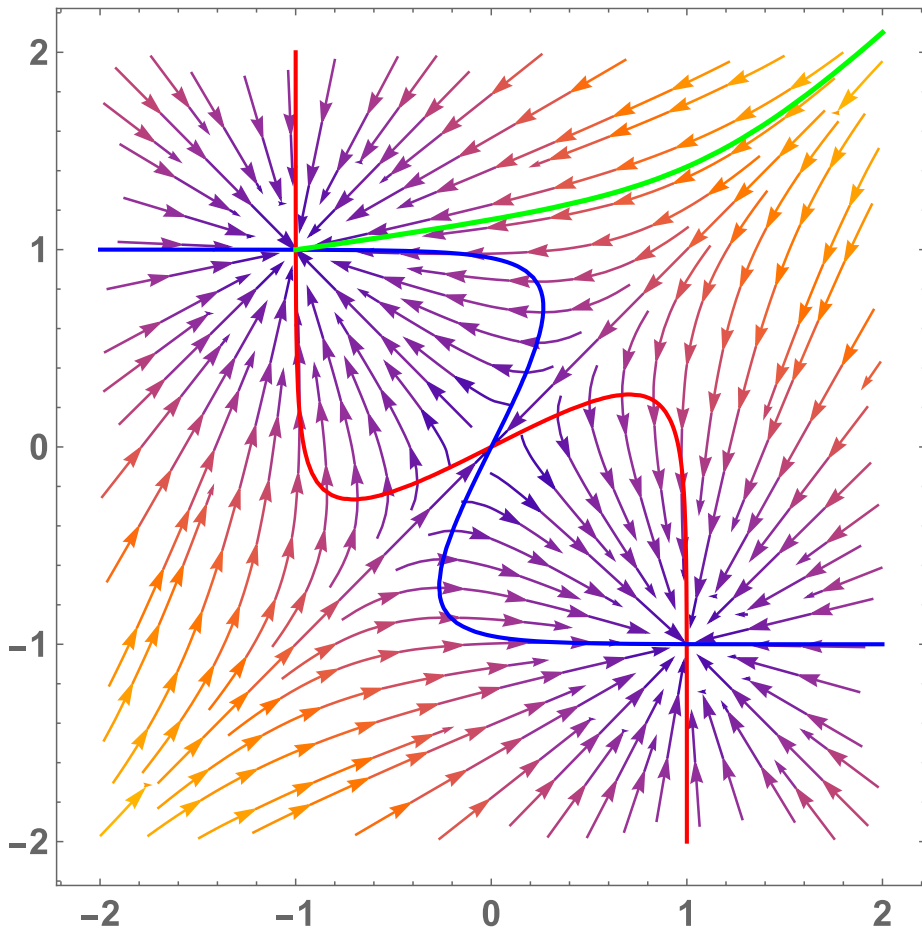


$$\begin{aligned} \dot{x}_1 &= -\delta x_1 + \tanh(u(\alpha_{self} x_1 + \alpha_{other} x_2)) & \alpha_{self} > 0 \\ \dot{x}_2 &= -\delta x_2 + \tanh(u(\alpha_{self} x_2 + \alpha_{other} x_1)) & \alpha_{other} < 0 \end{aligned}$$



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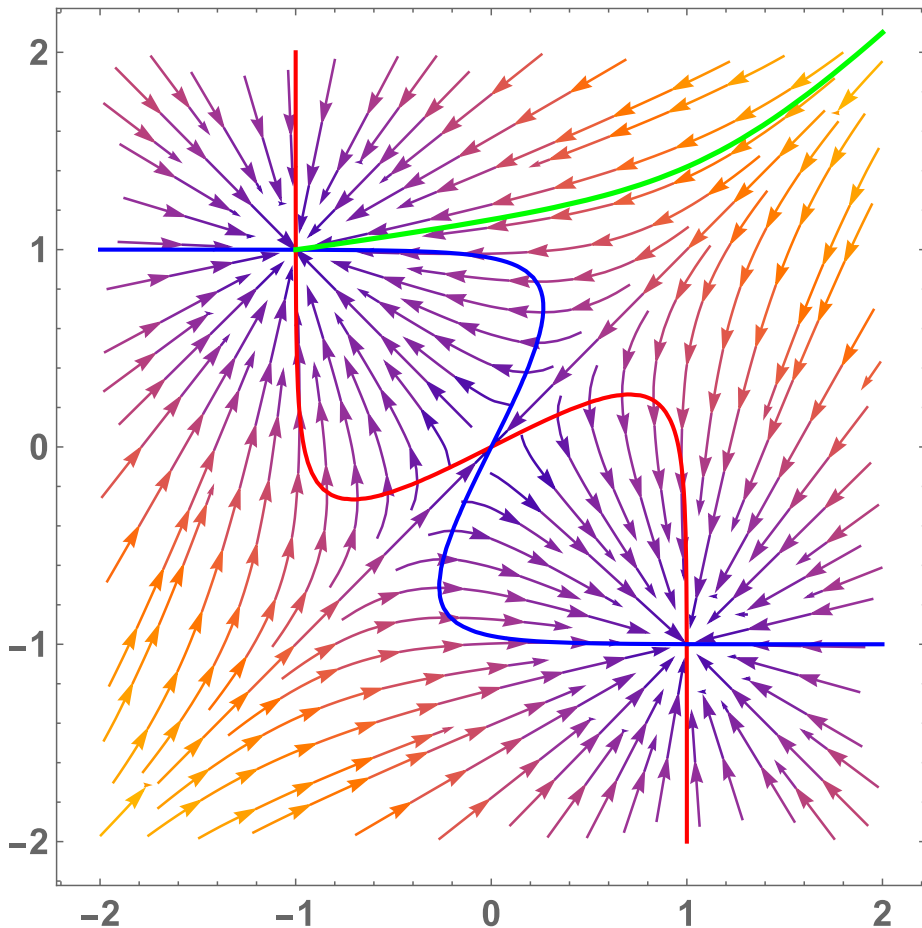
Two ways to establish symmetry:

1) Verify the criteria $\gamma f(\mathbf{x}, \mathbf{p}) = f(\gamma\mathbf{x}, \mathbf{p})$ Proof:

$$\dot{\mathbf{x}} = f(\mathbf{x}), y = \gamma\mathbf{x} \rightarrow \dot{y} = f(y) = f(\gamma\mathbf{x}), \gamma\dot{\mathbf{x}} = \gamma f(\mathbf{x}) \rightarrow \gamma f(\mathbf{x}) = f(\gamma\mathbf{x})$$

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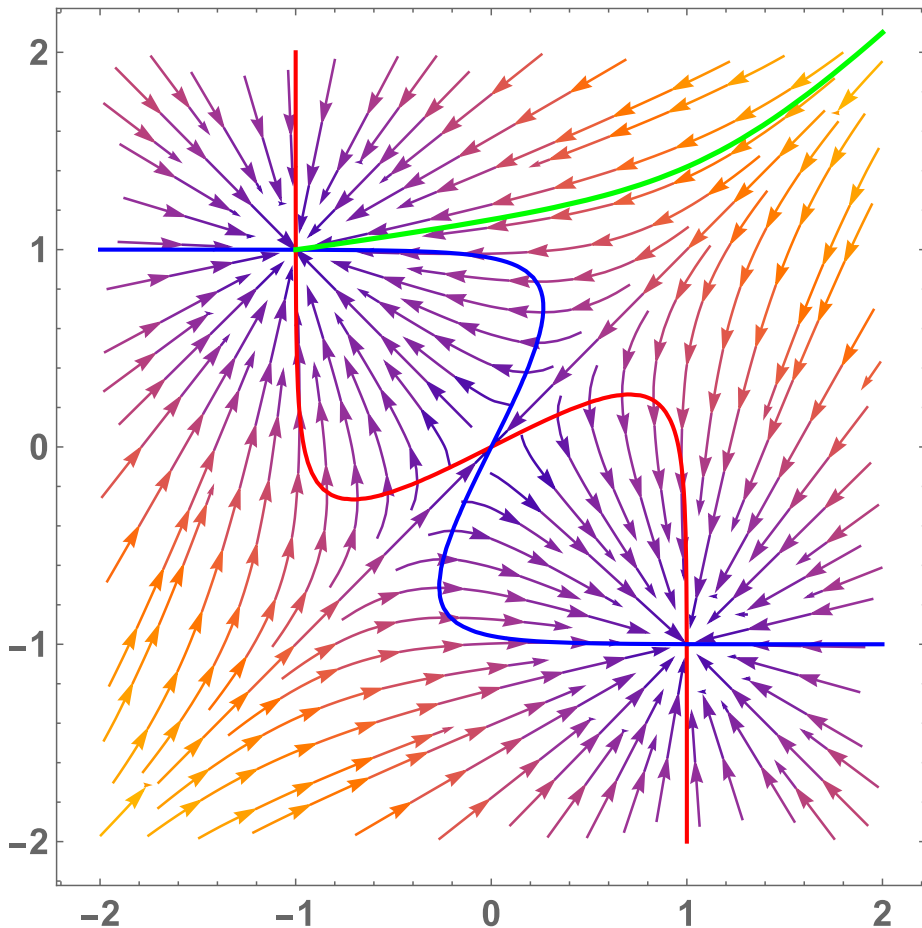
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Which equation does not hold for non equivariant systems?

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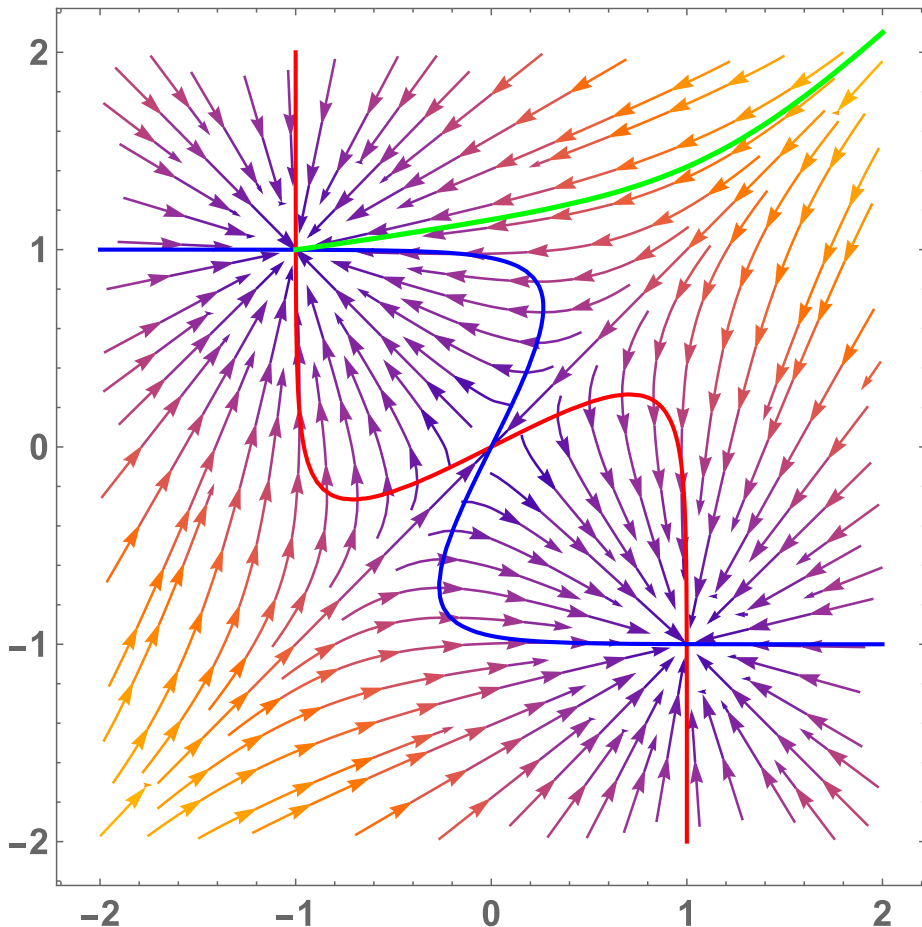
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Which equation does not hold for non equivariant systems?

$$\text{Correct change of variable: } \gamma^{-1}\dot{\mathbf{y}} = f(\gamma^{-1}\mathbf{y})$$

Symmetries in dynamical systems

Equivariance: given a system $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{p})$, if for every solution $\mathbf{x}(t)$, $\gamma\mathbf{x}(t)$ is also a solution, the system is symmetric with respect to γ (for our purposes γ is a matrix) (formal definition at the end).



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2) Just look at the vector field

**Under what transformation(s)
is the system equivariant:**

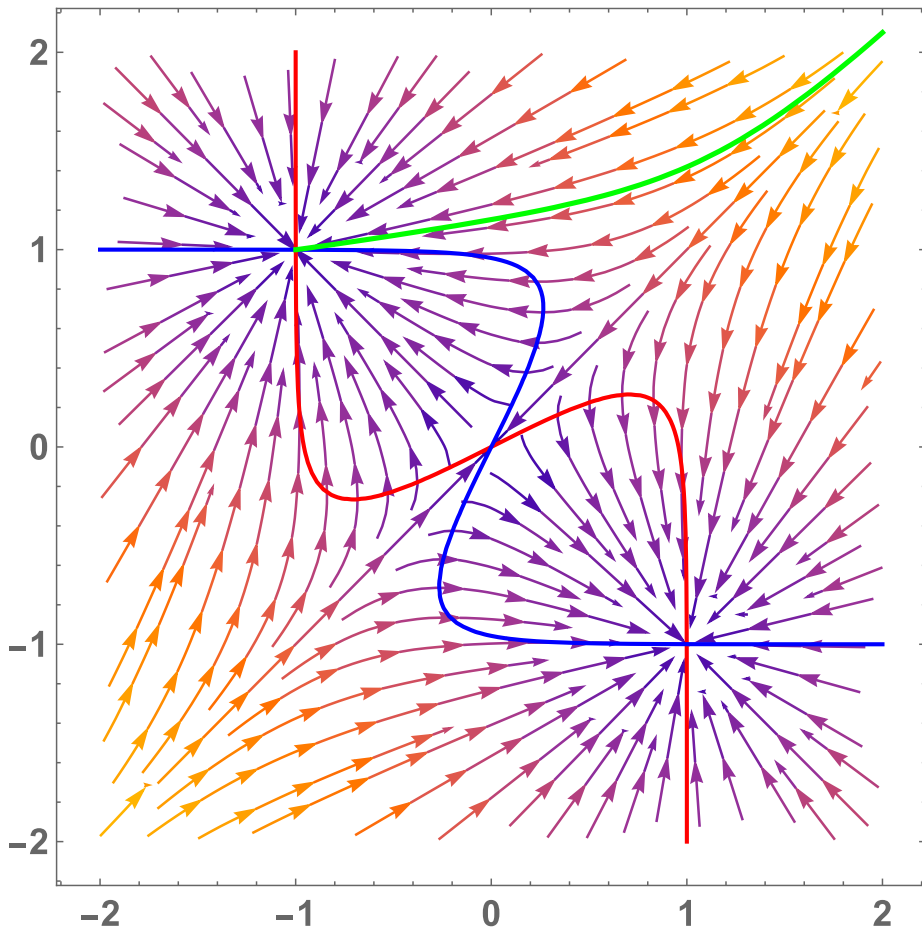
a) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ b) $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

c) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ d) $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

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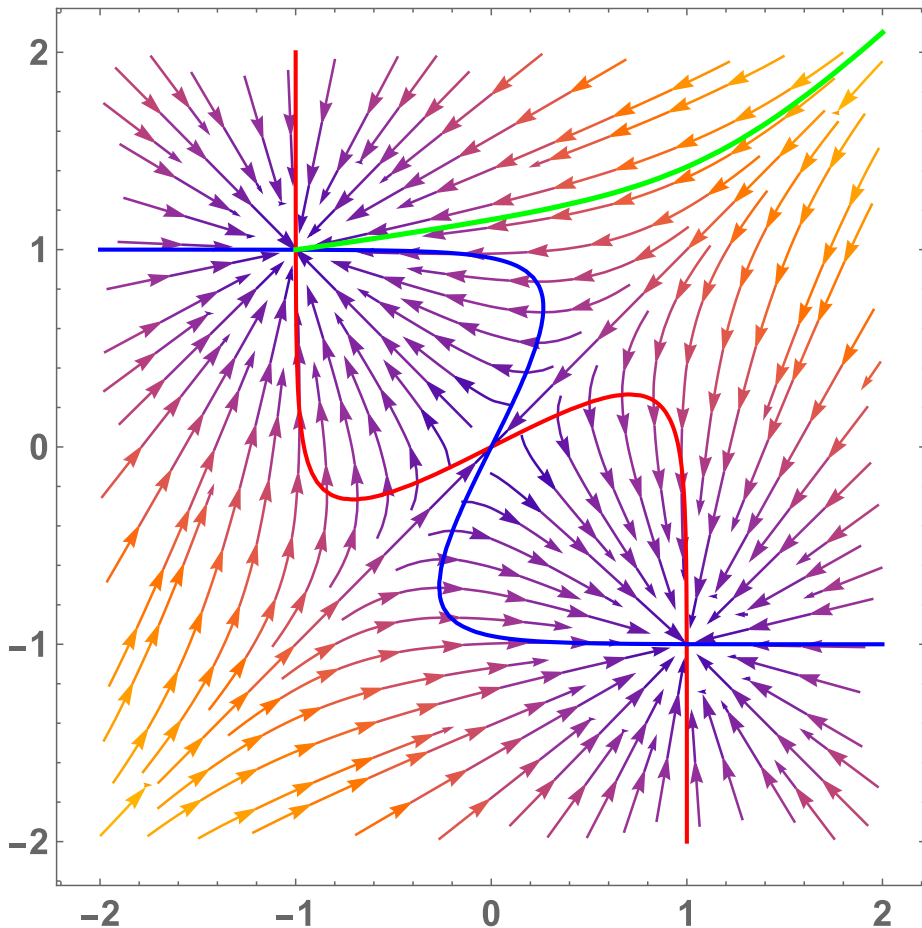
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S_n Symmetry group (**permutations** of n options) can be used to model multi-option decision-making

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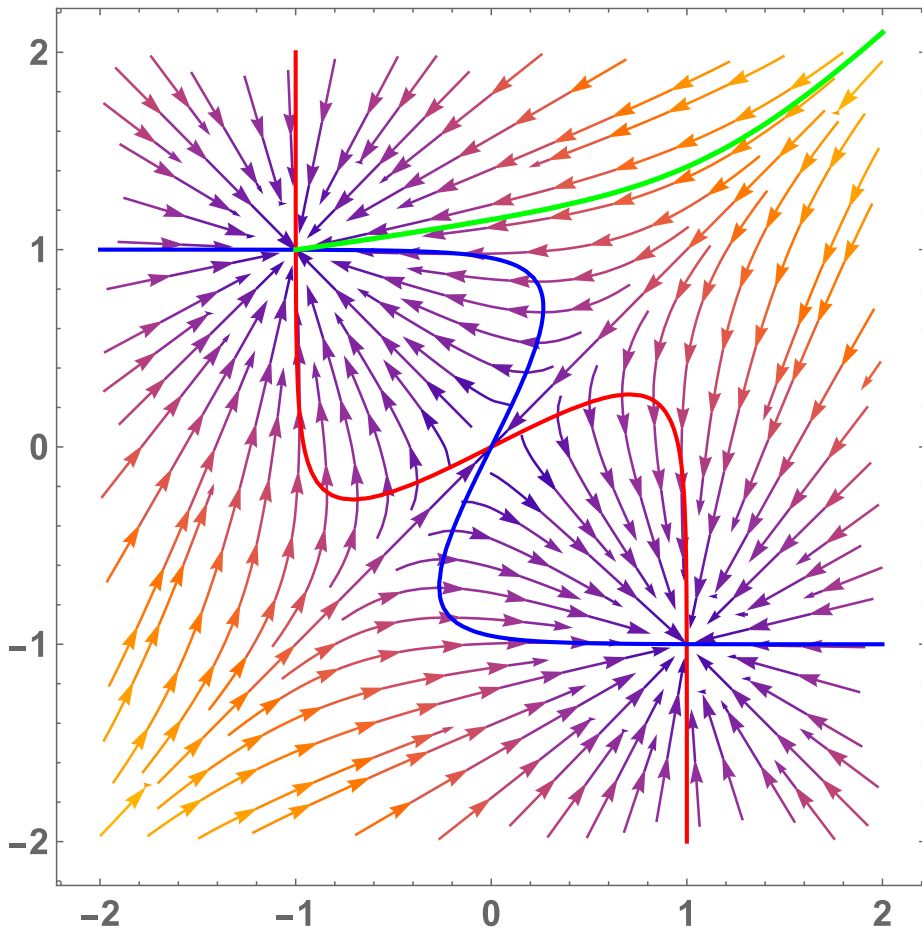
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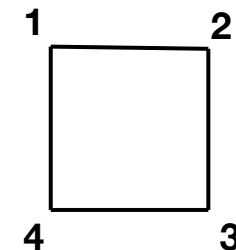


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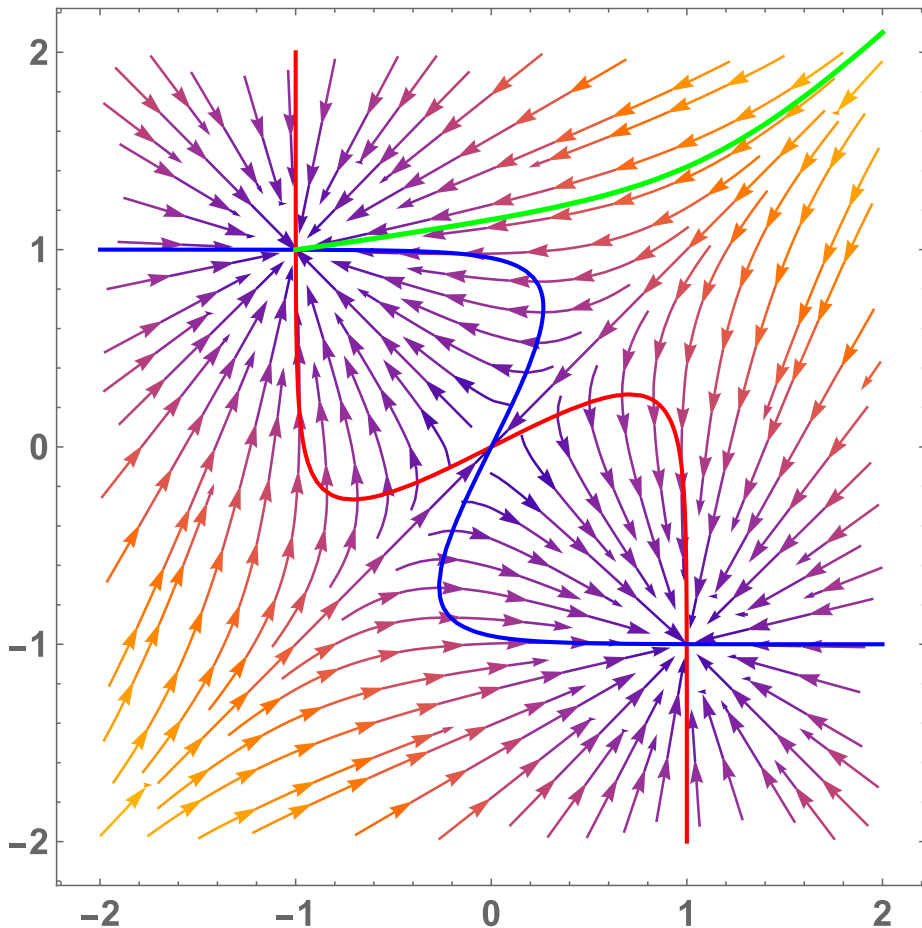
Is the shape symmetric under S_4 ?

- a) No
- b) Yes



Symmetries in dynamical systems

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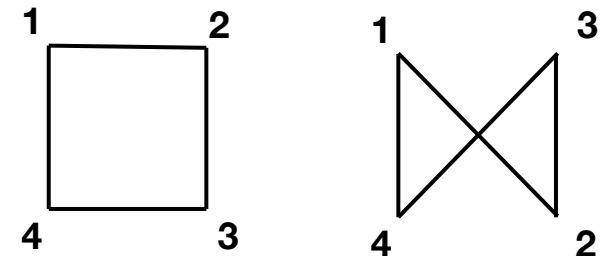


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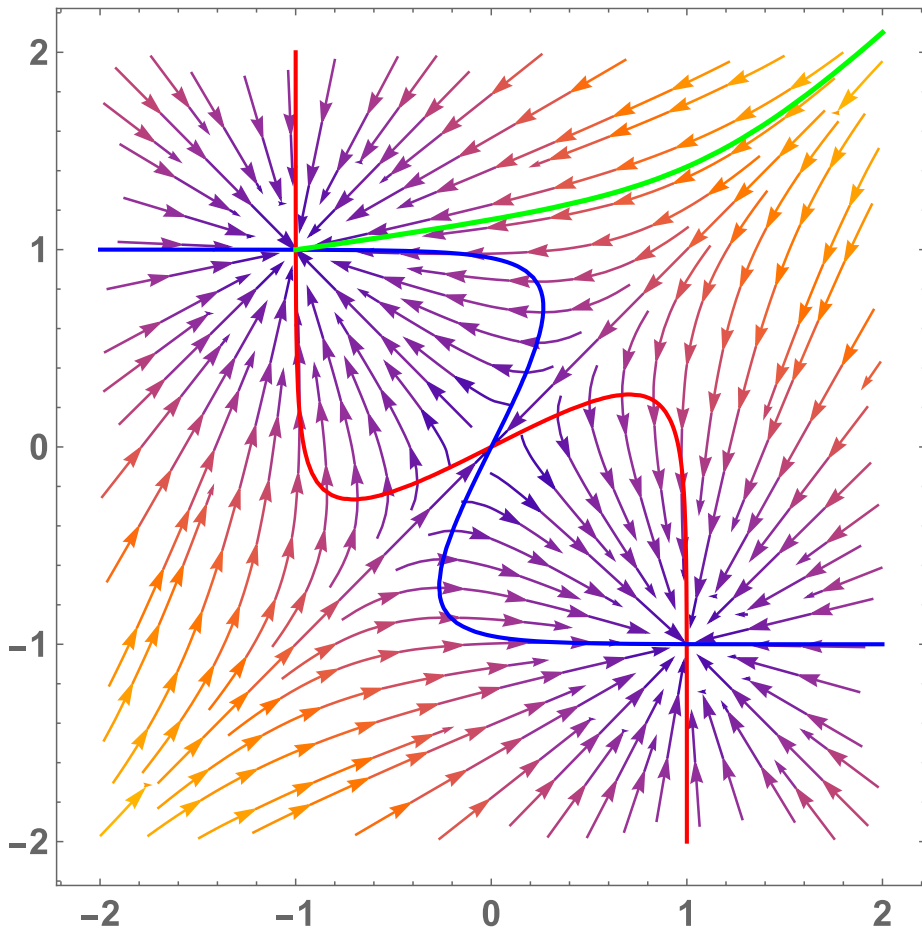
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Symmetric systems can have non-symmetric solutions: this is symmetry breaking

Part 6

The ring attractor



Ring attractor

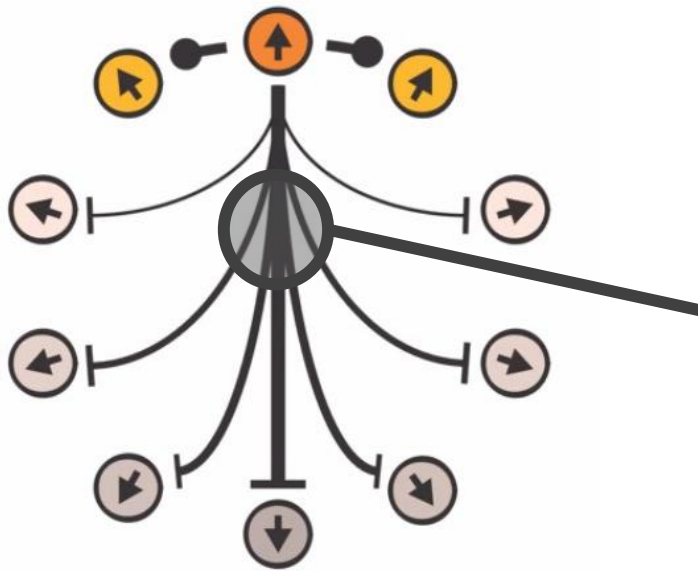
Direction

Interaction's kernel

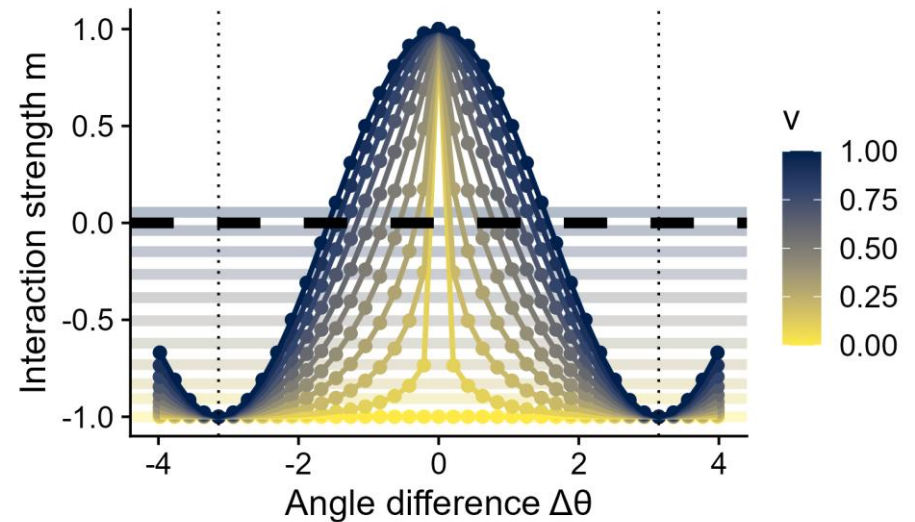
$$\dot{\mathbf{z}} = -\mathbf{z} + \tanh.(\mathbf{u} \mathbf{M} \mathbf{z} + \mathbf{b})$$

Network coupling $\mathbf{u}$

Targets' inputs



Short-range
excitation and
long-range
inhibition



Ring attractor

Direction

Interaction's kernel

$$\dot{\mathbf{z}} = -\mathbf{z} + \tanh. (\mathbf{u} \mathbf{M} \mathbf{z} + \mathbf{b})$$

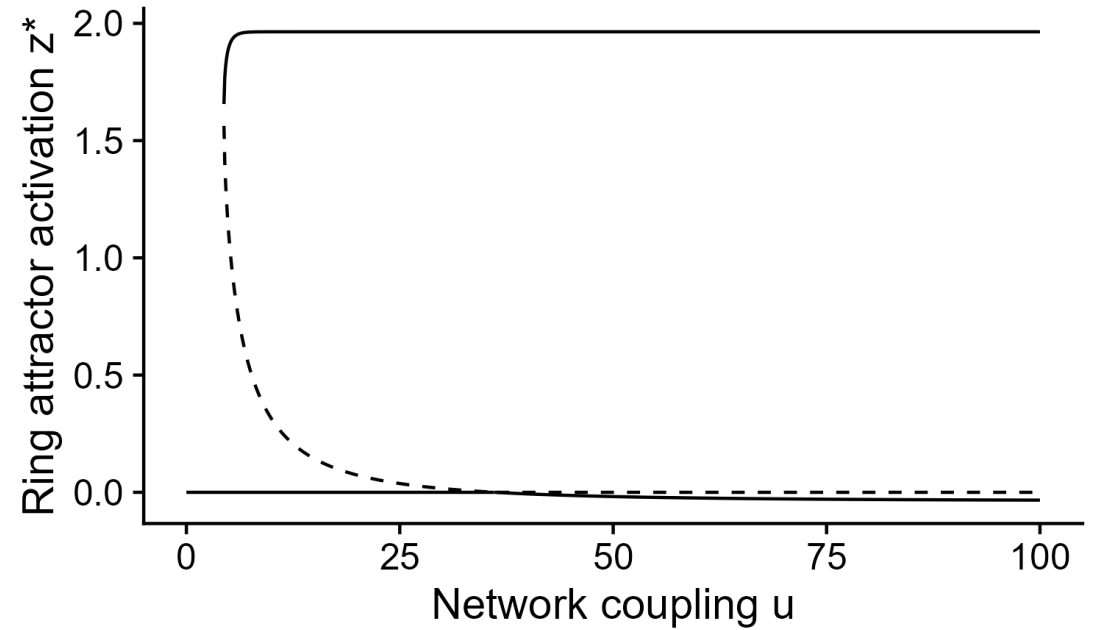
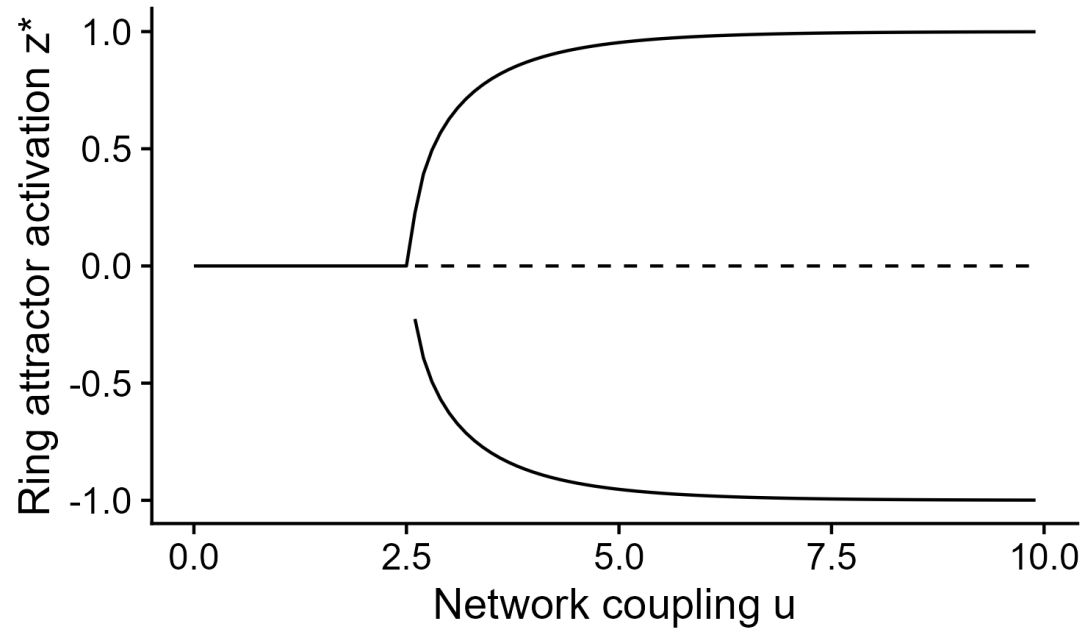
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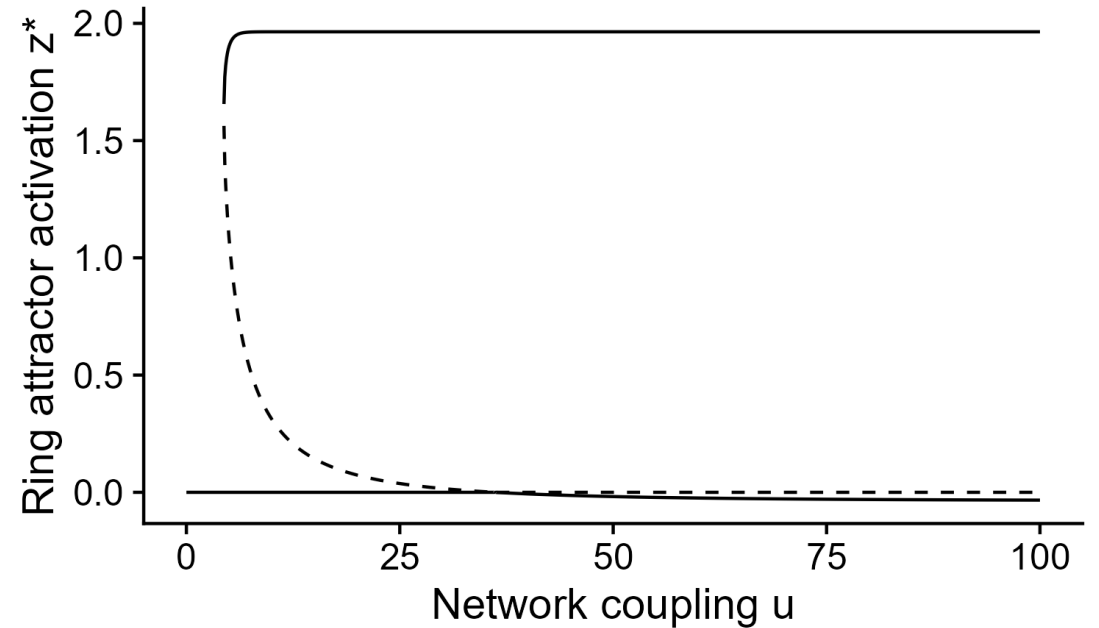
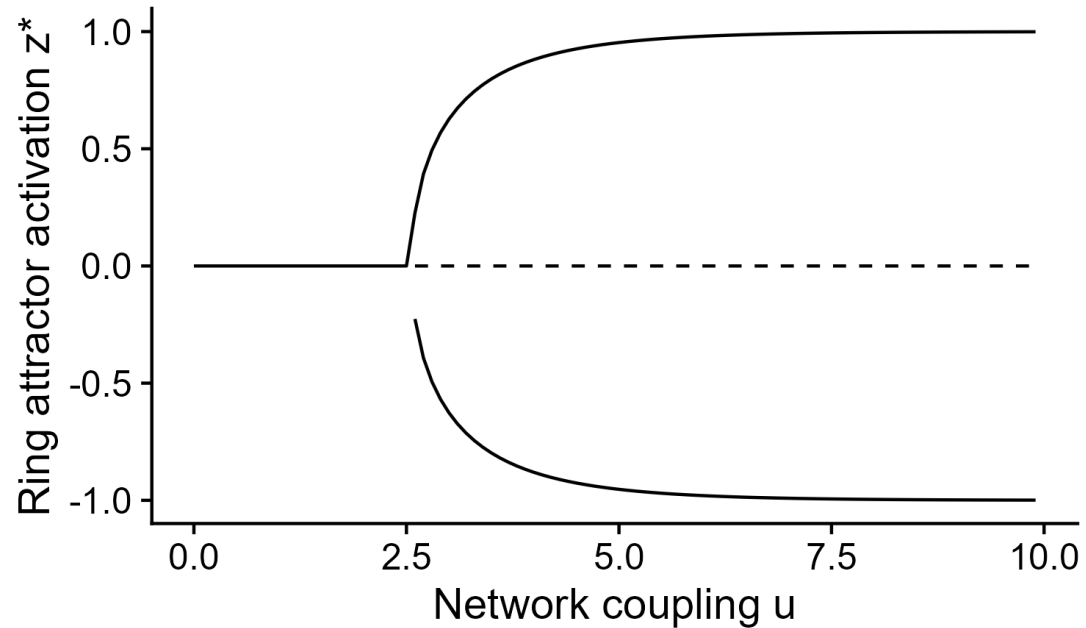
Multi-option decision-making

Circular symmetry $\mathcal{C}_n$

Reduced model dynamic

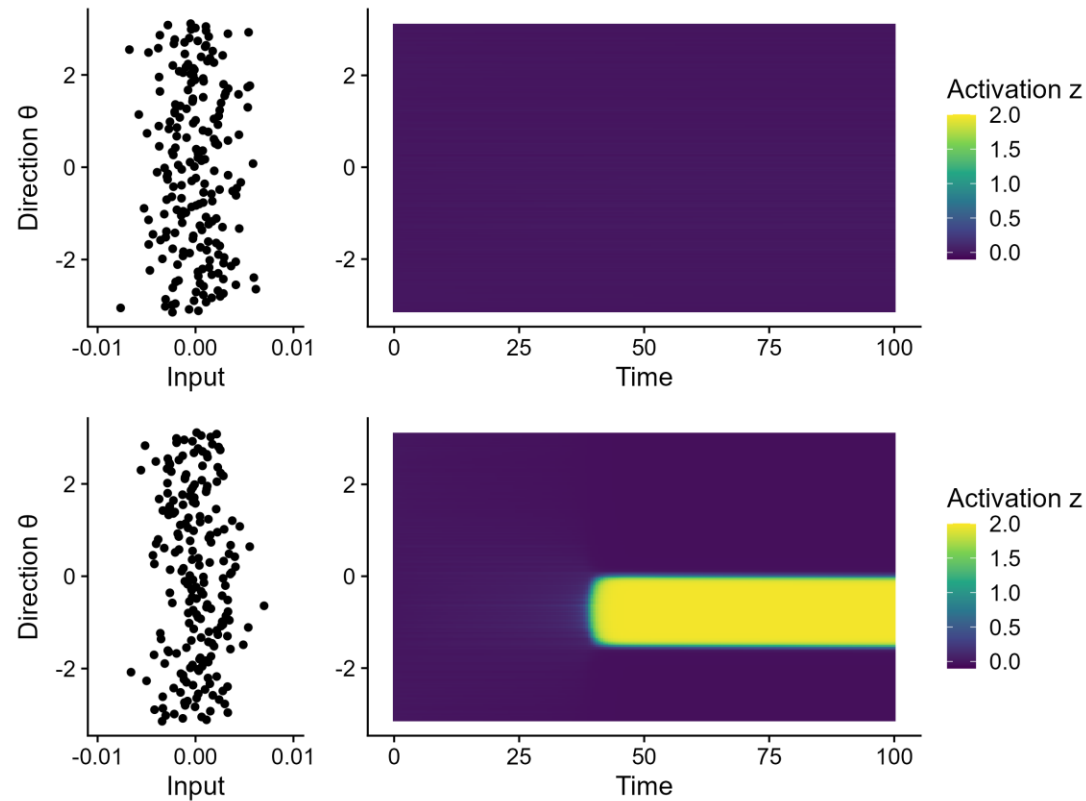


Reduced model dynamic



What bifurcations are these?

Selective ultra-sensitivity

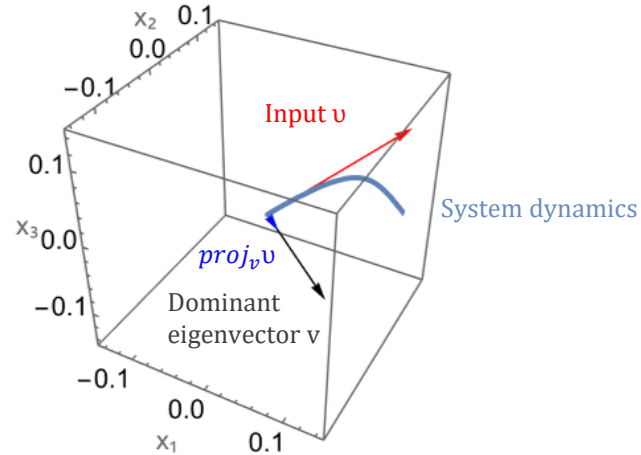
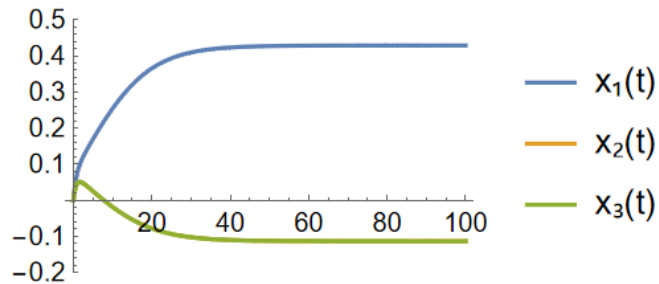


A ring attractor showing off his Fourier glasses

Mathematica notebook for designing input in frequency domain

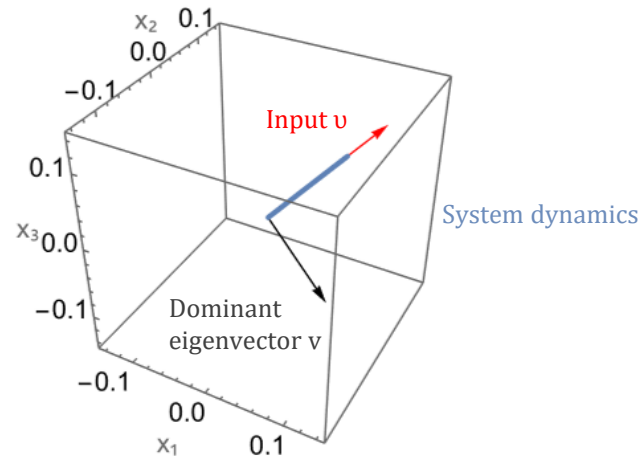
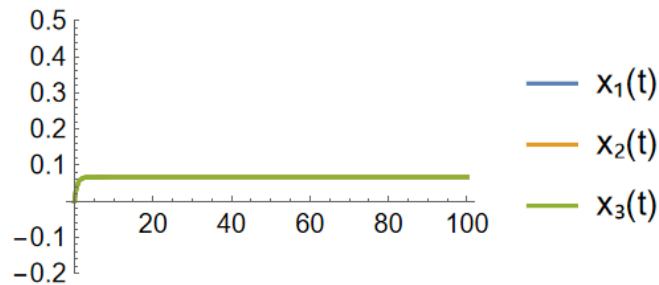
Selective ultra-sensitivity

Partially aligned input



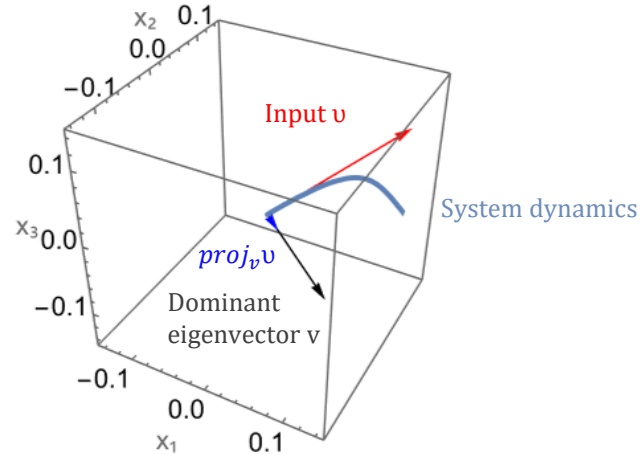
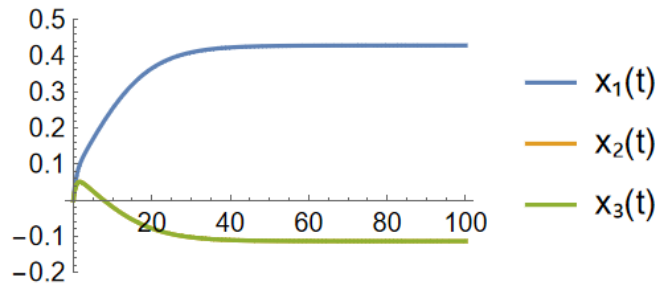
- At the critical point, a decision-maker responds to inputs that have a non-zero component when projected along the center manifold (one of the dynamics' dominant eigenvector)

Misaligned input

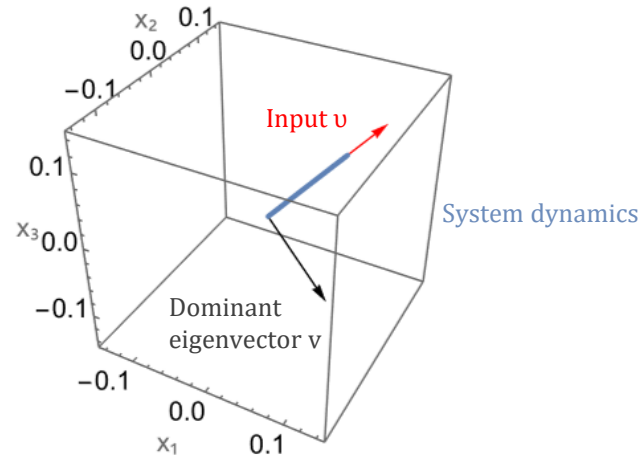
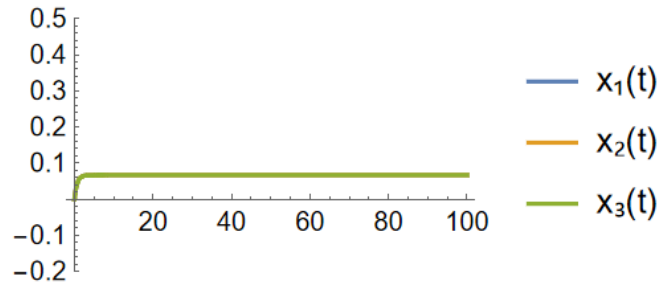


Selective ultra-sensitivity

Partially aligned input



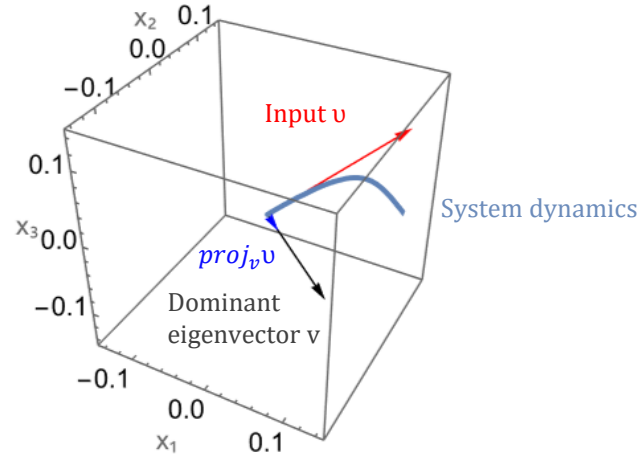
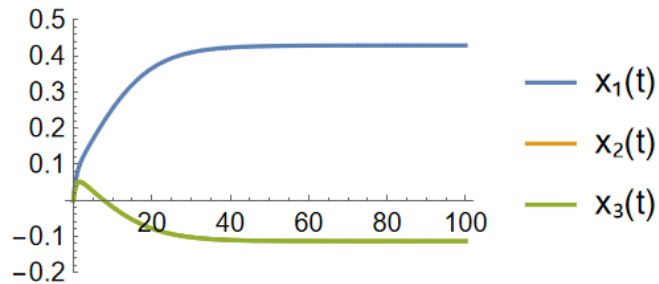
Misaligned input



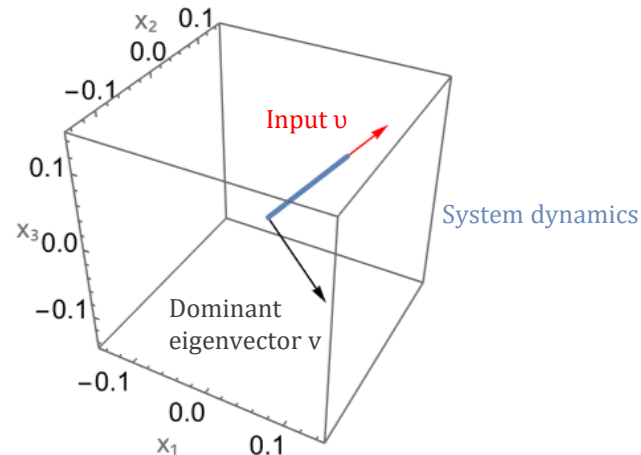
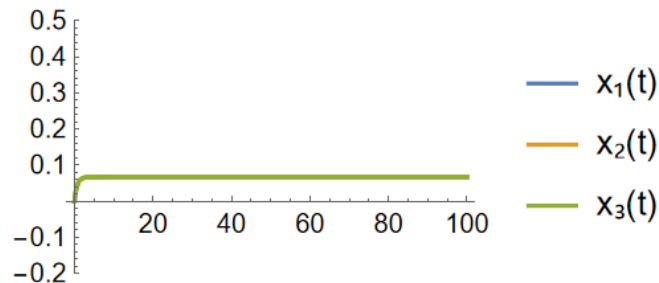
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- This is linked to infinite susceptibility
- For random inputs, dimensionality decreases probability of alignment

Selective ultra-sensitivity

Partially aligned input



Misaligned input



- At the critical point, a decision-maker responds to inputs that have a non-zero component when projected along the center manifold (one of the dynamics' dominant eigenvector)
- This is linked to infinite susceptibility
- For random inputs, dimensionality decreases probability of alignment
- For all circulant unimodal kernels, the center manifold is spanned by the Fourier basis of the first harmonic, which modes are the dominant eigenvalues. Every linear combination of the Fourier basis of a given harmonic is also one of its corresponding eigenvector (which preserves circulant symmetry of opinion formation by encoding phase). The center manifold for the ring attractor is a plane (2-d)

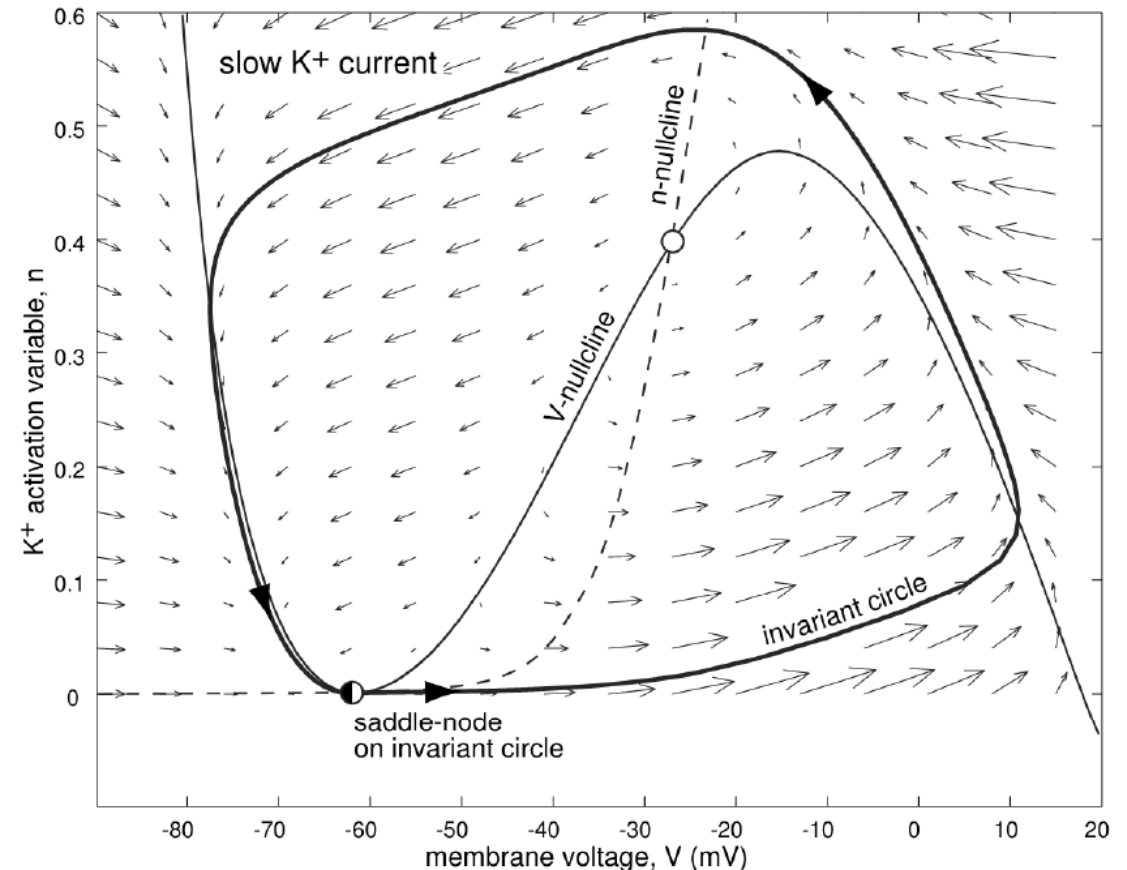
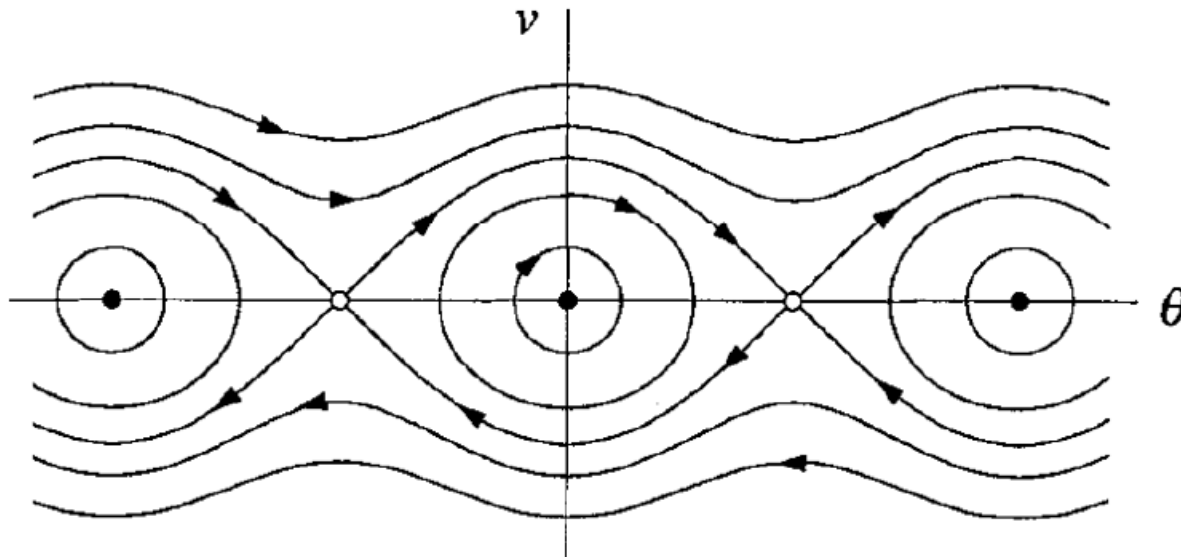
Caution

When linearization fails $\text{nullity}(J(\mathbf{x}^*, \mathbf{p}^*)) \neq 0$, different definitions of stability (Lyapunov) can be used and strange things can happen: a point can be *stable but not attractive* (e.g., Lotka-Volterra predator-prey, undamped pendulum) or *attractive but unstable* (e.g., in excitable systems). See end if curious

Lyapunov stable: if for every distance $\varepsilon > 0$ from the equilibrium there exists a starting condition within $\delta > 0$ for which the trajectory remains within ε , the equilibrium is Lyapunov stable

Why is δ needed in the definition?

Which points are stable but not attractive and attractive but not stable?



Caution

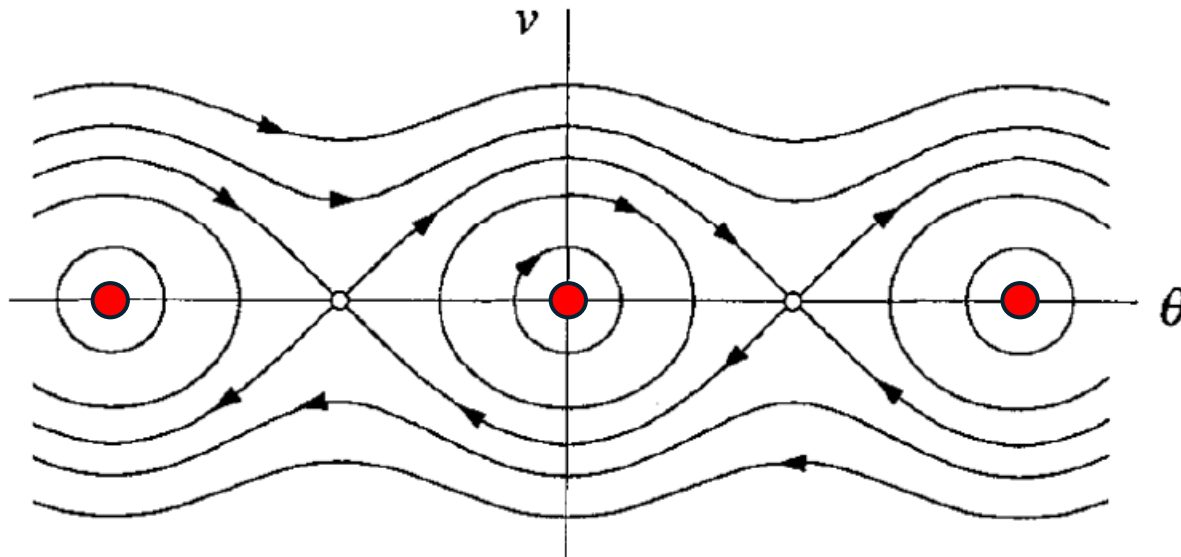
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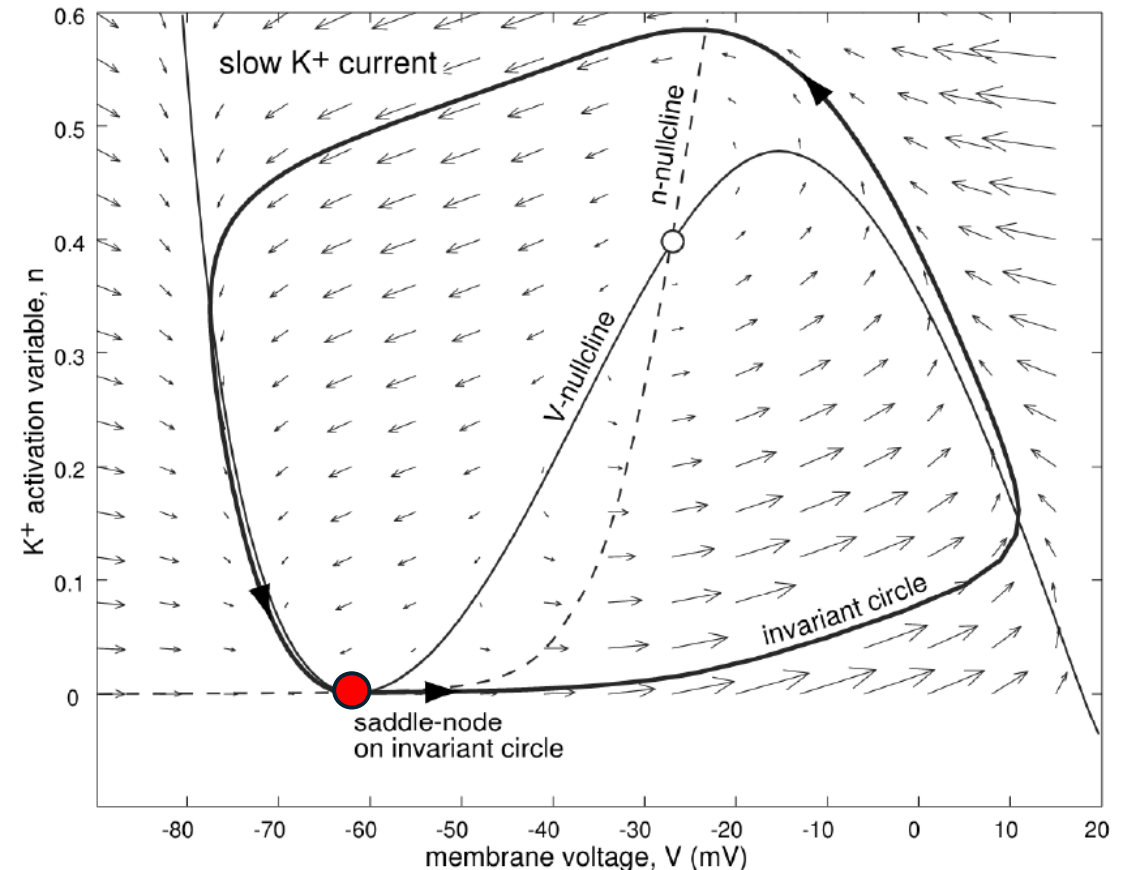
Why is δ needed in the definition?

Which points are stable but not attractive and attractive but not stable?

Lyapunov stable but not attractive



Lyapunov unstable but attractive



Equivariance: given a system $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{p})$, if for every solution $\mathbf{x}(t)$, $\gamma\mathbf{x}(t)$ is also a solution, the system is Γ -equivariant (symmetric under the action of a member of the group Γ). A group is always defined in respect to 1) an operation and 2) its members. A group is a collection of elements that are closed in respect to an operation that has associative property, and identity element, and an inverse. For our purposes, γ is a matrix and the operation is the dot product.